

Learning Objectives

1. Understand the properties of ideal amplifiers and the concepts of gain, input impedance, output impedance, and feedback. *Section 8.1.*
2. Understand the difference between open-loop and closed-loop op-amp configurations; and compute the gain of (or complete the design of) simple inverting, non-inverting, summing, and differential amplifiers using ideal op-amp analysis. Analyze more advanced op-amp circuits, using ideal op-amp analysis; and identify important performance parameters in op-amp data sheets. *Section 8.2.*
3. Analyze and design simple active filters. Analyze and design ideal integrator and differentiator circuits. *Sections 8.3–8.4.*
4. Understand the principal physical limitations of an op-amp. *Section 8.5.*

8.1 IDEAL AMPLIFIERS

Amplifiers are an essential aspect of many electronic applications. Perhaps the most familiar use of an amplifier is to convert the low-voltage, low-power signal from a digital audio player (e.g., iPhone or MP3 player) to a level suitable for driving a pair of earbuds or headphones, as shown in Figure 8.1. Amplifiers have important applications in practically every field of engineering because the vast majority of transducers and sensors used for measurement produce electrical signals, which are then amplified, filtered, sampled, and processed by analog and digital electronic instrumentation. For example, mechanical engineers use thermistors, accelerometers, and strain gauges to convert temperature, acceleration, and strain into electrical signals. These signals must be amplified prior to transmission and then filtered (a function carried out by amplifiers) prior to sampling the data in preparation for producing a digital version of the original analog signal. Other, less obvious, functions such as impedance isolation are also performed by amplifiers. It should now be clear that amplifiers do more than simply produce an enlarged replica of a signal although that function is certainly very important. This chapter explores the general features of amplifiers and focuses on the characteristics and applications of a particularly important integrated-circuit amplifier, the **operational amplifier**.



Figure 8.1 Typical digital audio player
(© Jim Kearns)

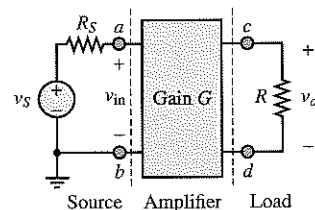


Figure 8.2 Amplifier between source and load

Ideal Amplifier Characteristics

The simplest model for an amplifier is depicted in Figure 8.2, where a signal v_S is amplified by a constant factor G , called the *voltage gain* of the amplifier. Ideally, the *input impedance* of the amplifier is infinite such that $v_{in} = v_S$; if its *output impedance* is zero, v_o will be determined by the amplifier independent of R such that:

$$v_o = G v_{in} = G v_S \quad \text{Ideal amplifier} \quad (8.1)$$

Note that the input seen by the amplifier is a Thévenin source (v_S in series with R_S), while the output seen by the amplifier is a single equivalent resistance R .

A more realistic (but still quite simple) amplifier model is shown in Figure 8.3. In this figure the concepts of input and output impedance of the amplifier are incorporated as single resistances R_{in} and R_{out} , respectively. That is, from the perspective

of the load R the amplifier acts as a Thévenin source ($A v_{in}$ in series with R_{out}), while from the perspective of the external source (v_S in series with R_S) the amplifier acts as an equivalent resistance R_{in} . The constant A is the multiplier associated with the dependent (controlled) voltage source and is known as the *open-loop gain*.¹

Using the amplifier model of Figure 8.3 and applying voltage division, the input voltage to the amplifier is now:

$$v_{ab} = v_{in} = \frac{R_{in}}{R_S + R_{in}} v_S \quad (8.2)$$

The output voltage of the amplifier can also be found by applying voltage division, where:

$$v_o = A v_{in} \frac{R}{R_{out} + R} \quad (8.3)$$

Substitute for v_{in} and divide both sides by v_S to obtain:

$$\frac{v_o}{v_S} = A \frac{R_{in}}{R_{in} + R_S} \frac{R}{R_{out} + R} \quad (8.4)$$

which is the overall voltage gain from v_S to v_o . The voltage gain G of the amplifier itself is:

$$G \equiv \frac{v_o}{v_{in}} = A \frac{R}{R_{out} + R} \quad (8.5)$$

For this model, the voltage gain G is dependent upon the external resistance R , which means that the amplifier performs differently for different loads. Moreover, the input voltage v_{in} to the amplifier is a modified version of v_S . Neither of these results seem desirable. Rather, it stands to reason that the gain of a “quality” amplifier would be independent of its load and would not impact its source signal. These attributes are achieved when $R_{out} \ll R$ and $R_{in} \gg R_S$. In the limit that $R_{out} \rightarrow 0$:

$$\lim_{R_{out} \rightarrow 0} \frac{R}{R + R_{out}} = 1 \quad (8.6)$$

such that:

$$G \equiv \frac{v_o}{v_{in}} \approx A \quad \text{when } R_{out} \rightarrow 0 \quad (8.7)$$

Also, in the limit that $R_{in} \rightarrow \infty$:

$$\lim_{R_{in} \rightarrow \infty} \frac{R_{in}}{R_{in} + R_S} = 1 \quad (8.8)$$

such that

$$v_{in} \approx v_S \quad \text{when } R_{in} \rightarrow \infty \quad (8.9)$$

In general, a “quality” voltage amplifier will have a very small output impedance and a very large input impedance.

¹The voltage gain G and the open-loop gain A may also be designated as A_V and A_{VOL} , respectively. Electrical conductance is also designated as G ; as always, it is important to correctly interpret a symbol from the context in which it is used. Happily, conductance G is rarely used in engineering work. Its inverse, resistance R , is preferred instead.

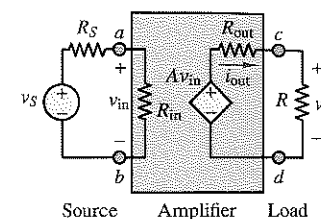


Figure 8.3 Simple voltage amplifier model

Input and Output Impedance

In general, the input impedance R_{in} and the output impedance R_{out} of an amplifier are defined as:

$$R_{in} = \frac{v_{in}}{i_{in}} \quad \text{and} \quad R_{out} = \frac{v_{OC}}{i_{SC}} \quad (8.10)$$

where v_{OC} is the open-circuit voltage and i_{SC} is the short-circuit current at the output of the amplifier. An ideal voltage amplifier has zero output impedance and infinite input impedance so that the amplifier does not suffer from loading effects at its input or output terminals. In practice, voltage amplifiers are designed to have large input impedance and small output impedance.

It is a worthwhile exercise to show that an ideal *current amplifier* has zero input impedance and infinite output impedance. Also, as was suggested in Chapter 7, an ideal *power amplifier* is designed so that its input impedance matches its source network and its output impedance matches its load impedance.

Feedback

Feedback, which is the process of using the output of an amplifier to reinforce or inhibit its input, plays a critical role in many amplifier applications. Without feedback an amplifier is said to be in *open-loop* mode; with feedback an amplifier is said to be in *closed-loop* mode. The output of the amplifier model shown in Figure 8.3 does not affect its input (because there is no path from output to input), so feedback is not present, and the model is open-loop. As suggested earlier, the most basic characteristic of an amplifier is its *gain*, which is simply the ratio of the output to the input. The open-loop gain A of a practical amplifier (e.g., an operational amplifier) is usually very large, whereas the closed-loop gain G is a reduced version of the open-loop gain. The relationship between A and G is developed and explored in the rest of this chapter.

There are two types of feedback possible in closed-loop mode: *positive feedback*, which tends to reinforce the amplifier input, and *negative feedback*, which tends to inhibit the amplifier input. Both positive and negative feedback have useful applications; however, negative feedback is by far the most common type of feedback found in applications. In general, negative feedback causes the large open-loop gain A of an amplifier to be exchanged for a smaller closed-loop gain G . While this exchange may seem undesirable at first glance, several key benefits accompany the exchange. These benefits to the amplifier are:

1. Decreased sensitivity to variations in circuit and environmental parameters, most notably temperature.
2. Increased bandwidth.
3. Increased linearity.
4. Increased signal-to-noise ratio.

In addition, negative feedback is implemented by establishing one or more paths from the output to the input of the amplifier. The impedance of each feedback path can be adjusted to produce improved input and output impedances of the overall amplifier circuit. These input and output impedances are key characteristics for understanding the *loading effects* of other circuits attached to an amplifier.

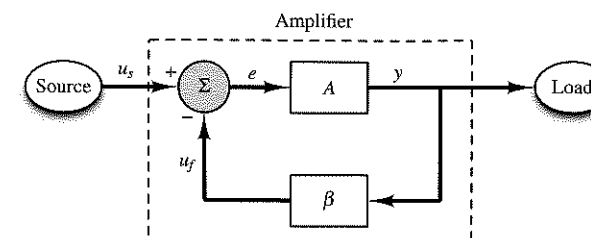


Figure 8.4 Signal-flow diagram of generic amplifier

Figure 8.4 shows a *signal-flow diagram* of an amplifier situated between a source and a load. The arrows indicate the direction of signal flow. The signals shown are u_s , u_f , e , and y . The output signal of each rectangle is a multiple of its input signal, where the two constants, A and β , are both positive such that:

$$y = Ae \quad \text{and} \quad u_f = \beta y \quad (8.11)$$

The circle sums its inputs, u_s and u_f , to produce one output, e . The polarity signs ($\pm$) indicate that u_s and u_f make positive and negative contributions to the sum, respectively. That is:

$$e = u_s - u_f = u_s - \beta y \quad (8.12)$$

Because the feedback signal u_f makes a negative contribution to the sum, the signal flow diagram of Figure 8.4 is said to employ *negative feedback*.

Equations 8.11 and 8.12 can be combined to yield:

$$y = Ae = A(u_s - \beta y) = A(u_s - \beta y) \quad (8.13)$$

which can be rearranged to solve for y . Then, the closed-loop gain of the amplifier is:

$$G \equiv \frac{y}{u_s} = \frac{A}{1 + A\beta} \quad (8.14)$$

The quantity $A\beta$ is known as the *loop gain*. Implicit in the derivation of equation 8.14 is that the behavior of the blocks within the amplifier is not affected by the other blocks nor by the external source and load. In other words, the blocks are *ideal* such that *loading effects* are zero.

Two important observations can be made at this point:

1. The closed-loop gain G depends upon β , which is known as the *feedback factor*.
2. Since $A\beta$ is positive, the closed-loop gain G is smaller than the open-loop gain A .

Furthermore, for most practical amplifiers, $A\beta$ is quite large such that:

$$G \approx \frac{1}{\beta} \quad (8.15)$$

This result is particularly important (and probably surprising!) because it indicates that the closed-loop gain G of the amplifier is largely *independent* of the open-loop gain A , as long as $A\beta \gg 1$, and that G is, in turn, determined largely by the feedback factor, β .

When $A\beta \gg 1$, the closed-loop gain G of an amplifier is determined largely by the feedback factor, β .

Furthermore, equation 8.14 can be used to find the ratio of the two inputs, u_s and u_f .

$$\frac{u_f}{u_s} = \frac{y}{u_s} \frac{u_f}{y} = \frac{A}{1 + A\beta} \beta = \frac{A\beta}{1 + A\beta} \quad (8.16)$$

Thus, when $A\beta \gg 1$, another important result is:

$$\frac{u_f}{u_s} \rightarrow 1 \quad \text{or} \quad u_s - u_f \rightarrow 0 \quad (8.17)$$

This result indicates that when the loop gain $A\beta$ is large, the *difference* between the input signal u_s and the feedback signal u_f is driven toward zero.

When $A\beta \gg 1$, the *difference* between the input signal u_s and the feedback signal u_f is driven toward zero.

Both of the results of equations 8.15 and 8.17 will show up repeatedly in the analysis of operational amplifier circuits in closed-loop mode.

Benefits of Negative Feedback

As mentioned in the previous section, negative feedback provides several benefits in exchange for a reduced gain. For example, take the derivative of both sides of equation 8.14 to find:

$$dG = \frac{dA}{1 + A\beta} - \frac{A\beta dA}{(1 + A\beta)^2} = \frac{dA}{(1 + A\beta)^2} \quad (8.18)$$

Divide the left side by G and the right side by $A/(1 + A\beta)$ to obtain:

$$\frac{dG}{G} = \frac{1}{1 + A\beta} \frac{dA}{A} \quad (8.19)$$

When $A\beta \gg 1$, this result indicates that the percentage change in G due to a percentage change in A is relatively small. In other words, the closed-loop gain G is relatively insensitive to changes in the open-loop gain A .

When $A\beta \gg 1$, the closed-loop gain G is relatively insensitive to changes in the open-loop gain A .

For any amplifier, the open-loop gain A is a function of frequency. For example, the open-loop gain $A(\omega)$ of an op-amp is characterized by a simple pole such that:

$$A(\omega) = \frac{A_0}{1 + j\omega/\omega_o} \quad (8.20)$$

where ω_o is its 3-dB break frequency. The Bode magnitude characteristic plot is shown in Figure 8.5. Equation 8.20 can be substituted into equation 8.14 to obtain:

$$G(\omega) = \frac{A(\omega)}{1 + A(\omega)\beta} = \frac{A_0/(1 + j\omega/\omega_o)}{1 + A_0\beta(1 + j\omega/\omega_o)} \quad (8.21)$$

Multiply the numerator and denominator on the right side of equation 8.21 by $1 + j\omega/\omega_o$ and then factor out $1 + A_0\beta$ from the denominator to obtain:

$$G(\omega) = \frac{A_0}{1 + A_0\beta} \frac{1}{1 + j\omega/\omega_g} = G_o \frac{1}{1 + j\omega/\omega_g} \quad (8.22)$$

where $\omega_g = \omega_o(1 + A_0\beta)$. Thus, the closed-loop 3-dB break frequency is $(1 + A_0\beta)$ larger than the open-loop 3-dB break frequency.

The closed-loop 3-dB break frequency is $(1 + A_0\beta)$ larger than the open-loop 3-dB break frequency.

Likewise, if the amplifier is characterized by a simple zero, its 3-dB break frequency will be $(1 + A_0\beta)$ smaller than the open-loop 3-dB break frequency. It is a worthwhile exercise to work out this result.

Similar analyses can be performed to show the increased linearity and increased signal-to-noise ratio resulting from negative feedback. All these benefits are acquired at the expense of amplifier gain. Finally, all of the features of a generic amplifier with negative feedback outlined in this section also occur in closed-loop amplifiers constructed using operational amplifiers and other basic components.

8.2 THE OPERATIONAL AMPLIFIER

An **operational amplifier** (op-amp) is an **integrated circuit (IC)** that contains a large number of microscopic electrical and electronic components integrated on a single silicon wafer. An op-amp can be used in conjunction with other common components to create circuits that perform amplification and filtering, as well as mathematical operations, such as addition, subtraction, multiplication, differentiation, and integration, on electrical signals. Op-amps are found in most measurement and instrumentation applications, serving as a versatile building block for many applications.

The behavior of an op-amp is well described by fairly simple models, which permit an understanding of its effects and applications without delving into its internal details. Its simplicity and versatility make the op-amp an appealing electronic device with which to begin understanding electronics and integrated circuits. The lower right portion of Figure 8.6 shows a standard single op-amp IC chip pin layout. It has two input pins (2 and 3) and one output pin (6). Also notice the two DC power supply pins (4 and 7) that provide external power to the chip and thus *enable* the op-amp. Operational amplifiers are *active* devices; that is, they need an external power source to function. Pin 4 is held at a low DC voltage V_s^- , while pin 7 is held at a high DC voltage V_s^+ . These two DC voltages are well below and above, respectively, the op-amp's reference voltage and bound the output of the op-amp.

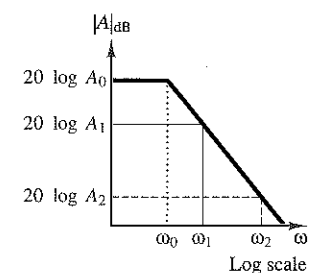


Figure 8.5 Typical amplifier Bode magnitude characteristic

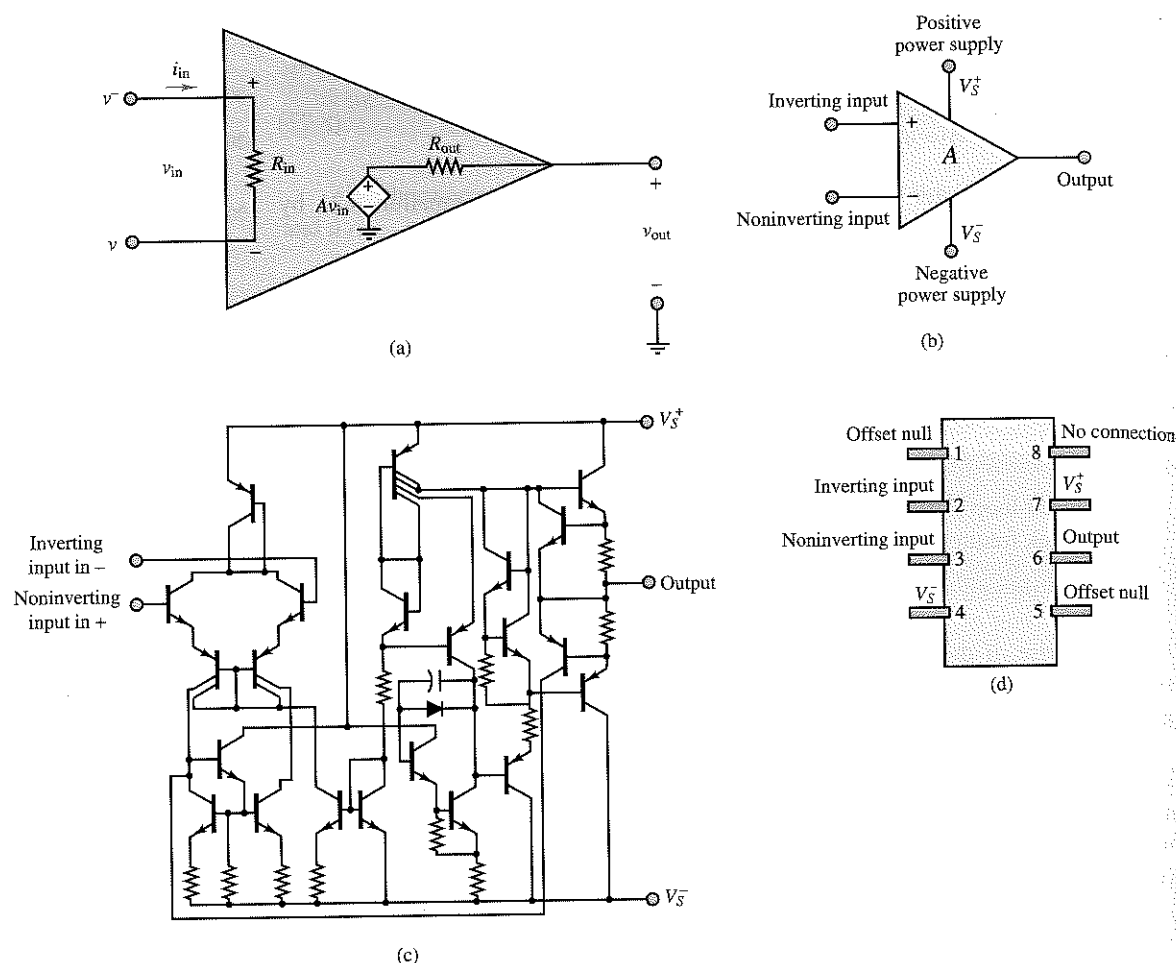


Figure 8.6 (a) Small-signal op-amp model; (b) simplified op-amp circuit symbol; (c) generic op-amp IC schematic; (d) single op-amp IC chip pin layout.

The upper left portion of Figure 8.6 shows the so-called small-signal, low-frequency model of an op-amp, which is exactly the same amplifier model shown in Figure 8.3. For this model, the input impedance is R_{in} and the output impedance is R_{out} . The op-amp itself is a *differential amplifier* because its output is a function of the difference between two input voltages, v^+ and v^- , which are known as the *non-inverting* and *inverting* inputs, respectively. Notice that the value of the internal dependent voltage source is $A(v^+ - v^-)$, where A is the *open-loop gain* of the op-amp. In a practical op-amp, A is quite large by design, typically on the order of 10^5 to 10^7 . As discussed in the previous section, this large open-loop gain can be exchanged, by design, for a smaller *closed-loop gain* G to acquire various beneficial characteristics for an amplifier circuit, of which the op-amp is just one component.²

²The operational amplifier of Figure 8.6 is a *voltage amplifier*; another type of operational amplifier, called a *current* or *transconductance amplifier*, is described in the homework problems.

The Ideal Op-Amp

Practical op-amps have a large open-loop gain A , as noted above. The input impedance R_{in} is also large, typically on the order of 10^6 to $10^{12} \Omega$, while the output impedance R_{out} is small, typically on the order of 10^0 or $10^1 \Omega$. In the ideal case, the open-loop gain and the input impedance would be infinite, while the output impedance would be zero. When the output impedance is zero, the output voltage of an ideal op-amp is simply:

$$v_{out} = A(v^+ - v^-) = A\Delta v \quad (8.23)$$

But is this relationship practical when the open-loop gain A approaches infinity? The implication for a practical op-amp is that one of the two following possibilities will hold:

1. In the case that $\Delta v \neq 0$, the output voltage *saturates* near either the positive or negative DC power supply value, V_S^+ or V_S^- , as shown in Figure 8.7. These external DC power supply *rails* enable a practical op-amp to function but also bound the op-amp output voltage v_{out} . This case applies to all practical applications of an op-amp used in *open-loop mode*; that is, when there is no feedback from v_{out} to v^- .
2. In the case that $\Delta v = 0$, the output voltage is not determined by the op-amp itself but by whatever other circuitry is attached to it. Recall from section 8.1 that when $A\beta \gg 1$ the closed-loop gain of an amplifier is approximately equal to $1/\beta$ and largely independent of A itself. Thus, this case applies to all practical applications of an op-amp in *closed-loop mode*; that is, when negative feedback is present from v_{out} to v^- .

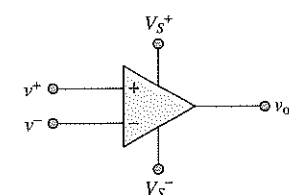


Figure 8.7 Ideal operational amplifier

By far the most prominent open-loop mode application of an op-amp is the *comparator*, which is discussed in Chapter 15. Many of the practical applications of the op-amp in closed-loop mode are explored in this chapter.

Notice in Figure 8.7 that the letter “A” does not appear within the triangle symbol, thus implying that the open-loop gain is infinite. Also implied by the ideal op-amp symbol is that the current into or out of either input terminal is zero. This result is a consequence of the infinite input impedance of an ideal op-amp and is known as the *first golden rule* of ideal op-amps:

$$i^+ = i^- = 0 \quad \text{First golden rule} \quad (8.24)$$

Also recall from the discussion of negative feedback in section 8.1 that when $A\beta \gg 1$ the difference between the two amplifier inputs, u_s and u_f , approaches zero. In the context of ideal op-amps, where $A \rightarrow \infty$, the difference between the two amplifier inputs, v^+ and v^- , will be zero exactly as long as there is a *feedback path* from v_{out} to v^- .

$$v^+ = v^- \quad \text{Second golden rule; negative feedback required} \quad (8.25)$$

The Golden Rules of Ideal Op-Amps:

1. $i^+ = i^- = 0$.
2. $v^+ = v^-$ (when negative feedback is present).



Amplifier Archetypes

There are three fundamental amplifiers that utilize the operational amplifier and employ *negative feedback*. They are:

- The inverting amplifier.
- The non-inverting amplifier.
- The isolation buffer (or voltage follower).

These archetypes have many important applications and are the building blocks for other important amplifiers. Understanding and recognizing these archetypes is an essential first step in the study of amplifiers based upon the op-amp. It is worth emphasizing that the op-amp is rarely used as a stand-alone amplifier; rather it is used along with other components to form specialized amplifiers.

The Inverting Amplifier

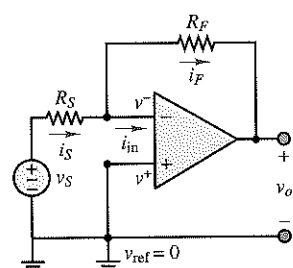


Figure 8.8 Inverting amplifier

Figure 8.8 shows a basic *inverting amplifier* circuit. The name derives from the fact that the input signal v_S “sees” the inverting terminal ($-$) and that, as is shown below, the output signal v_o is an inverted (negative) version of the input signal. The goal of the following analysis is to determine the relationship between the output and the input signals. To begin, assume the op-amp is ideal and apply KCL at the inverting node marked v^- .

$$i_S = i_F + i_{in} \quad (8.26)$$

However, the first golden rule of ideal op-amps states that $i_{in} = i^- = 0$. Thus, $i_S = i_F$ such that R_S and R_F form a *virtual series connection*. Ohm’s law can be applied to each resistor to yield:

$$i_S = \frac{v_S - v^-}{R_S} \quad i_F = \frac{v^- - v_o}{R_F} \quad (8.27)$$

These expressions can be simplified by noting that $v^+ = 0$ and then applying the second golden rule of ideal op-amps to realize $v^- = v^+ = 0$. Thus:

$$i_S = i_F$$

or

$$\frac{v_S}{R_S} = -\frac{v_o}{R_F}$$

Cross-multiply to find the closed-loop gain G :

$$G = \frac{v_o}{v_S} = -\frac{R_F}{R_S} \quad \text{Inverting amplifier}$$

Note that the magnitude of G can be greater or less than 1.



An alternate approach is to apply voltage division across the virtual series connection of R_S and R_F .

$$\frac{v_S - v_o}{v_S - 0} = \frac{R_S + R_F}{R_S}$$

or

$$1 - \frac{v_o}{v_S} = 1 + \frac{R_F}{R_S} \quad (8.30)$$

Subtract 1 from each side of this expression to find the same result as equation 8.29.

Notice that the closed-loop gain G of an inverting amplifier is determined solely by the choice of resistors. This result was derived for an ideal op-amp. For a practical op-amp the result is only slightly different as long as the open-loop gain A is large. It is important to remember that this result depends upon both golden rules of ideal op-amps and that, in particular, the second golden rule is valid only when negative feedback is present.

As long as the open-loop gain A is large, the presence of negative feedback from the output to the inverting input drives the voltage difference between the two input terminals to zero.



The input impedance of the inverting amplifier is simply:

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_S - 0}{i_S} = R_S \quad (8.31)$$

Notice the important role played by the virtual ground at the inverting terminal in making this calculation so easy. This result also reveals a shortcoming of the inverting amplifier. In general, an ideal amplifier would have an infinite input impedance so as to not load the source network. It is tempting to correct this problem by choosing R_S to be very large; however, in so doing, the closed-loop gain (equation 8.29) will be reduced. Thus, it is not possible to design an inverting amplifier to have a large gain and also a large input impedance. Alas, there is no such thing as a free lunch!

The Non-Inverting Amplifier

Figure 8.9 shows a basic *non-inverting amplifier* circuit. The name derives from the fact that the input signal v_S “sees” the non-inverting terminal ($+$) and that, as is shown below, the output signal v_o is a non-inverted (positive) version of the input signal. The goal of the following analysis is to determine the relationship between the output and the input signals. As was done for the inverting amplifier circuit, assume the op-amp is ideal and apply KCL at the inverting node marked v^- .

$$i_F = i_1 + i_{in} \quad (8.32)$$

However, the first golden rule of ideal op-amps states that $i_{in} = i^- = i^+ = 0$. Thus, $i_F = i_1$ such that R_F and R_1 form a *virtual series connection*. Ohm’s law can be applied to each resistor to yield:

$$i_1 = \frac{v^- - 0}{R_1} \quad i_F = \frac{v_o - v^-}{R_F}$$

or

$$\frac{v^-}{R_1} = \frac{v_o - v^-}{R_F} \quad (8.33)$$

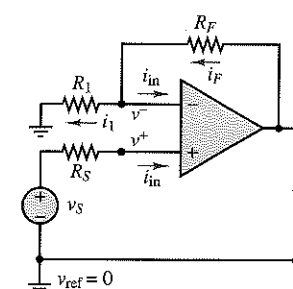


Figure 8.9 Non-inverting amplifier

Since there is negative feedback present, the second golden rule of ideal op-amps can be applied such that $v^- = v^+$. Notice that because $i_{in} = 0$, the voltage drop across R_S is zero with the result that $v^- = v^+ = v_S$. Substitute this result into equation 8.33 and rearrange terms to yield the closed-loop gain G :



$$G = \frac{V_o}{v_S} = 1 + \frac{R_F}{R_1} \quad \text{Non-inverting amplifier} \quad (8.34)$$

Note that $G \geq 1$.

An alternate approach is to apply voltage division across the virtual series connection of R_1 and R_F .

$$\frac{V_o - 0}{v^- - 0} = \frac{R_1 + R_F}{R_1} \quad (8.35)$$

Since $v^- = v_S$:

$$\frac{V_o}{v_S} = 1 + \frac{R_F}{R_1} \quad (8.36)$$

which is the same result as that found in equation 8.34.

Notice that the closed-loop gain G of a non-inverting amplifier is also determined solely by the choice of resistors. This result was derived for an ideal op-amp. For a practical op-amp the result is only slightly different as long as the open-loop gain A is large. It is important to remember that this result depended upon both golden rules of ideal op-amps and that, in particular, the second golden rule is valid only when negative feedback is present.



As long as the open-loop gain A is large, the presence of negative feedback from the output to the inverting input drives the voltage difference between the two input terminals to zero.

The input impedance of the non-inverting amplifier is simply:

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_S - 0}{i_{in}} \rightarrow \infty \quad (8.37)$$

In practice, the input impedance of a non-inverting amplifier is very large due to the very large input impedance of the op-amp, which limits i_{in} to very small values. Notice that the closed-loop gain of the non-inverting amplifier is independent of its input impedance. Thus, the non-inverting amplifier does not suffer from a trade-off between gain and input impedance, as does the inverting amplifier. However, the gain of a non-inverting amplifier is limited to values greater than one, whereas the gain of the inverting amplifier can take on any value. Alas, again there is no such thing as a free lunch!

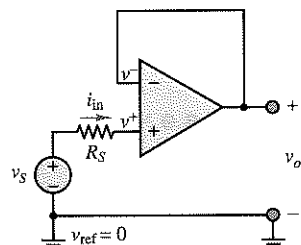


Figure 8.10 Isolation Buffer or Voltage Follower

Isolation Buffer or Voltage Follower

Figure 8.10 shows an *isolation buffer*, which is also known as a *voltage follower*. Notice that the input signal v_S “sees” the non-inverting terminal (+) such that the output signal v_o should be a non-inverted (positive) version of v_S . The analysis of

this circuit is as simple as the circuit itself. Assume that the op-amp is ideal. Since negative feedback is present, both golden rules are valid. That is:

$$i^+ = i^- = 0 \quad \text{and} \quad v^+ = v^- \quad (8.38)$$

By observation, $v^+ = v_S$ and $v^- = v_{out}$ with the result that the closed-loop gain G is:

$$G = \frac{v_o}{v_S} = 1 \quad \text{Isolation buffer (voltage follower)} \quad (8.39)$$



The reason this circuit is called a voltage follower should now be obvious; the output voltage v_o “follows” (equals) the input voltage v_S . On the other hand, the reason this circuit is also known as an isolation buffer is not obvious. However, since $i^+ = 0$, the ideal op-amp is said to possess an infinite *input resistance* or *input impedance* such that the voltage source experiences no loading from the op-amp. Yet the circuit still reproduces v_S at the output. Any loading effects at the output are experienced by the op-amp rather than the voltage source, such that the source is *isolated* or *buffered* from the output.

The input impedance of an isolation buffer is simply:

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_S - 0}{i_{in}} \rightarrow \infty \quad (8.40)$$

In practice, the input impedance of an isolation buffer is very large due to the very large input impedance of the op-amp, which limits i_{in} to very small values. The closed-loop gain is fixed at unity as long as the open-loop gain A is large such that v^- will be driven to v^+ by negative feedback.

Application of Thévenin's theorem

Notice in Figures 8.8 and 8.9 that the input source is represented as a Thévenin source. The implication is that the previous results for inverting and non-inverting amplifiers can be applied to any case where the input source of the amplifier circuit is linear and can be simplified to an equivalent Thévenin source. In other words, R_S and v_S are the Thévenin equivalent resistance and the open-circuit voltage, respectively, of any arbitrary linear input circuit.

For example, consider the inverting amplifier circuit shown in Figure 8.11. It does not have the same form as the archetype of Figure 8.8. However, the voltage source v_{in} “sees” the inverting terminal; therefore, the output voltage v_o will be an inverted version of v_{in} . The circuit is an inverting amplifier. To solve for v_o replace the entire linear circuit to the left of terminals a and b with its Thévenin equivalent.

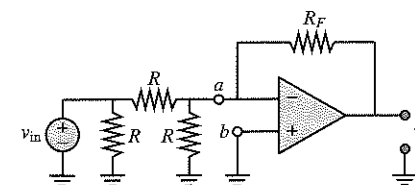


Figure 8.11 Inverting amplifier before simplification to archetype

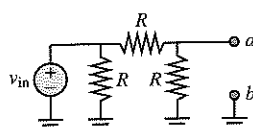


Figure 8.12 Source network detached at terminals a and b

Figure 8.12 shows the source circuit detached at terminals a and b . To find the Thévenin equivalent resistance of the input circuit set the voltage source to zero and replace it with a short-circuit. Then:

$$R_T = R_{ab} = R \parallel R = \frac{R}{2} \quad (8.41)$$

The Thévenin (open-circuit) voltage across terminals a and b can be found from voltage division:

$$V_T = V_{ab} = \frac{R}{R + R} v_{in} = \frac{v_{in}}{2} \quad (8.42)$$

The Thévenin equivalent source network attached to the rest of the amplifier circuit is shown in Figure 8.13. Notice that the simplified amplifier is now identical in form to the inverting amplifier archetype of Figure 8.8. Thus, using equation 8.29:

$$\frac{v_o}{V_T} = -\frac{R_F}{R_T} = -\frac{2R_F}{R} \quad (8.43)$$

To obtain the closed-loop gain G of the original amplifier circuit, write:

$$\begin{aligned} G = \frac{v_o}{v_{in}} &= \frac{v_o}{V_T} \cdot \frac{V_T}{v_{in}} \\ &= -\frac{2R_F}{R} \cdot \frac{1}{2} \\ &= -\frac{R_F}{R} \end{aligned} \quad (8.44)$$

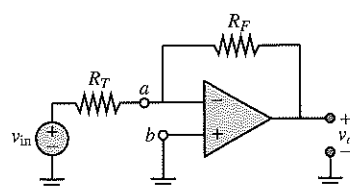


Figure 8.13 Inverting amplifier after simplification to archetype

Figure 8.13 generalizes Figure 8.8 by representing explicitly the source network as the Thévenin equivalent network of any linear input source network. The same approach can be taken to generalize the non-inverting amplifier and isolation buffer circuits shown in Figures 8.9 and 8.10, respectively, where v_s and R_s are now the Thévenin (open-circuit) voltage and the Thévenin equivalent resistance, respectively, of the input source network.

Multiple Sources and the Principle of Superposition

There are many situations that call for an amplifier to accommodate multiple input source networks. The analysis of these amplifiers can be accomplished using basic principles, such as KCL, KVL, and Ohm's law. However, it is often useful to apply the principle of superposition to simplify the overall amplifier circuit into multiple component amplifiers, each with only one independent source still turned on. Thévenin's theorem can often be used to transform these component amplifiers into

one of the amplifier archetypes: the inverting amplifier, the non-inverting amplifier, and the isolation buffer. Two important examples of amplifiers with multiple input sources are the summing amplifier and the differential amplifier.

The Summing Amplifier

A useful op-amp circuit that is based on the inverting amplifier is the **op-amp summer**, or **summing amplifier**, as shown in Figure 8.14. Assume the op-amp is ideal. The first golden rule of op-amps states that $i^+ = i^- = 0$. Thus, when KCL is applied at the inverting node, the result is:

$$\sum_{n=1}^N i_n = i_1 + i_2 + \cdots + i_N = i_F \quad (8.45)$$

Since negative feedback is present, the second golden rule is also valid such that $v^- = v^+ = 0$. Ohm's law can then be applied at each resistor to obtain:

$$i_n = \frac{v_{S_n} - 0}{R_{S_n}} \quad n = 1, 2, \dots, N$$

and

$$i_F = \frac{0 - v_o}{R_F}$$

The results of equation 8.46 can be plugged into equation 8.45 to find:

$$\sum_{n=1}^N \frac{v_{S_n}}{R_{S_n}} = -\frac{v_o}{R_F} \quad (8.47)$$

or

$$v_o = -\sum_{n=1}^N \frac{R_F}{R_{S_n}} v_{S_n}$$

The output of the summing amplifier is the weighted sum of N input signal sources, where the weighting factor for each source v_{S_n} equals the ratio of the feedback resistance R_F to the source resistance R_{S_n} . Notice that if $R_S = R_{S_1} = R_{S_2} = \cdots = R_{S_N}$, then:

$$v_o = -\frac{R_F}{R_S} \sum_{n=1}^N v_{S_n} \quad \text{Summing amplifier} \quad (8.48)$$

The summing amplifier can also be analyzed using the principle of superposition. Consider turning off all the voltage sources except v_{S_1} . The result is that the voltage drop across the resistors $R_2, \dots, R_N$ is zero since a zero voltage source is equivalent to a short-circuit. Thus, for this case, $i_2 = i_3 = \cdots = i_N = 0$ as shown in Figure 8.15. Assume the op-amp is ideal such that $i^+ = i^- = 0$. Then KCL applied

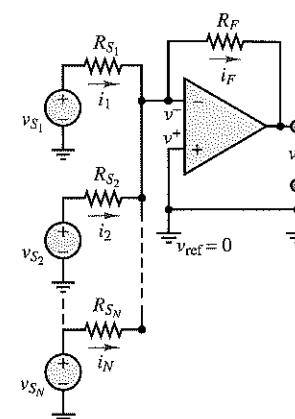


Figure 8.14 Summing amplifier



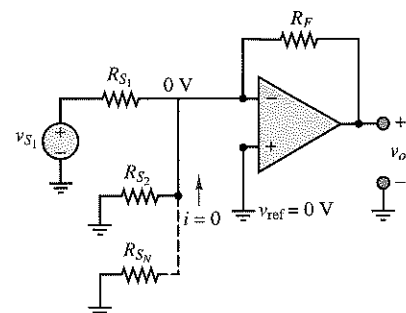


Figure 8.15 Summing amplifier with only one source turned on

at the inverting terminal node yields simply:

$$i_1 = i_F \quad (8.49)$$

Again, because negative feedback is present, the second golden rule is valid such that $v^- = v^+ = 0$. Ohm's law can then be applied to R_{S1} and R_F to obtain:

$$i_1 = \frac{v_{S1} - 0}{R_{S1}} \quad \text{and} \quad i_F = \frac{0 - v_{o1}}{R_F} \quad (8.50)$$

Plug these two results into equation 8.49 and rearrange to yield:

$$v_{o1} = -\frac{R_F}{R_{S1}} v_{S1} \quad (8.51)$$

where v_{o1} is the component of v_o due to the voltage source v_{S1} . It is worth noting that this result is equivalent to what would be obtained for the inverting amplifier archetype shown in Figure 8.16. This equivalence is due to the fact that the currents $i_2, i_3, \dots, i_N$ are all zero such that R_{S1} and R_F are still in a virtual series connection as in the inverting amplifier archetype.

Since the Thévenin source pairs v_{S_n} and R_{S_n} in Figure 8.14 are all in parallel, the component of v_o due to v_{S_n} is:

$$v_{on} = -\frac{R_F}{R_{S_n}} v_{S_n} \quad n = 1, 2, \dots, N \quad (8.52)$$

Summing all these component contributions yields:

$$v_o = -\sum_{n=1}^N \frac{R_F}{R_{S_n}} v_{S_n} \quad (8.53)$$

which is the same result as that found in equation 8.47.

The Differential Amplifier

A useful op-amp circuit that is based on the inverting and non-inverting amplifier archetypes is the **differential amplifier** shown in Figure 8.17. This amplifier finds frequent use in situations where the difference between two signals needs to be amplified. The two sources v_1 and v_2 may be independent of each other or may originate from the same process, as they do in the Focus on Measurements box "Electrocardiogram (EKG) Amplifier."

The analysis of the differential amplifier can be accomplished by applying basic principles (e.g., KCL, Ohm's law) or by applying the principle of superposition. Both

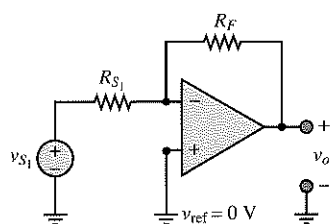


Figure 8.16 Equivalent inverting amplifier circuit for summing amplifier with only one source turned on

approaches will assume an ideal op-amp, and since negative feedback is present, both golden rules are valid. The former approach begins by noting that $i^+ = i^- = 0$ such that R_1 and R_F are in a virtual series connection as are R_2 and R_3 . Thus, the voltage at the non-inverting terminal v^+ can be computed from voltage division.

$$v^+ = \frac{R_3}{R_3 + R_2} v_2 = \frac{R_3/R_2}{1 + (R_3/R_2)} v_2 \quad (8.54)$$

Likewise, voltage division along the other virtual series connection yields:

$$i = \frac{v_1 - v_o}{R_1 + R_F} = \frac{v^- - v_o}{R_F} \quad (8.55)$$

Solving for v^- yields:

$$v^- = \frac{R_F v_1 + R_1 v_o}{R_1 + R_F} = \frac{(R_F/R_1) v_1 + v_o}{1 + (R_F/R_1)} \quad (8.56)$$

The second golden rule is $v^+ = v^-$ such that:

$$\frac{R_3/R_2}{1 + (R_3/R_2)} v_2 = \frac{(R_F/R_1) v_1 + v_o}{1 + (R_F/R_1)} \quad (8.57)$$

or

$$v_o = \frac{1 + (R_F/R_1)}{1 + (R_3/R_2)} \frac{R_3}{R_2} v_2 - \frac{R_F}{R_1} v_1$$

In this form the expression for v_o is too complicated to leave much of an impression. However, it is greatly simplified by choosing the resistor values to satisfy:

$$\frac{R_F}{R_1} = \frac{R_3}{R_2} \quad (8.58)$$

such that:

$$v_o = \frac{R_F}{R_1} (v_2 - v_1) \quad \text{Differential amplifier} \quad (8.59)$$

Figure 8.18 shows one particular version of a differential amplifier where equation 8.58 is satisfied by setting $R_3 = R_F$ and $R_2 = R_1$.

The circuit in Figure 8.17 can also be analyzed using the principle of superposition. The op-amp is still assumed to be ideal, and since negative feedback is present, both golden rules are valid. To begin, set $v_2 = 0$ and find the component of v_o due to v_1 as shown in Figure 8.19. Since $i^+ = 0$, there can be no voltage drop across R_2 and R_3 with the result that $v^+ = 0$. Thus, the circuit is an inverting amplifier with the output given by equation 8.29 as:

$$v_{o1} = -\frac{R_F}{R_1} v_1 \quad (8.60)$$

Now set $v_1 = 0$ and find the component of V_o due to v_2 as shown in Figure 8.20. Since $i^+ = 0$, v_2 , R_2 , and R_3 are in a virtual series connection. Apply voltage division to yield:

$$v^+ = \frac{R_3}{R_3 + R_2} v_2 \quad (8.61)$$

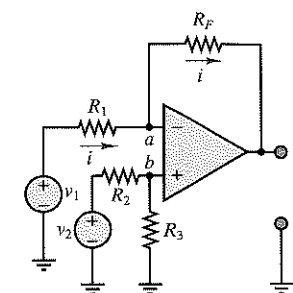


Figure 8.17 Amplifier with input sources at the inverting and non-inverting terminals

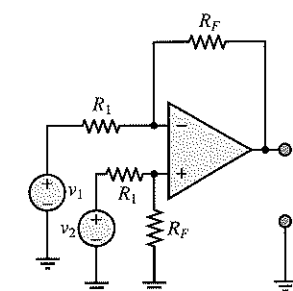


Figure 8.18 Differential amplifier

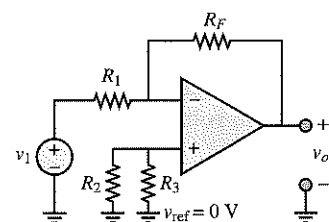


Figure 8.19 Inverting amplifier appears when $v_2 = 0$

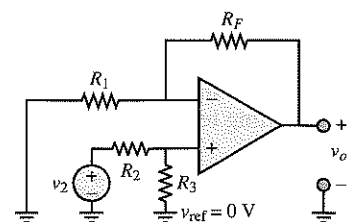


Figure 8.20 Amplifier of Figure 8.17 but with $v_1 = 0$.

Thus, the equivalent circuit is a non-inverting amplifier as shown in Figure 8.21 with the output given by equation 8.34 as:

$$v_{o2} = \left(1 + \frac{R_F}{R_1}\right) v^+ = \left(1 + \frac{R_F}{R_1}\right) \frac{R_3}{R_3 + R_2} v_2 \quad (8.62)$$

Finally, apply the principle of superposition to obtain:

$$v_o = v_{o1} + v_{o2} = -\frac{R_F}{R_1} v_1 + \left(1 + \frac{R_F}{R_1}\right) \frac{R_3}{R_3 + R_2} v_2 \quad (8.63)$$

As before, this expression is greatly simplified by choosing the resistor values such that:

$$\frac{R_F}{R_1} = \frac{R_3}{R_2} \quad (8.64)$$

The result is (of course!) equation 8.59.

Both of the solution methods shown above are completely valid. Neither is particularly easier than the other although tastes do vary! However, the principle of superposition has the added appeal of determining the individual contributions of each voltage source and therefore allows for a quick recalculation of the solution when only one of the voltage sources is changed.

It is important to realize that if the linear source network seen by either terminal is more complicated than those shown in Figure 8.17 it is possible to simplify them using Thévenin's theorem. For example, the source network seen by the non-inverting terminal is shown in Figure 8.22, where:

$$v_T = \frac{R_3}{R_3 + R_2} v_2 \quad \text{and} \quad R_T = R_2 \parallel R_3 \quad (8.65)$$

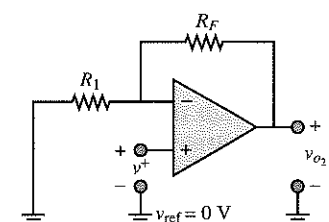


Figure 8.21 Non-inverting amplifier appears when $v_1 = 0$.

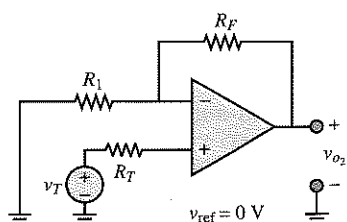


Figure 8.22 Figure 8.20 but with the network seen by the non-inverting terminal replaced with its Thévenin equivalent network

Common and Differential Modes

It is often necessary to amplify the difference between two signals that may both be corrupted by noise or interference. The two input signals v_1 and v_2 can be decomposed be into two parts: the **common mode** (CM) and the **differential mode** (DM). These two modes are defined mathematically as:

$$v_{CM} = \frac{v_1 + v_2}{2} \quad \text{and} \quad v_{DM} = v_2 - v_1 \quad (8.66)$$

where the common mode v_{CM} is the average value of v_1 and v_2 .

$$v_1 = v_{CM} - \frac{v_{DM}}{2} \quad \text{and} \quad v_2 = v_{CM} + \frac{v_{DM}}{2} \quad (8.67)$$

With these definitions, the output of an ideal differential amplifier is simply:

$$v_o = \frac{R_F}{R_1} (v_2 - v_1) = \frac{R_F}{R_1} v_{DM} \quad (8.68)$$

In other words, the common mode of the two input signals is *rejected* by the differential amplifier. In many situations, the noise and interference of one input is identical to (or nearly the same as) that of the other input. Thus, a differential amplifier can be used to eliminate noise and interference that is common to both inputs. In practice, the output of a differential amplifier is given by:

$$v_o = A_{DM} (v_2 - v_1) + A_{CM} \left(\frac{v_2 + v_1}{2} \right) \quad (8.69)$$

where A_{DM} and A_{CM} are the differential-mode and common-mode gains, respectively. In the ideal case, $A_{CM} = 0$, such as for the circuit of Figure 8.18 when the op-amp is ideal and the external resistors satisfy equation 8.58 exactly. The extent to which a practical differential amplifier rejects the common mode is known as the **common-mode rejection ratio (CMRR)**:

$$\text{CMRR} = 20 \log \left| \frac{A_{DM}}{A_{CM}} \right| \quad (\text{in dB}) \quad (8.70)$$

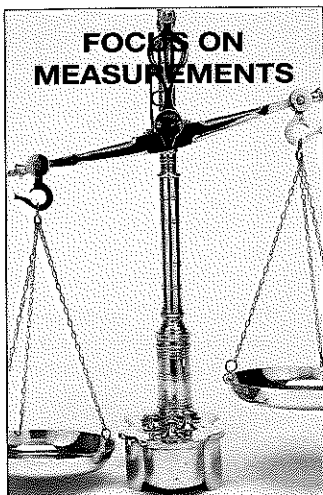
For example, op-amps themselves are differential amplifiers. A particular op-amp known as the 741 has a typical CMRR of 90 dB. The Focus on Measurements box "Electrocardiogram (EKG) Amplifier" provides a realistic look at a common application of a differential amplifier.

Table 8.1 summarizes the basic op-amp circuits presented in this section.

Table 8.1 Summary of basic amplifiers

Configuration	Circuit diagram	Output voltage (ideal op-amp)
Inverting amplifier	Figure 8.8	$-\frac{R_F}{R_S} v_S$
Non-inverting amplifier	Figure 8.9	$\left(1 + \frac{R_F}{R_S}\right) v_S$
Isolation buffer	Figure 8.10	v_S
Summing amplifier	Figure 8.14	$-\frac{R_F}{R_S} \sum_{n=1}^N v_{S_n}$
Differential amplifier	Figure 8.18	$= \frac{R_F}{R_1} (v_2 - v_1)$

FOCUS ON MEASUREMENTS



Electrocardiogram (EKG) Amplifier

This example illustrates the principle behind a two-lead electrocardiogram (EKG) measurement. The desired cardiac waveform is given by the difference between the potentials measured by two electrodes suitably placed on the patient's chest, as shown in Figure 8.23. A healthy, noise-free EKG waveform $v_1 - v_2$ is shown in Figure 8.24.

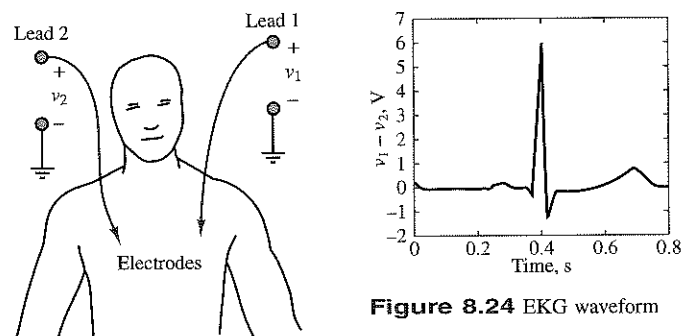


Figure 8.23 Two-lead electrocardiogram

Unfortunately, noise present on the 60-Hz, 110-V AC line used to power the equipment may appear in the EKG itself, due to capacitive coupling. Ambient electromagnetic interference can also interact with the closed-loop formed by the lead wires to generate another source of noise. Other sources of noise include changes at the electrode-skin interface due to respiration, muscle contractions, and other displacements. In addition, different DC offsets due to the electrodes complicate the signals. The signal processing associated with an actual EKG involves instrumentation amplifiers (see Example 8.2) and active filters (see section 8.3). In this example, the focus is limited to the role of a differential amplifier in rejecting common-mode 60-Hz noise found in a typical EKG. With that limitation in mind, assume that the EKG signals v_1 and v_2 indicated in Figure 8.23 are represented by:

Lead 1:

$$v_1(t) + v_n(t) = v_1(t) + V_n \cos(377t + \phi_n)$$

Lead 2:

$$v_2(t) + v_n(t) = v_2(t) + V_n \cos(377t + \phi_n)$$

As shown in Figure 8.25, the interference signal $V_n \cos(377t + \phi_n)$ is approximately the same at both leads because the electrodes are designed to be identical and are used in close proximity to each other. If the resistors of the differential amplifier are properly matched, the voltage output will be:

$$v_o = \frac{R_2}{R_1} [(v_1 + v_n) - (v_2 + v_n)] = \frac{R_2}{R_1} (v_1 - v_2)$$

Thus, common-mode 60-Hz noise is eliminated, or greatly reduced, while the desired EKG waveform is amplified. Great! In practice, the common-mode rejection ratio is not infinite but can be made quite large to satisfy the design specifications required for a proper diagnosis.

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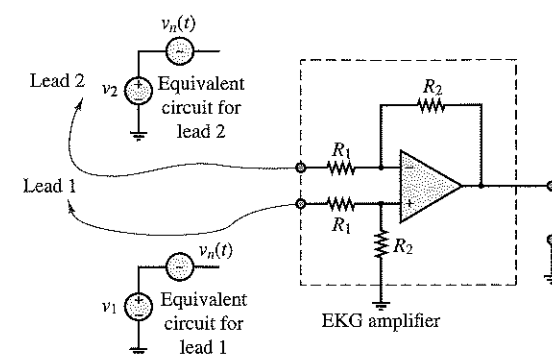


Figure 8.25 EKG amplifier

Sensor Calibration Circuit

In many practical instances, the output of a sensor is related to the physical variable we wish to measure in a form that requires some signal conditioning. The most desirable form of a sensor output is one in which the electrical output of the sensor (e.g., voltage) is related to the physical variable by a constant factor. Such a relationship is depicted in Figure 8.26(a), where k is the calibration constant relating voltage to temperature. Note that k is a positive number, and that the calibration curve passes through the (0, 0) point. On the other hand, the sensor characteristic of Figure 8.26(b) is best described by the following equation:

$$v_{\text{sensor}} = -\beta T + V_0$$

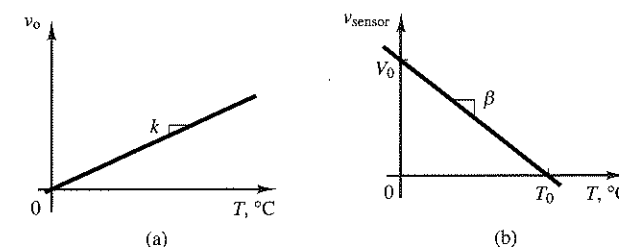
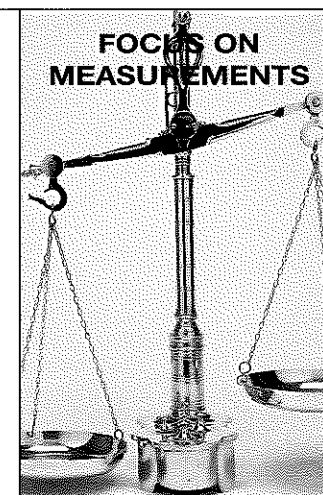


Figure 8.26 Sensor calibration curves

It is possible to modify the sensor calibration curve of Figure 8.26(b) to the more desirable one of Figure 8.26(a) by means of the simple circuit displayed in Figure 8.27. This circuit provides the desired calibration constant k by a simple gain adjustment, while the zero (or bias) offset is adjusted by means of a potentiometer connected to the voltage supplies. The detailed operation of the circuit is described in the following paragraphs.

(Continued)

FOCUS ON MEASUREMENTS



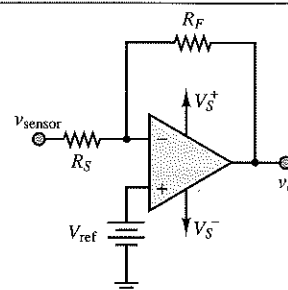


Figure 8.27 Sensor calibration circuit

This non-ideal characteristic is described by:

$$v_{\text{sensor}} = -\beta T + V_0$$

When $V_{\text{ref}} = 0$, the sensor voltage input sees an inverting amplifier such that:

$$(v_o)_{\text{sensor}} = -\frac{R_F}{R_S} v_{\text{sensor}}$$

Likewise, when $v_{\text{sensor}} = 0$, the battery voltage sees a non-inverting amplifier such that:

$$(v_o)_{\text{ref}} = 1 + \frac{R_F}{R_S} V_{\text{ref}}$$

Thus, the total output of the op-amp circuit of Figure 8.27 may be determined from the principle of superposition:

$$\begin{aligned} v_o &= (v_o)_{\text{sensor}} + (v_o)_{\text{ref}} \\ &= -\frac{R_F}{R_S} v_{\text{sensor}} + \left(1 + \frac{R_F}{R_S}\right) V_{\text{ref}} \\ &= -\frac{R_F}{R_S} (-\beta T + V_0) + \left(1 + \frac{R_F}{R_S}\right) V_{\text{ref}} \end{aligned}$$

The requirement for a linear response such as that shown in Figure 8.26(a) is $v_o = kT$, where k is the constant slope of the linear response. This requirement is satisfied by suitable choices of R_F , R_S , and V_{ref} such that:

$$kT = -\frac{R_F}{R_S} (-\beta T + V_0) + \left(1 + \frac{R_F}{R_S}\right) V_{\text{ref}}$$

For this equation to hold, the coefficients of T on both sides must be equal and the sum of the constant terms on the right side must equal zero. That is:

$$k = \frac{R_F}{R_S} \beta$$

and

$$\frac{R_F}{R_S} V_0 = \left(1 + \frac{R_F}{R_S}\right) V_{\text{ref}}$$

or

$$V_{\text{ref}} = \frac{R_F}{R_S + R_F} V_0$$

It is worth noting that $V_{\text{ref}} \approx V_0$ when $R_F \gg R_S$. Thus, when this condition holds, the appropriate battery voltage for the sensor calibration circuit can be determined directly from the sensor calibration curve of Figure 8.26(b). One should also attempt to pick a large enough value of R_S such that the sensor is not loaded by the calibration circuit.

It is also worth noting that the effect of the inverting aspect of the amplifier is to invert (change the sign of) the slope, while the effect of the reference battery voltage is to raise or lower the inverted calibration curve so that it passes through the origin. For this reason, the sensor calibration circuit is known more generally as a *level shifter*. See Example 8.3 for further discussion.

EXAMPLE 8.1 Inverting Amplifier Circuit

Problem

Determine the voltage gain and output voltage for the inverting amplifier circuit of Figure 8.8. What will the uncertainty in the gain be if 5 and 10 percent tolerance resistors are used, respectively?

Solution

Known Quantities: Feedback and source resistances, source voltage.

Find: $G = v_{\text{out}}/v_{\text{in}}$; maximum percent change in G for 5 and 10 percent tolerance resistors.

Schematics, Diagrams, Circuits, and Given Data: $R_S = 1 \text{ k}\Omega$; $R_F = 10 \text{ k}\Omega$; $v_s(t) = A \cos(\omega t)$; $A = 0.015 \text{ V}$; $\omega = 50 \text{ rad/s}$.

Assumptions: The amplifier behaves ideally; that is, the input current into the op-amp is zero, and negative feedback forces $v^+ = v^-$.

Analysis: Using equation 8.29, the output voltage is:

$$v_o(t) = G \times v_s(t) = -\frac{R_F}{R_S} \times v_s(t) = -10 \times 0.015 \cos(\omega t) = -0.15 \cos(\omega t)$$

The input and output waveforms are sketched in Figure 8.28.

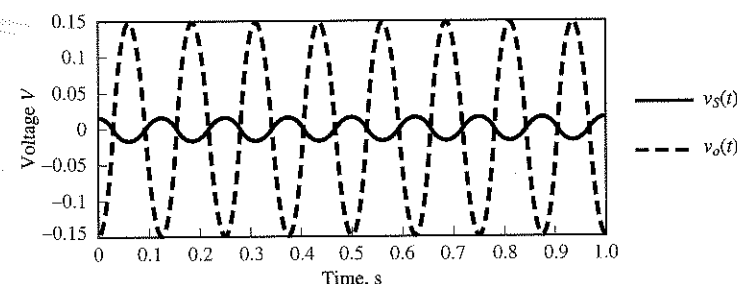


Figure 8.28

The nominal gain of the amplifier is $G_{\text{nom}} = -10$. If 5 percent tolerance resistors are employed, the worst-case error will occur at the extremes:

$$G_{\min} = -\frac{R_{F\min}}{R_{S\max}} = -\frac{9,500}{1,050} = 9.05 \quad G_{\max} = -\frac{R_{F\max}}{R_{S\min}} = -\frac{10,500}{950} = 11.05$$

The percentage error is therefore computed as

$$100 \times \frac{G_{\text{nom}} - G_{\min}}{G_{\text{nom}}} = 100 \times \frac{10 - 9.05}{10} = 9.5\%$$

$$100 \times \frac{G_{\text{nom}} - G_{\max}}{G_{\text{nom}}} = 100 \times \frac{10 - 11.05}{10} = -10.5\%$$

Thus, the amplifier gain could vary by as much as ± 10 percent (approximately) when 5 percent resistors are used. If 10 percent resistors were used, we would calculate a percent error of approximately ± 20 percent, as shown below.

$$G_{\min} = -\frac{R_{F\min}}{R_{S\max}} = -\frac{9,000}{1,100} = 8.18 \quad G_{\max} = -\frac{R_{F\max}}{R_{S\min}} = -\frac{11,000}{900} = 12.2$$

$$100 \times \frac{G_{\text{nom}} - G_{\min}}{G_{\text{nom}}} = 100 \times \frac{10 - 8.18}{10} = 18.2\%$$

$$100 \times \frac{G_{\text{nom}} - G_{\max}}{G_{\text{nom}}} = 100 \times \frac{10 - 12.2}{10} = -22.2\%$$

Comments: Note that the worst-case percent error in the closed-loop gain G is approximately double the resistor tolerance. This result can be calculated by assuming a resistor tolerance x and noting that the worst case is:

$$|\Delta G| = \frac{R_F(1+x)}{R_S(1-x)} - \frac{R_F}{R_S}$$

Let $G_{\text{nom}} = -R_F/R_S$ such that:

$$\begin{aligned} \frac{|\Delta G|}{|G_{\text{nom}}|} &= \frac{1+x}{1-x} - 1 = \frac{2x}{1-x} \\ &= 2x(1+x+x^2+\dots) \approx 2x \quad (x \ll 1) \end{aligned}$$



EXAMPLE 8.2 Instrumentation Amplifier

Problem

Determine the closed-loop voltage gain of the instrumentation amplifier circuit of Figure 8.29.

Solution

Known Quantities: Feedback and source resistances.

Find:

$$G = \frac{v_o}{v_1 - v_2}$$

Assumptions: Assume ideal op-amps.

Analysis: Often, to provide impedance isolation between bridge transducers and the differential amplifier stage, the signals v_1 and v_2 are amplified separately. This technique gives rise to the **instrumentation amplifier (IA)**, shown in Figure 8.29.

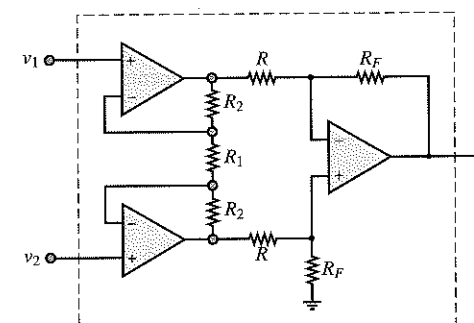


Figure 8.29 Instrumentation amplifier

Because the instrumentation amplifier has widespread application—and to ensure the best possible match between resistors—the entire circuit of Figure 8.29 is often packaged as a single integrated circuit. The advantage of this configuration is that resistors R_1 and R_2 can be matched much more precisely in an integrated circuit than would be possible by using discrete components.

Consider the input circuit first. Thanks to the symmetry of the circuit, we can represent one-half of the circuit as illustrated in Figure 8.30(a), depicting the lower half of the first stage of the instrumentation amplifier. We next recognize that the circuit of Figure 8.30(a) is a non-inverting amplifier (see Figure 8.9), such that the closed-loop voltage gain is (equation 8.34):

$$A = 1 + \frac{R_2}{R_1/2} = 1 + \frac{2R_2}{R_1}$$

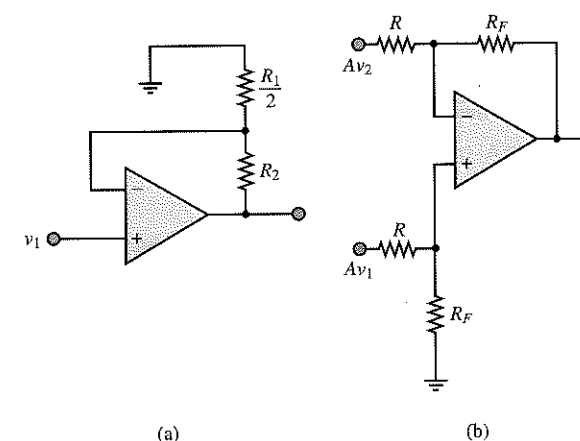


Figure 8.30 Input (a) and output (b) stages of instrumentation amplifier

Each of the two inputs v_1 and v_2 is therefore an input to the second stage of the instrumentation amplifier, shown in Figure 8.30(b). We recognize the second stage to be a differential amplifier (see Figure 8.18), and can therefore write the output voltage using equation 8.59:

$$v_o = \frac{R_F}{R} (Av_1 - Av_2) = \frac{R_F}{R} \left(1 + \frac{2R_2}{R_1}\right) (v_1 - v_2)$$

from which we can compute the closed-loop voltage gain of the instrumentation amplifier:



$$G = \frac{v_o}{v_1 - v_2} = \frac{R_F}{R} \left(1 + \frac{2R_2}{R_1}\right) \quad \text{Instrumentation amplifier}$$



EXAMPLE 8.3 Level Shifter

Problem

The level shifter of Figure 8.31 has the ability to add or subtract a DC offset to or from a signal. Analyze the circuit, and design it so that it can remove a 1.8-V DC offset from a sensor signal.

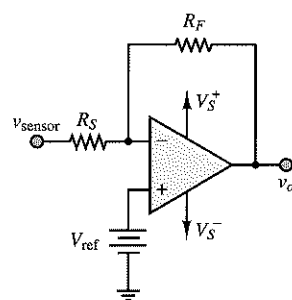


Figure 8.31 Level shifter

Solution

Known Quantities: Sensor (input) voltage; feedback and source resistors.

Find: Value of V_{ref} required to remove DC bias.

Schematics, Diagrams, Circuits, and Given Data: $v_{sensor}(t) = 1.8 + 0.1 \cos(\omega t)$; $R_F = 220 \text{ k}\Omega$; $R_S = 10 \text{ k}\Omega$.

Assumptions: Assume an ideal op-amp.

Analysis: The output voltage can be computed quite easily using the principle of superposition. When the reference voltage source V_{ref} is set to zero and replaced by a short-circuit, the sensor input voltage v_{sensor} sees an inverting amplifier such that:

$$\frac{v_{o1}}{v_{sensor}} = -\frac{R_F}{R_S}$$

When the sensor input voltage source is set to zero and replaced by a short-circuit, the reference voltage source (the battery) sees a non-inverting amplifier such that:

$$\frac{v_{o2}}{V_{ref}} = 1 + \frac{R_F}{R_S}$$

Thus, the total output voltage is the sum of contributions from the two sources:

$$v_o = v_{o1} + v_{o2} = -\frac{R_F}{R_S} v_{sensor} + \left(1 + \frac{R_F}{R_S}\right) V_{ref}$$

Substitute the expression for v_{sensor} into the previous equation to find:

$$\begin{aligned} v_o &= -\frac{R_F}{R_S} [1.8 + 0.1 \cos(\omega t)] + \left(1 + \frac{R_F}{R_S}\right) V_{ref} \\ &= -\frac{R_F}{R_S} [0.1 \cos(\omega t)] - \frac{R_F}{R_S} (1.8) + \left(1 + \frac{R_F}{R_S}\right) V_{ref} \end{aligned}$$

To remove the DC offset, require:

$$-\frac{R_F}{R_S} (1.8) + \left(1 + \frac{R_F}{R_S}\right) V_{ref} = 0$$

or

$$V_{ref} = 1.8 \frac{R_F/R_S}{1 + R_F/R_S} = 1.722 \text{ V}$$

Comments: The presence of a precision voltage source in the circuit is undesirable because it may add considerable expense to the circuit design and, in the case of a battery, it is not adjustable. The circuit of Figure 8.32 illustrates how an adjustable voltage reference can be produced from the DC supplies already used by the op-amp, two fixed resistors R , and a potentiometer R_p . The fixed resistors are included to guarantee a minimum resistance R from the wiper to either power supply at all times and thus prevent possible overheating of the potentiometer. An expression for V_{ref} is obtained from voltage division:

$$\frac{V_{ref} - V_S^-}{V_S^+ - V_S^-} = \frac{R + \Delta R}{2R + R_p}$$

If the voltage supplies are symmetric, as is usually the case, $V_S^+ = -V_S^-$ such that:

$$\frac{V_{ref} + V_S^+}{2V_S^+} = \frac{R + \Delta R}{2R + R_p}$$

Rearrange terms to find:

$$V_{ref} = \frac{2\Delta R - R_p}{2R + R_p} V_S^+$$

The value of V_{ref} is determined by the position of the wiper ΔR . Also, when $R_p \gg R$, the range of V_{ref} is approximately $\pm V_S^+$.

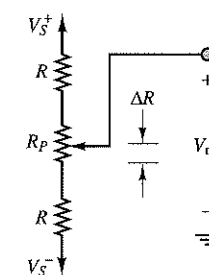


Figure 8.32

EXAMPLE 8.4 Temperature Control Using Op-Amps

Problem

Op-amps often serve as building blocks in analog control systems. The objective of this example is to illustrate the use of op-amps in a temperature control circuit. Figure 8.33(a) depicts a system for which the temperature is to be maintained constant at 20°C in a variable temperature environment. The system temperature is measured via a thermocouple (see the "Temperature Measurement" section in Chapter 15). Heat is added to the system by a coil of resistance R_{coil} . The heat flux is $q_{in} = i^2 R_{coil}$, where i is the current provided by a power amplifier. The system is insulated on three sides. The fourth side is not insulated such that heat is transferred across the boundary by convection, which is represented by an equivalent thermal resistance R_t . The system has mass m , specific heat c , and thermal capacitance $C_t = mc$ (see



the Make the Connection boxes "Thermal Capacitance" and "Thermal System Dynamics" in Chapter 5).

Solution

Known Quantities: Sensor (input) voltage; feedback and source resistors, thermal system component values.

Find: Select desired value of proportional gain K_p to achieve automatic temperature control.

Schematics, Diagrams, Circuits, and Given Data: $R_{\text{coil}} = 5 \Omega$; $R_t = 2^\circ\text{C}/\text{W}$; $C_t = 50 \text{ J}/^\circ\text{C}$; $\alpha = 1 \text{ V}/^\circ\text{C}$. Figure 8.33(a) to (e).

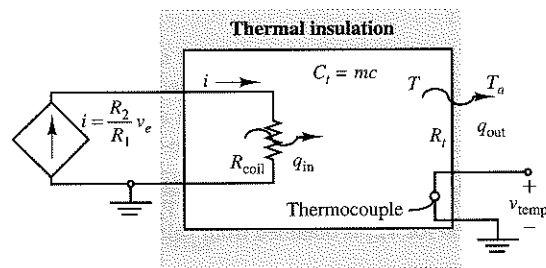


Figure 8.33 (a) Thermal system

Assumptions: Assume ideal op-amps.

Analysis: Conservation of energy requires that:

$$q_{\text{in}} - q_{\text{out}} = \frac{dE_{\text{stored}}}{dt}$$

where q_{in} represents the heat added to the system by the electrical heater, q_{out} represents the heat lost from the system through convection to the surrounding air, and E_{stored} represents the energy stored in the system due to its thermal capacitance. The system temperature T is measured by a thermocouple whose output voltage is proportional to temperature: $v_{\text{temp}} = \alpha T$. Further, assume that the power amplifier is modeled by a voltage-controlled current source (VCCS) such that:

$$i = K_v K_p v_e = \frac{R_2}{R_1} v_e = \frac{R_2}{R_1} (v_{\text{ref}} - v_{\text{temp}}) = \frac{R_2}{R_1} \alpha (T_{\text{ref}} - T)$$

where v_e is the error or difference between the reference voltage and the measured voltage. The negative feedback system shown in Figure 8.33(b) tends to drive v_e to zero. When v_e

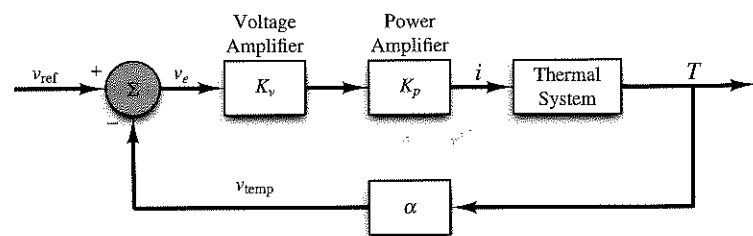


Figure 8.33 (b) Block diagram of control system

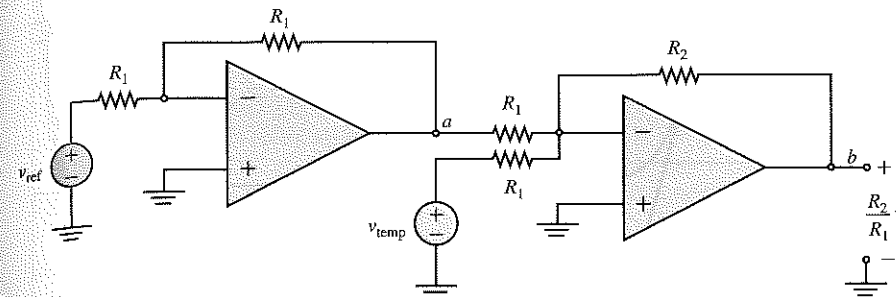


Figure 8.33 (c) Circuit for generating proportional gain of error voltage

is positive, $v_{\text{ref}} > v_{\text{temp}}$ and the system calls for heating; on the other hand, when v_e is negative, $v_{\text{ref}} < v_{\text{temp}}$ and the system calls for cooling. The power amplifier outputs a positive current for a positive v_e . Thus, the block diagram shown in Figure 8.33(b) corresponds to an automatic control system that increases or decreases the heating coil current to maintain the system temperature at the desired (reference) value. The proportional gain K_p of the power amplifier determines how much to increase coil current and allows the user to optimize the response of the system for a specific design requirement. For example, a system specification could require that the automatic temperature control system be designed so as to maintain the temperature to within 1 degree of the reference temperature for external temperature disturbances as large as 10 degrees. The response of the system can be adjusted by varying the proportional gain.

The voltage amplifier can be realized by a two-stage amplifier using two op-amps as shown in Figure 8.33(c). The first stage is an inverting amplifier with closed-loop gain $G_1 = -1$ such that the voltage at node a is $-v_{\text{ref}}$. The second stage is a summing amplifier with a closed-loop gain of $G_2 = -R_2/R_1$ for each input. Thus, the output voltage at node b is:

$$v_b = -\frac{R_2}{R_1} (v_a + v_{\text{temp}}) = \frac{R_2}{R_1} (v_{\text{ref}} - v_{\text{temp}}) = \frac{R_2}{R_1} (v_e)$$

The coefficient R_2/R_1 is the voltage gain K_v . In other words, selecting the feedback resistor R_2 is equivalent to choosing K_v .

The thermal system itself is described by the conservation of energy equation given above. The rate of energy added to the system by the heating coil is simply $i^2 R_{\text{coil}}$. The rate of energy subtracted from the system by convective heat transfer is defined as $(T - T_a)/R_t$, where R_t is a lumped parameter called the thermal resistance. Small values of R_t correspond to large values of the convective heat transfer coefficient, and vice versa. Finally, the net rate at which energy is stored in the system is proportional to the rate at which the system temperature T is changing, where the constant of proportionality C_t is known as the thermal capacitance. With these definitions in place, the conservation of energy equation can be rewritten as:

$$i^2 R_{\text{coil}} - \frac{T - T_a}{R_t} = C_t \frac{dT}{dt}$$

or

$$R_t C_t \frac{dT}{dt} + T = R_t R_{\text{coil}} i^2 + T_a$$

where $i = K_p K_v v_e = K_p K_v \alpha (T_{\text{ref}} - T)$. Notice this equation is, in general, a nonlinear first-order ordinary differential equation. The time constant is $\tau = R_t C_t = 2^\circ\text{C}/\text{W} \times 50 \text{ J}/^\circ\text{C} = 100 \text{ s}$.

When $K_p = 0$, no current is supplied to the heating coil and the thermal system response is simply its own natural response; that is, no automatic control is active when $K_p = 0$ and the

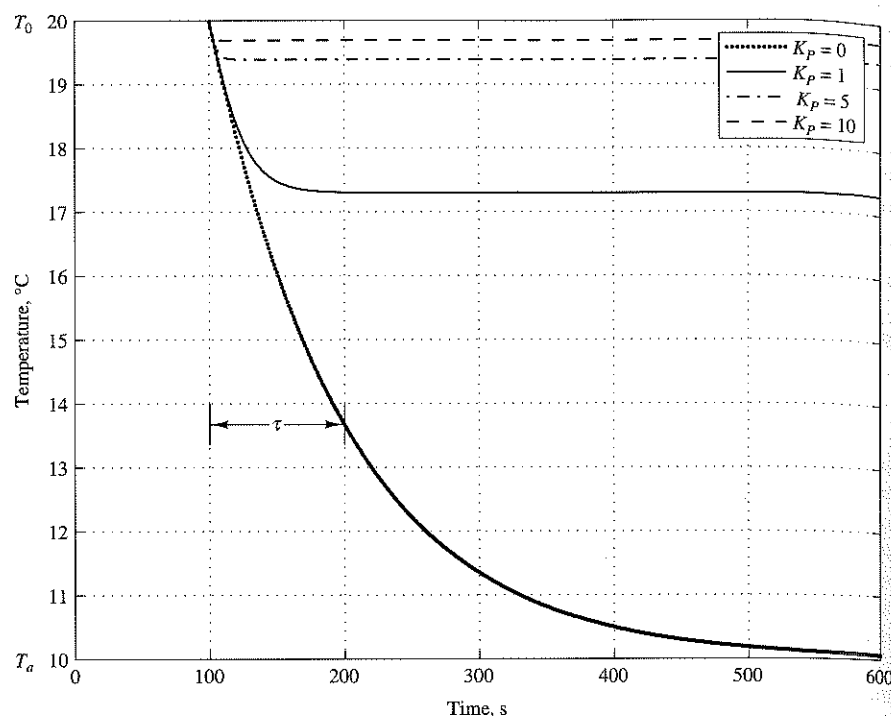


Figure 8.33 (d) Response of thermal system for various values of proportional gain K_p

system response is the *open-loop* response. In that case, the governing differential equation is:

$$\tau \frac{dT}{dt} + T = T_a$$

The solution is (see Chapter 5):

$$T = (T_0 - T_a)e^{-t/\tau} + T_a$$

where T_0 is the initial value of the system temperature. For example, assume $T_0 = 20^\circ\text{C}$ and $T_a = 10^\circ\text{C}$. The time constant τ is $R_t C_t = 100$ s, using the data given for R_t and C_t . Thus:

$$T = (10^\circ\text{C})e^{-t/100} + 10^\circ\text{C} \quad \text{Open-loop response}$$

When the gain K_p is increased to 1, v_e increases as soon as the temperature drops below the reference value. The transduction constant of the thermocouple was given as $\alpha = 1$ such that the voltage v_{temp} is numerically equal to the system temperature. Figure 8.33(d) shows the temperature response for values of K_p ranging from 1 to 10. As the gain increases, the error between the desired and actual temperatures decreases very quickly. Observe that the error is less than 1 degree (recall the design specification) for $K_p = 5$. To better understand the workings of the complete control system, it is helpful to observe the heater current, which is an amplified version of the error voltage. Figure 8.33(e) shows that when $K_p = 1$ the current increases to a final value of roughly 2.7 A; when $K_p = 5$ and 10, the current increases more rapidly, and eventually settles to values of 3 and 3.1 A, respectively. The steady-state value of the current is reached in about 17 s for $K_p = 5$, and in about 8 s for $K_p = 10$.

Comments: As K_p increases, the system's speed of response increases; however, the system's *steady-state error* also increases. The design specifications anticipate this effect by setting a tolerance of 1°C .

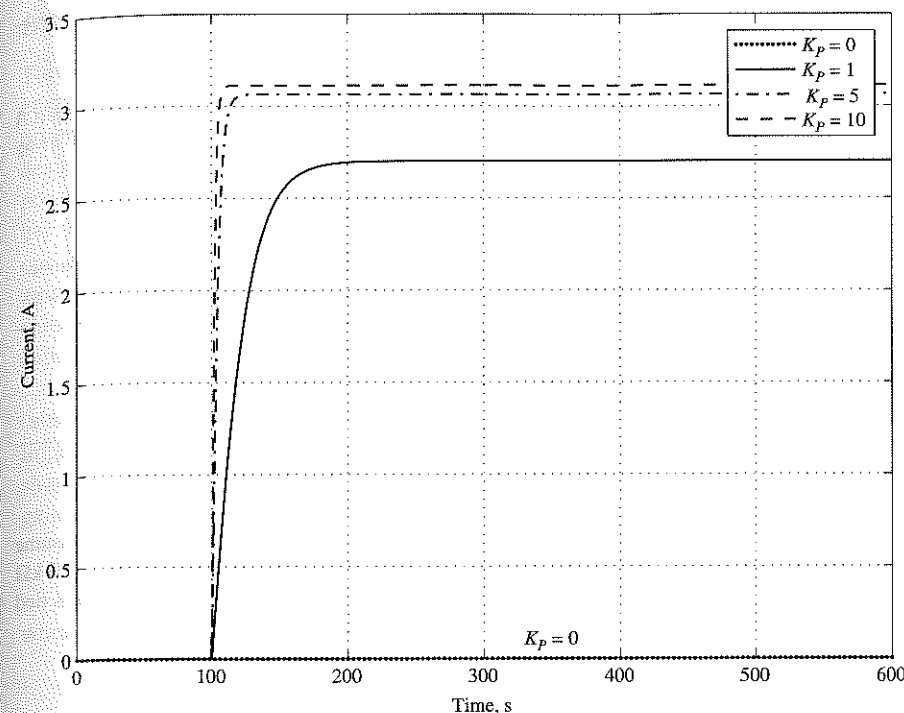


Figure 8.33 (e) Power amplifier output current for various proportional gain K_p

CHECK YOUR UNDERSTANDING

Consider an ideal inverting amplifier (see Figure 8.8) with a nominal closed-loop gain of $-1,000$. The impact of a non-ideal op-amp with a finite, but large, open-loop gain A on the closed-loop gain can be derived by assuming that the voltage v^- at the inverting terminal is only approximately equal to the voltage $v^+ = 0$ at the non-inverting terminal. Under this assumption, $v_{\text{out}} = -Av^-$. The first golden rule still applies such that $i_{\text{in}} = 0$ and R_S is virtually in series with R_F . Use this information to find an expression for the closed-loop gain as a function of the open-loop gain A . Compute the closed-loop gain when A equals 10^7 , 10^6 , 10^5 , 10^4 . How large is the open-loop gain when the closed-loop gain is less than 0.1 percent away from its nominal value?

Answers: 999.1; 999.0; 999.0; 999.1. For 0.1 percent accuracy, $A = 10^5$.

CHECK YOUR UNDERSTANDING

For Example 8.1, calculate the uncertainty in the gain when 1 percent “precision” resistors are used.

Answer: +1.98 to -2.02 percent

CHECK YOUR UNDERSTANDING

Derive an expression for the closed-loop gain of an isolation buffer when the open-loop gain A is finite. How large is the open-loop gain when the closed-loop gain is only 0.1 percent away from unity?

Answer: The expression for the closed-loop gain is $v_{out}/v_{in} = 1 + 1/A$; thus A should be equal 10^4 for 0.1 percent accuracy.

CHECK YOUR UNDERSTANDING

For Example 8.3, find the value ΔR that removes the DC bias from the sensor signal. Assume the supply voltages are symmetric at ± 15 V and a 10-k Ω potentiometer is tied to two 10-k Ω fixed resistors as in Figure 8.32. What is the range of V_{ref} when a 1-k Ω potentiometer is tied to two 10-k Ω fixed resistors?

Answers: $\Delta R = 6.722$ k Ω ; V_{ref} is between ± 0.714 V

CHECK YOUR UNDERSTANDING

How much steady-state power, in watts, will be input to the thermal system of Example 8.4 to maintain its temperature in the face of a 10°C ambient temperature drop for values of K_p of 1, 5, and 10?

Answers: $K_p = 1$: 36.5 W; $K_p = 5$: 45 W; $K_p = 10$: 48 W

CHECK YOUR UNDERSTANDING

With reference to the Focus on Measurements box "Sensor Calibration Circuit," find numerical values of R_F/R_S and V_{ref} if the temperature sensor has $\beta = 0.235$ and $V_0 = 0.7$ V and the desired relationship is $v_{out} = 10 T$.

Answers: $R_F/R_S = 42.55$; $V_{ref} = 0.684$ V

8.3 ACTIVE FILTERS

The range of useful applications of an operational amplifier is greatly expanded if energy storage elements are introduced into the design; the frequency-dependent properties of these elements, studied in Chapters 4 and 6, will prove useful in the design of various types of op-amp circuits. In particular, it will be shown that it is possible to shape the frequency response of an operational amplifier by appropriate use of complex impedances in the input and feedback circuits. The class of filters one can obtain by means of op-amp designs is called **active filters** because op-amps can provide amplification (gain) in addition to the filtering effects already studied in Chapter 6 for passive circuits (i.e., circuits comprising exclusively resistors, capacitors, and inductors).

The easiest way to see how the frequency response of an op-amp can be shaped (almost) arbitrarily is to replace the resistors R_F and R_S in Figures 8.8 and 8.9 with impedances Z_F and Z_S , as shown in Figure 8.34. It is a straightforward matter to show that in the case of the inverting amplifier, the expression for the closed-loop gain is given by

$$\frac{V_o}{V_S}(j\omega) = -\frac{Z_F}{Z_S} \quad (8.71)$$

whereas for the non-inverting case, the gain is

$$\frac{V_o}{V_S}(j\omega) = 1 + \frac{Z_F}{Z_S} \quad (8.72)$$

where Z_F and Z_S can be arbitrarily complex impedance functions and where V_S , V_o , I_F , and I_S are all phasors. Thus, it is possible to shape the frequency response of an ideal op-amp filter simply by selecting suitable ratios of feedback impedance to source impedance. By connecting a circuit similar to the low-pass filters studied in Chapter 6 in the feedback loop of an op-amp, the same filtering effect can be achieved and, in addition, the signal can be amplified.

The simplest op-amp low-pass filter is shown in Figure 8.35. Its analysis is quite simple if we take advantage of the fact that the closed-loop gain, as a function of frequency, is given by

$$G_{LP}(j\omega) = -\frac{Z_F}{Z_S} \quad (8.73)$$

where

$$Z_F = R_F \parallel \frac{1}{j\omega C_F} = \frac{R_F}{1 + j\omega C_F R_F} \quad (8.74)$$

and

$$Z_S = R_S \quad (8.75)$$

Note the similarity between Z_F and the low-pass characteristic of the passive RC circuit! The closed-loop gain $G_{LP}(j\omega)$ is then computed to be

$$G_{LP}(j\omega) = -\frac{Z_F}{Z_S} = -\frac{R_F/R_S}{1 + j\omega C_F R_F} \quad \text{Low-pass filter} \quad (8.76)$$

This expression can be factored into two terms. The first is an amplification factor analogous to the amplification that would be obtained with a simple inverting amplifier (i.e., the same circuit as that of Figure 8.35 with the capacitor removed); the second is a low-pass filter, with a cutoff frequency dictated by the parallel combination of

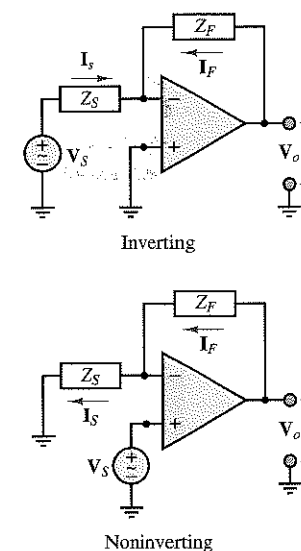


Figure 8.34 Op-amp circuits employing complex impedances

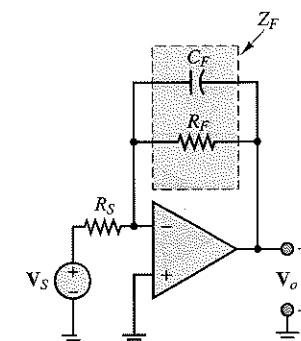


Figure 8.35 Active low-pass filter



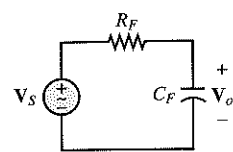


Figure 8.36 Passive low-pass filter

R_F and C_F in the feedback loop. The filtering effect is completely analogous to what would be attained by the passive circuit shown in Figure 8.36. However, the op-amp filter also provides amplification by a factor of R_F/R_S .

It should be apparent that the response of this op-amp filter is just an amplified version of that of the passive filter. Figure 8.37 depicts the amplitude response of the active low-pass filter (in the figure, $R_F/R_S = 10$ and $1/R_F C_F = 1$) in two different graphs; the first plots the amplitude ratio $V_o(j\omega)$ versus radian frequency ω on a logarithmic scale, while the second plots the amplitude ratio $20 \log V_o(j\omega)$ (in units of decibels), also versus ω on a logarithmic scale. Recall from Chapter 6 that decibel frequency response plots are often encountered. Note that in the decibel plot, the slope of the filter response for frequencies significantly higher than the cutoff frequency,

$$\omega_0 = \frac{1}{R_F C_F} \quad (8.77)$$

is -20 dB/decade, while the slope for frequencies significantly lower than this cutoff frequency is equal to zero. The value of the response at the cutoff frequency is found to be, in units of decibel,

$$|G_{LP}(j\omega_0)|_{dB} = 20 \log_{10} \frac{R_F}{R_S} - 20 \log \sqrt{2} \quad (8.78)$$

where

$$-20 \log_{10} \sqrt{2} = -3 \text{ dB} \quad (8.79)$$

Thus, ω_0 is also called the 3-dB frequency.

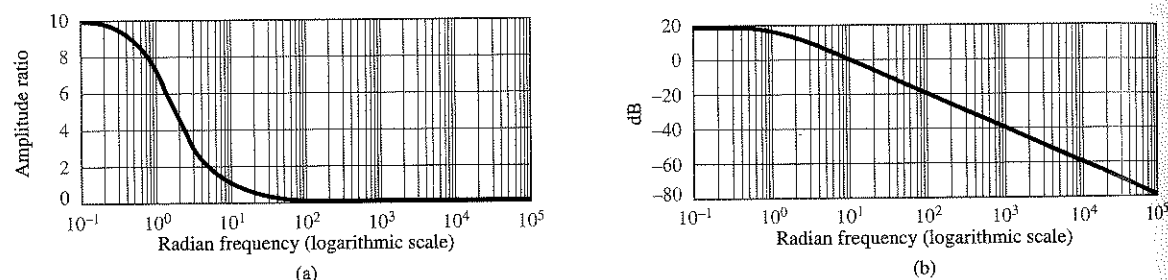


Figure 8.37 Normalized response of active low-pass filter (a) amplitude ratio response; (b) dB response

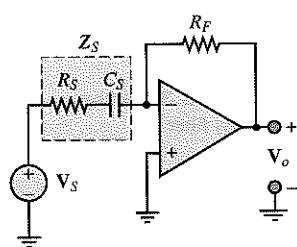


Figure 8.38 Active high-pass filter

Among the advantages of such active low-pass filters is the ease with which the gain and the bandwidth can be adjusted by controlling the ratios R_F/R_S and $1/R_F C_F$, respectively.

It is also possible to construct other types of filters by suitably connecting resistors and energy storage elements to an op-amp. For example, a high-pass active filter can easily be obtained by using the circuit shown in Figure 8.38. The impedance of the input path is:

$$Z_S = R_S + \frac{1}{j\omega C_S} \quad (8.80)$$

The impedance of the feedback path is:

$$Z_F = R_F \quad (8.81)$$

The closed-loop gain for this inverting amplifier is:

$$G_{HP}(j\omega) = -\frac{Z_F}{Z_S} = -\frac{j\omega C_S R_F}{1 + j\omega R_S C_S} \quad \text{High-pass filter} \quad (8.82)$$

Note that $G \rightarrow 0$ as $\omega \rightarrow 0$. Also note that as $\omega \rightarrow \infty$, the closed-loop gain G approaches a constant:

$$\lim_{\omega \rightarrow \infty} G_{HP}(j\omega) = -\frac{R_F}{R_S} \quad (8.83)$$

That is, above a certain frequency range, the circuit acts as a linear amplifier. This is exactly the behavior one would expect of a high-pass filter. The high-pass response is depicted in Figure 8.39, in both linear and decibel plots (in the figure, $R_F/R_S = 10$ and $1/R_S C_S = 1$). Note that in the decibel plot, the slope of the filter response for frequencies significantly lower than $\omega = 1/R_S C_S = 1$ is $+20$ dB/decade, while the slope for frequencies significantly higher than this cutoff (or 3 dB) frequency is equal to zero.

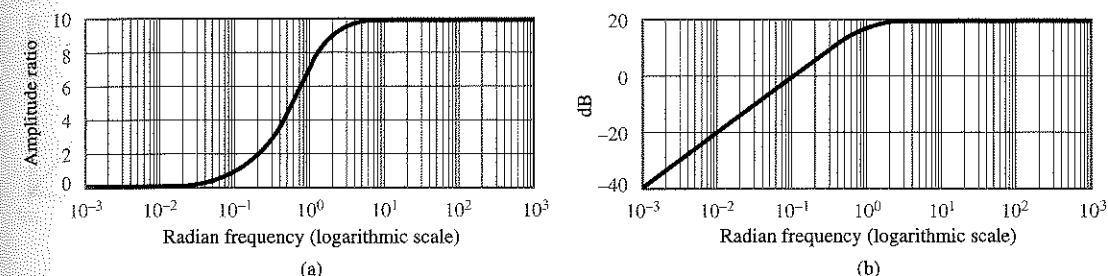


Figure 8.39 Normalized response of active high-pass filter (a) amplitude ratio response; (b) dB response

As a final example of active filters, let us look at a simple active bandpass filter configuration. This type of response may be realized simply by combining the high- and low-pass filters we examined earlier. The circuit is shown in Figure 8.40.

The analysis of the bandpass circuit follows the same structure used in previous examples. First we evaluate the feedback and input impedances:

$$Z_F = R_F \parallel \frac{1}{j\omega C_F} = \frac{R_F}{1 + j\omega C_F R_F} \quad (8.84)$$

$$Z_S = R_S + \frac{1}{j\omega C_S} = \frac{1 + j\omega C_S R_S}{j\omega C_S} \quad (8.85)$$

Next we compute the closed-loop frequency response of the op-amp, as follows:

$$G_{BP}(j\omega) = -\frac{Z_F}{Z_S} = -\frac{j\omega C_S R_F}{(1 + j\omega C_F R_F)(1 + j\omega C_S R_S)} \quad \text{Bandpass filter} \quad (8.86)$$

The form of the op-amp response we just obtained should not be a surprise. It is very similar (although not identical) to the product of the low-pass and high-pass responses of equations 8.76 and 8.82. In particular, the denominator of $G_{BP}(j\omega)$ is exactly the product of the denominators of $G_{LP}(j\omega)$ and $G_{HP}(j\omega)$. It is particularly enlightening

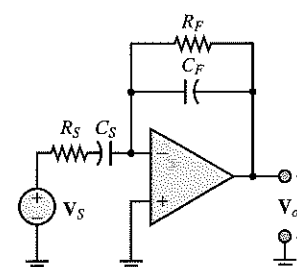


Figure 8.40 Active bandpass filter



to rewrite $G_{LP}(j\omega)$ in a slightly different form, after making the observation that each RC product corresponds to some "critical" frequency:

$$\omega_1 = \frac{1}{R_F C_S} \quad \omega_{LP} = \frac{1}{R_F C_F} \quad \omega_{HP} = \frac{1}{R_S C_S} \quad (8.87)$$

It is easy to verify that for the case where

$$\omega_{HP} > \omega_{LP} \quad (8.88)$$

the response of the op-amp filter may be represented as shown in Figure 8.41 in both linear and decibel plots (in the figure, $\omega_1 = 1$, $\omega_{HP} = 1,000$, and $\omega_{LP} = 10$). The decibel plot is very revealing, for it shows that, in effect, the bandpass response is the graphical superposition of the low-pass and high-pass responses shown earlier. The two 3-dB (or cutoff) frequencies are the same as in $G_{LP}(j\omega)$, $1/R_F C_F$; and in $G_{HP}(j\omega)$, $1/R_S C_S$. The third frequency, $\omega_1 = 1/R_F C_S$, represents the point where the response of the filter crosses the 0-dB axis (rising slope). Since 0 dB corresponds to a gain of 1, this frequency is called the **unity gain frequency**.

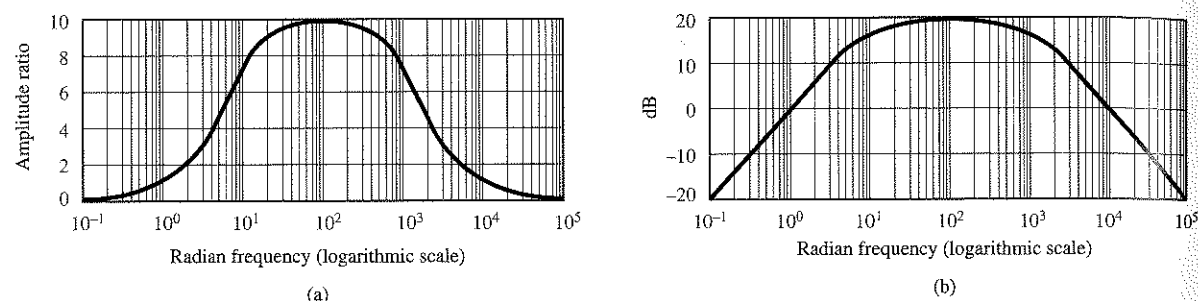


Figure 8.41 Normalized amplitude response of active bandpass filter (a) amplitude ratio response; (b) dB response

The ideas developed thus far can be employed to construct more complex functions of frequency. In fact, most active filters one encounters in practical applications are based on circuits involving more than one or two energy storage elements. By constructing suitable functions for Z_F and Z_S , it is possible to realize filters with greater frequency selectivity (i.e., sharpness of cutoff), as well as flatter bandpass or band-rejection functions (i.e., filters that either allow or reject signals in a limited band of frequencies). A few simple applications are investigated in the homework problems; advanced applications are explored in Chapter 15. One remark that should be made in passing, though, pertains to the exclusive use of capacitors in the circuits analyzed thus far. One of the advantages of op-amp filters is that it is not necessary to use both capacitors and inductors to obtain a bandpass response. Suitable connections of capacitors can accomplish that task in an op-amp. This seemingly minor fact is of great importance in practice because inductors are expensive to mass-produce to close tolerances and exact specifications and are often bulkier than capacitors with equivalent energy storage capabilities. On the other hand, capacitors are easy to manufacture in a wide variety of tolerances and values, and in relatively compact packages, including in integrated-circuit form.

Example 8.5 illustrates how it is possible to construct active filters with greater frequency selectivity by adding energy storage elements to the design.

EXAMPLE 8.5 Second-Order Low-Pass Filter

Problem

Determine the closed-loop voltage gain as a function of frequency for the op-amp circuit of Figure 8.42.

Solution

Known Quantities: Feedback and source impedances.

Find:

$$G(j\omega) = \frac{V_o(j\omega)}{V_s(j\omega)}$$

Schematics, Diagrams, Circuits, and Given Data: $1/R_2 C = R_1/L = \omega_0$.

Assumptions: Assume an ideal op-amp.

Analysis: The expression for the gain of the filter of Figure 8.42 can be determined by using equation 8.71:

$$G(j\omega) = \frac{V_o(j\omega)}{V_s(j\omega)} = -\frac{Z_F(j\omega)}{Z_S(j\omega)}$$

where

$$Z_F(j\omega) = R_2 \parallel \frac{1}{j\omega C} = \frac{R_2}{1 + j\omega C R_2} = \frac{R_2}{1 + j\omega/\omega_0}$$

$$Z_S(j\omega) = R_1 + j\omega L = R_1 \left(1 + j\omega \frac{L}{R_1} \right) = R_1 \left(1 + \frac{j\omega}{\omega_0} \right)$$

Thus, the closed-loop gain G of the filter is:

$$\begin{aligned} G(j\omega) &= -\frac{R_2/(1 + j\omega/\omega_0)}{R_1(1 + j\omega/\omega_0)} \\ &= -\frac{R_2/R_1}{(1 + j\omega/\omega_0)^2} \end{aligned}$$

Note that it is possible to simplify the circuit in Figure 8.42 at very low and very high frequencies. The solutions for these simplified forms can be used to validate the previous expression. For example, at very low frequencies, the inductor acts like a short-circuit and the capacitor acts like an open-circuit. Using these approximations, the circuit becomes a simple inverting amplifier with closed-loop gain:

$$G(j\omega) \approx -\frac{R_2}{R_1} \quad \omega \ll \omega_0$$

This approximation matches the complete solution when $\omega \ll \omega_0$ because $1 + j\omega/\omega_0 \approx 1$. Likewise, at very high frequencies, the inductor acts as an open-circuit and the capacitor acts as a short-circuit. Using these approximations, the circuit experiences a virtual short-circuit from its output to the inverting terminal and the source V_s sees a very large input impedance. The result is that $V_o \approx 0$ due to the virtual ground at the inverting terminal and thus the closed-loop

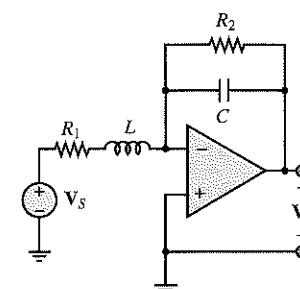


Figure 8.42

gain is:

$$G(j\omega) \approx 0 \quad \omega \gg \omega_0$$

This approximation matches the complete solution when $\omega \gg \omega_0$ because $1 + j\omega/\omega_0 \rightarrow j\omega/\omega_0 \rightarrow \infty$ as $\omega \rightarrow \infty$. Validations are an important aspect of good problem solving because they add confidence to solutions and, if done smartly, usually expose erroneous solutions.

Comments: Note the similarity between the expression for the gain of the filter of Figure 8.42 and that given in equation 8.76 for the gain of a first-order low-pass filter. Clearly, the circuit analyzed in this example is also a second-order low-pass filter, as indicated by the quadratic term in the denominator. Figure 8.43 compares the two responses in both linear and decibel (Bode) magnitude plots. The slope of the decibel plot for the second-order filter at higher frequencies is twice that of the first-order filter (-40 versus -20 dB/decade). We should also remark that the use of an inductor in the filter design is not recommended in practice, as explained in the above section, and that we have used it in this example only because of the simplicity of the resulting gain expressions. Chapter 15 introduces design methods for practical high-order filters.

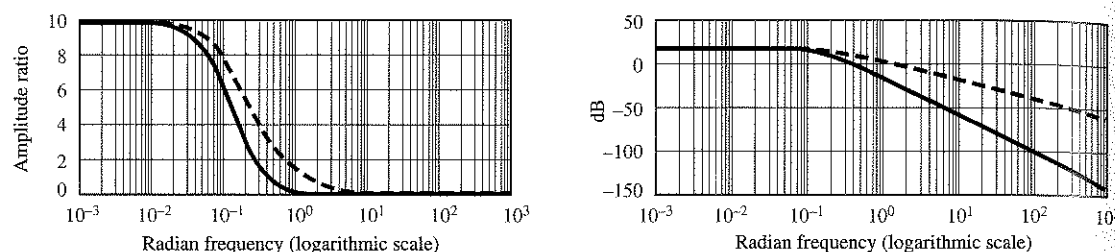


Figure 8.43 Comparison of first- and second-order active low-pass filters: (a) amplitude ratio response; (b) dB response

CHECK YOUR UNDERSTANDING

- Design a low-pass filter with a closed-loop gain of 100 and cutoff (3-dB) frequency equal to 800 Hz. Assume that only $0.01\text{-}\mu\text{F}$ capacitors are available. Find R_F and R_S .
- Repeat the design of the exercise above for a high-pass filter with a cutoff frequency of 2,000 Hz. This time, however, assume that only standard values of resistors are available (see the table of standard values in Chapter 2). Select the nearest component values, and calculate the percent error in cutoff frequency.
- Find the frequencies corresponding to 1-dB attenuation from the low-frequency gains of the filters of parts a and b.
- What is the decibel gain for the filter of Example 8.5 at the cutoff frequency ω_0 ? Find the 3-dB frequency for this filter in terms of the cutoff frequency ω_0 , and note that the two are not the same.

Answers: Part a: $R_F = 19.9\text{ k}\Omega$, $R_S = 199\text{ }\Omega$; part b: $R_F = 8.2\text{ k}\Omega$, $R_S = 82\text{ }\Omega$, error: gain = 0 percent, $\omega_{3\text{dB}} = 2.95$ percent; part c: 407 Hz and 3.8 kHz; part d: -6 dB; $\omega_{3\text{dB}} = 0.642\omega_0$

8.4 INTEGRATORS AND DIFFERENTIATORS

In the preceding sections, we examined the frequency response of op-amp circuits for sinusoidal inputs. However, certain op-amp circuits containing energy storage elements reveal some of their more general properties if we analyze their response to inputs that are time varying but not necessarily sinusoidal. Among such circuits are the commonly used integrator and differentiator; the analysis of these circuits is presented in the following paragraphs.

The Ideal Integrator

Consider the circuit of Figure 8.44, where $v_S(t)$ is an arbitrary function of time (e.g., a pulse train, a triangular wave, or a square wave). The op-amp circuit shown provides an output that is proportional to the integral of $v_S(t)$. The analysis of the integrator circuit is, as always, based on the observation that

$$i_S(t) = -i_F(t) \quad (8.89)$$

where

$$i_S(t) = \frac{v_S(t)}{R_S} \quad (8.90)$$

It is also known that

$$i_F(t) = C_F \frac{dv_o(t)}{dt} \quad (8.91)$$

from the fundamental definition of the capacitor. The source voltage can then be expressed as a function of the derivative of the output voltage:

$$\frac{1}{R_S C_F} v_S(t) = -\frac{dv_o(t)}{dt} \quad (8.92)$$

Integrate both sides of equation 8.92 to obtain:

$$v_o(t) = -\frac{1}{R_S C_F} \int_{-\infty}^t v_S(t') dt' \quad \text{Integrator} \quad (8.93)$$

There are numerous applications for integrators,

The Ideal Differentiator

Using an argument similar to that employed for the integrator, we can derive a result for the ideal differentiator circuit of Figure 8.45. The relationship between input and output is obtained by observing that

$$i_S(t) = C_S \frac{dv_S(t)}{dt} \quad (8.94)$$

and

$$i_F(t) = \frac{V_o(t)}{R_F} \quad (8.95)$$

so that the output of the differentiator circuit is proportional to the derivative of the input:

$$V_o(t) = -R_F C_S \frac{dv_S(t)}{dt} \quad \text{Op-amp differentiator} \quad (8.96)$$

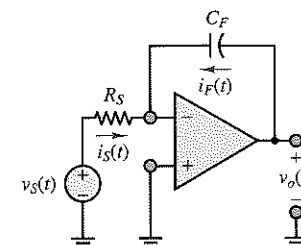


Figure 8.44 Op-amp integrator

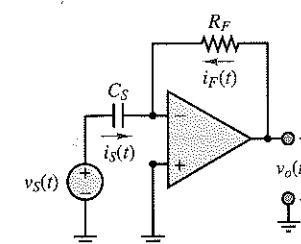
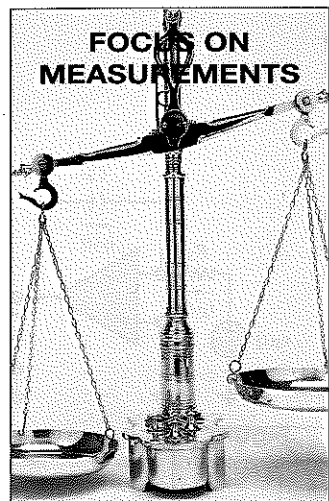


Figure 8.45 Op-amp differentiator



Although mathematically attractive, the differentiation property of this op-amp circuit is seldom used in practice because differentiation tends to amplify any noise that may be present in a signal.

FOCUS ON MEASUREMENTS



Charge Amplifiers

One of the most common families of transducers for the measurement of force, pressure, and acceleration is that of **piezoelectric transducers**. These transducers contain a piezoelectric crystal that generates an electric charge in response to deformation. Thus, if a force is applied to the crystal (leading to a displacement), a charge is generated within the crystal. If the external force generates a displacement x_i , then the transducer will generate a charge q according to the expression:

$$q = K_P x_i$$

Figure 8.46 depicts the basic structure of the piezoelectric transducer, and a simple circuit model. The model consists of a current source in parallel with a capacitor, where the current source represents the rate of change of the charge generated in response to an external force; and the capacitance is a consequence of the structure of the transducer, which consists of a piezoelectric crystal (e.g., quartz or Rochelle salt) sandwiched between conducting electrodes (in effect, this is a parallel-plate capacitor).

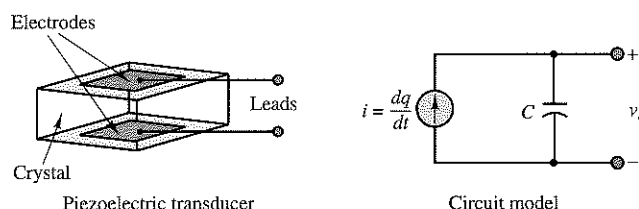


Figure 8.46 Piezoelectric transducer

Although it is possible, in principle, to employ a conventional voltage amplifier to amplify the transducer output voltage v_i , given by

$$v_i = \frac{1}{C} \int i \, dt = \frac{1}{C} \int \frac{dq}{dt} \, dt = \frac{q}{C} = \frac{K_P x_i}{C}$$

it is often advantageous to use a **charge amplifier**. The charge amplifier is essentially an integrator circuit, as shown in Figure 8.47, characterized by an extremely high input impedance.³ The high impedance is essential; otherwise, the charge generated by the transducer would leak to ground through the input impedance of the amplifier.

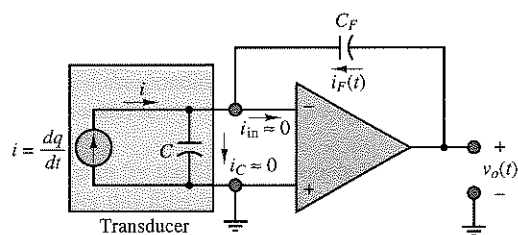


Figure 8.47 Charge amplifier

(Continued)

³Special op-amps are employed to achieve extremely high input impedance, through FET input circuits. See Chapter 11.

(Concluded)

Because of the high input impedance, the input current to the amplifier is negligible; further, because of the high open-loop gain of the amplifier, the inverting-terminal voltage is essentially at ground potential. Thus, *the voltage across the transducer is effectively zero*. As a consequence, to satisfy KCL, the feedback current $i_F(t)$ must be equal and opposite to the transducer current i :

$$i_F(t) = -i$$

and since

$$V_o(t) = \frac{1}{C_F} \int i_F(t) \, dt$$

it follows that the output voltage is proportional to the charge generated by the transducer, and therefore to the displacement:

$$V_o(t) = \frac{1}{C_F} \int -i \, dt = \frac{1}{C_F} \int -\frac{dq}{dt} \, dt = -\frac{q}{C_F} = -\frac{K_P x_i}{C_F}$$

Since the displacement is caused by an external force or pressure, this sensing principle is widely adopted in the measurement of force and pressure.

EXAMPLE 8.6 Integrating a Square Wave



Problem

Determine the output voltage for the integrator circuit of Figure 8.44 if the input is a square wave of amplitude $\pm A$ and period T , as shown in figure 8.48.

Solution

Known Quantities: Feedback and source impedances; input waveform characteristics.

Find: $v_o(t)$.

Schematics, Diagrams, Circuits, and Given Data: $T = 10 \text{ ms}$; $C_F = 1 \mu\text{F}$; $R_F = 10 \text{ k}\Omega$.

Assumptions: The op-amp is ideal and $v_o = 0$ at $t = 0$.

Analysis: Equation 8.93 expresses the output of an integrator as:

$$\begin{aligned} v_o(t) &= -\frac{1}{R_F C_S} \int_{-\infty}^t v_S(t') \, dt' = -\frac{1}{R_F C_S} \left[\int_{-\infty}^0 v_S(t') \, dt' + \int_0^t v_S(t') \, dt' \right] \\ &= v_o(0) - \frac{1}{R_F C_S} \int_0^t v_S(t') \, dt' \end{aligned}$$

The square wave can be integrated in a piecewise fashion by observing that $v_S(t) = A$ for $0 \leq t < T/2$ and $v_S(t) = -A$ for $T/2 \leq t < T$. Thus, for the two half periods of the

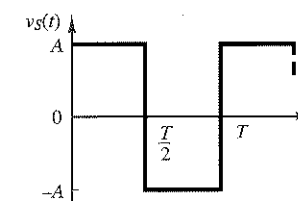


Figure 8.48

waveform:

$$v_o(t) = -\frac{1}{R_F C_S} \int_0^t v_S(t') dt' = -100 \int_0^t A dt'$$

$$= -100At \quad 0 \leq t < \frac{T}{2}$$

$$v_o(t) = v_o\left(\frac{T}{2}\right) - \frac{1}{R_F C_S} \int_{T/2}^t v_S(t') dt' = -100A \frac{T}{2} - 100 \int_{T/2}^t (-A) dt'$$

$$= -100A \frac{T}{2} + 100A \left(t - \frac{T}{2}\right) = -100A(T - t) \quad \frac{T}{2} \leq t < T$$

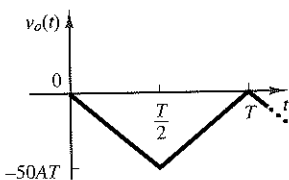


Figure 8.49

Since the waveform is periodic, the above result will repeat with period T , as shown in Figure 8.49. Note also that the average value of the output voltage is not zero.

Comments: The integral of a square wave is thus a triangular wave. This is a useful fact to remember. Note that the effect of the initial condition is very important since it determines the starting point of the triangular wave.



EXAMPLE 8.7 Proportional-Integral Control with Op-Amps

Problem

The aim of this example is to illustrate the very common practice of *proportional-integral*, or *PI*, control. Consider the temperature control circuit of Example 8.4, shown again in Figure 8.50(a), where it was discovered that the proportional control implemented with the gain K_P could still give rise to a steady-state error in the final temperature of the system. This error can be eliminated by using an automatic control system that feeds back a component that is proportional to the *integral of the error voltage*, in addition to the proportional term. Figure 8.50(b) depicts the block diagram of such a PI controller. Now, the design of the control system requires selecting two gains, the *proportional gain* K_P and the *integral gain* K_I .

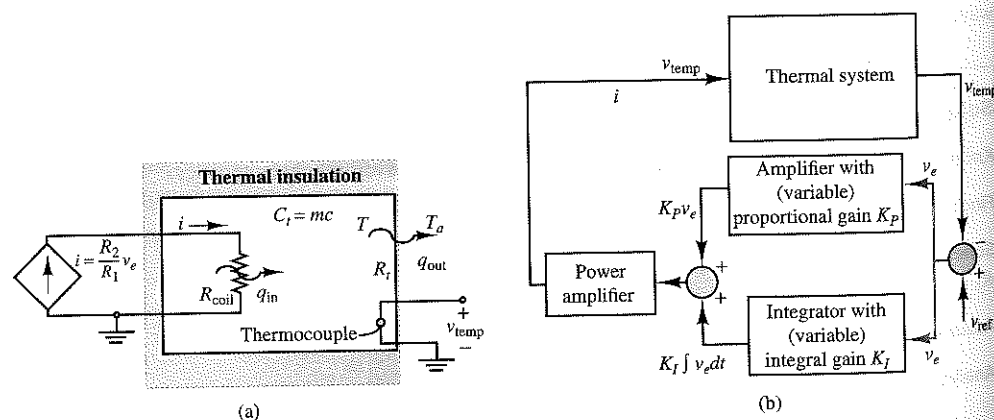


Figure 8.50 (a) Thermal system and (b) block diagram of control system

Solution

Known Quantities: Sensor (input) voltage; feedback and source resistors, thermal system component values.

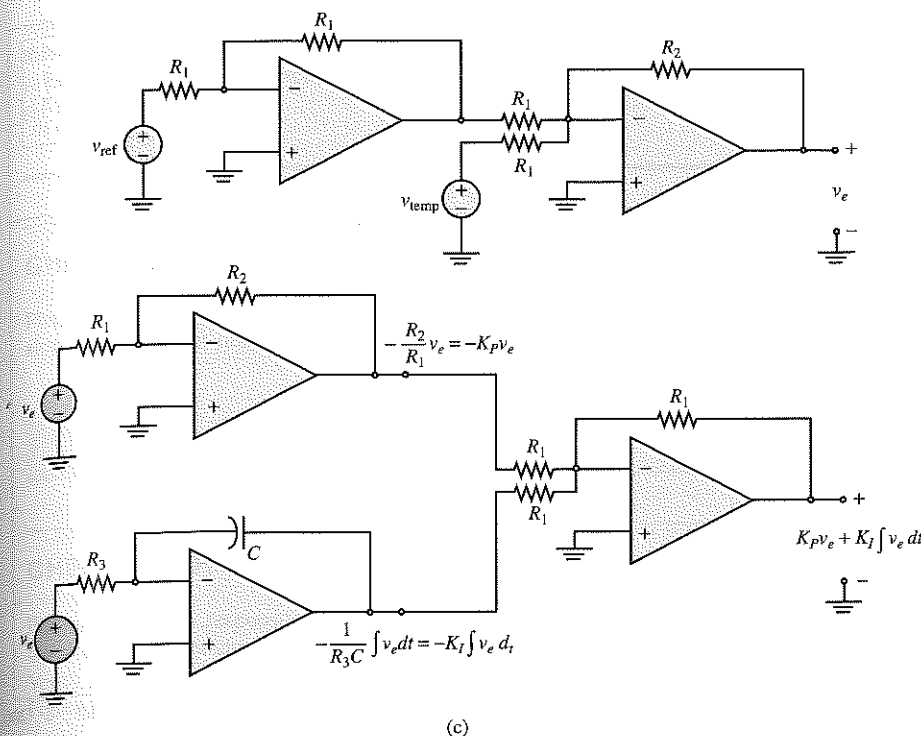


Figure 8.50 (c) Circuit for generating error voltage and proportional gain

Find: Select desired value of proportional gain K_P and integral gain K_I to achieve automatic temperature control with zero steady-state error.

Schematics, Diagrams, Circuits and Given Data: $R_{\text{coil}} = 5 \Omega$; $R_t = 2^\circ\text{C}/\text{W}$; $C_t = 50 \text{ J}/^\circ\text{C}$; $\alpha = 1 \text{ V}/^\circ\text{C}$.

Assumptions: Assume ideal op-amps.

Analysis: The circuit of Figure 8.50(c) shows two op-amp circuits — the top circuit generates the error voltage v_e . The only difference is that in this case the circuit does not provide any gain. The bottom circuit amplifies v_e by the proportional gain $-K_P = -R_2/R_1$ and also computes the integral of v_e times the integral gain $-K_I = -1/R_3C$. These two quantities are then summed through another inverting summer circuit, which takes care of the sign change as well.

Figure 8.50(d) depicts the temperature response of the system for $K_P = 5$ (as selected in Example 8.5) and different values of K_I . Note that the steady-state error is now zero! This is a property of controllers that incorporate an integral term. Figure 8.50(e) shows the current supplied to the heater coil. Note that the response is quite fast and that the temperature deviation is minimal.

Comments: The addition of the integral term in the controller causes the system temperature to oscillate in response to the -10°C temperature disturbance described in Example 8.4 (for sufficiently high values of K_I). This oscillation is a characteristic of an underdamped second-order system (see Chapter 5)—but we originally started out with a first-order thermal system! The addition of the integral term has increased the order of the system, and now it is possible for the system to display oscillatory behavior, that is, to have complex conjugate roots (poles). To those familiar with thermal systems, this behavior should cause a raised eyebrow! It is

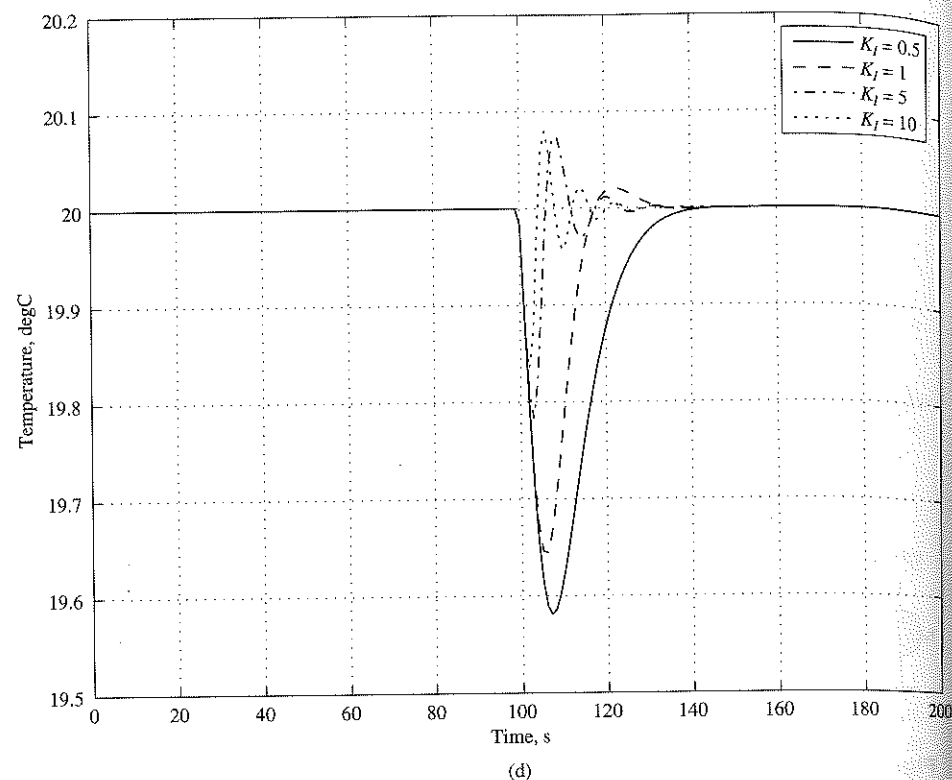


Figure 8.50 (d) Response of thermal system for various values of integral gain, K_I ($K_P = 5$)

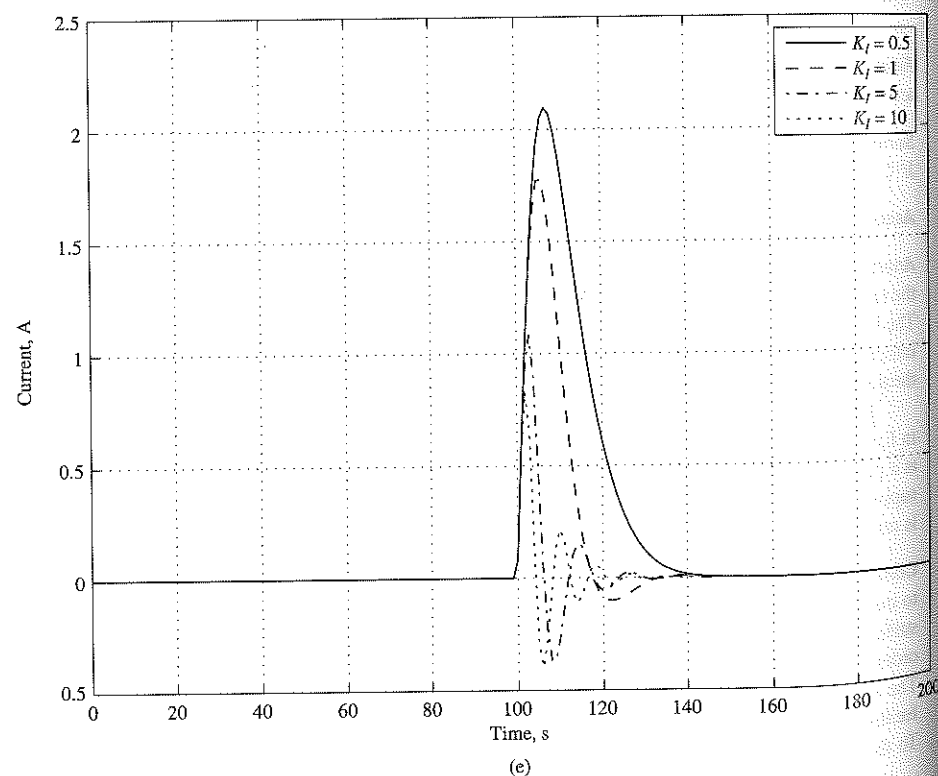


Figure 8.50 (e) Power amplifier current system for various values of integral gain K_I ($K_P = 5$)

well known that thermal systems cannot display underdamped behavior (that is, there is no thermal system property analogous to inductance). The introduction of the integral gain can, in fact, cause temperature oscillations as if an artificial "thermal inductor" were introduced in the system.

CHECK YOUR UNDERSTANDING

Plot the frequency response of an ideal integrator in the form of a Bode plot. Determine the slope of the straight-line segments in dB/decade. Assume $R_S C_F = 10$ s.

Answer: -20 dB/decade

CHECK YOUR UNDERSTANDING

Plot the frequency response of an ideal differentiator in the form of a Bode plot. Determine the slope of the straight-line segments in dB/decade. Assume $R_F C_S = 100$ s.

Verify that, if the triangular wave of Example 8.6 is the input to the ideal differentiator of Figure 8.45, the resulting output is a square wave.

Answer: $+20$ dB/decade

EXAMPLE 8.8 Using Cascaded Amplifiers to Simulate a Differential Equation

Problem

Derive the differential equation corresponding to the circuit shown in Figure 8.51.

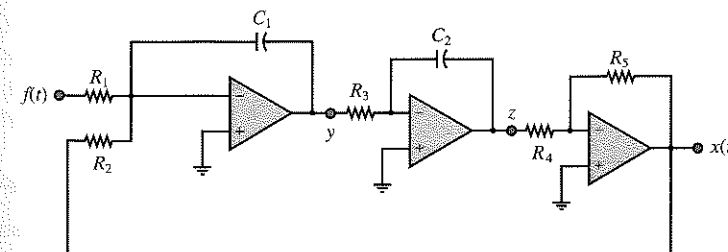


Figure 8.51 Analog computer simulation of unknown system

Solution

Known Quantities: Resistor and capacitor values.

Find: Differential equation in $x(t)$.

Schematics, Diagrams, Circuits, and Given Data: $R_1 = 0.4 \text{ M}\Omega$; $R_2 = R_3 = R_5 = 1 \text{ M}\Omega$; $R_4 = 2.5 \text{ k}\Omega$; $C_1 = C_2 = 1 \text{ }\mu\text{F}$.

Assumptions: Assume ideal op-amps.

Analysis: Begin the analysis from the right-hand side of the circuit to determine the intermediate variable z as a function of x :

$$x = -\frac{R_5}{R_4}z = -400z$$

Moving to the left, next determine the relationship between y and z :

$$z = -\frac{1}{R_3 C_2} \int y(t') dt' \quad \text{or} \quad y = -\frac{dz}{dt}$$

Finally, determine y as a function of x and f :

$$y = -\frac{1}{R_2 C_1} \int x(t') dt' - \frac{1}{R_1 C_1} \int f(t') dt' = -\int [x(t') + 2.5f(t')] dt'$$

or

$$\frac{dy}{dt} = -x - 2.5f$$

Substitute the expressions into one another and eliminate y and z to obtain:

$$\begin{aligned} x &= -400z \\ \frac{dx}{dt} &= -400 \frac{dz}{dt} = 400y \\ \frac{d^2x}{dt^2} &= 400 \frac{dy}{dt} = 400(-x - 2.5f) \end{aligned}$$

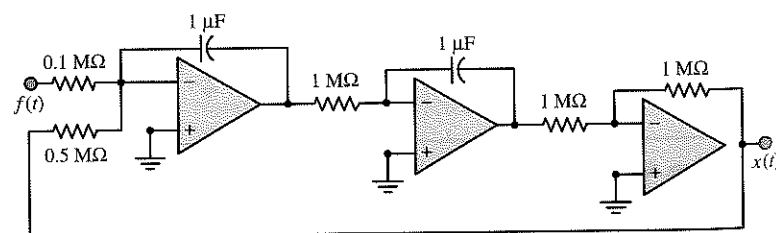
and

$$\frac{d^2x}{dt^2} + 400x = -1,000f$$

Comments: Note that the summing and integrating functions have been combined into a single block in the first amplifier.

CHECK YOUR UNDERSTANDING

Derive the differential equation corresponding to the circuit shown in the figure.



$$\text{Answer: } d^2x/dt^2 + 2x = -10f(t)$$

8.5 PHYSICAL LIMITATIONS OF OPERATIONAL AMPLIFIERS

In nearly all the discussion and examples so far, the op-amp has been treated as an ideal device, characterized by infinite input impedance, zero output resistance, and infinite open-loop voltage gain. Although this model is adequate to represent its behavior in a large number of applications, practical op-amps are not ideal but exhibit limitations that should be considered in the design of instrumentation. In particular, in dealing with relatively large voltages and currents, and in the presence of high-frequency signals, it is important to be aware of the non-ideal properties of the op-amp.

Voltage Supply Limits

As indicated in Figure 8.6, operational amplifiers (and all amplifiers, in general) are powered by external DC voltage supplies V_S^+ and V_S^- , which are usually symmetric and on the order of ± 10 to ± 20 V. Some op-amps are especially designed to operate from a single voltage supply, but for the sake of simplicity from here on we shall consider only symmetric supplies. The effect of limiting supply voltages is that amplifiers are capable of amplifying signals *only within the range of their supply voltages*; it would be physically impossible for an amplifier to generate a voltage greater than V_S^+ or less than V_S^- . This limitation may be stated as follows:

$$V_S^- < v_o < V_S^+ \quad \text{Voltage supply limitation} \quad (8.97)$$

For most op-amps, the limit is actually approximately 1.5 V less than the supply voltages. How does this practically affect the performance of an amplifier circuit? An example will best illustrate the idea.

Note how the voltage supply limit actually causes the peaks of the sine wave to be clipped in an abrupt fashion. This type of hard nonlinearity changes the characteristics of the signal quite radically and could lead to significant errors if not taken into account. Just to give an intuitive idea of how such clipping can affect a signal, have you ever wondered why rock guitar has a characteristic sound that is very different from the sound of classical or jazz guitar? The reason is that the “rock sound” is obtained by overamplifying the signal, attempting to exceed the voltage supply limits, and causing clipping similar in quality to the distortion introduced by voltage supply limits in an op-amp. This clipping broadens the spectral content of each tone and causes the sound to be distorted.

One of the circuits most directly affected by supply voltage limitations is the op-amp integrator.

Frequency Response Limits

Another property of all amplifiers that may pose severe limitations to the op-amp is their finite bandwidth. We have so far assumed, in our ideal op-amp model, that the open-loop gain is a very large constant. In reality, A is a function of frequency and is characterized by a low-pass response. For a typical op-amp,

$$A(j\omega) = \frac{A_0}{1 + j\omega/\omega_0} \quad \text{Finite bandwidth limitation} \quad (8.98)$$

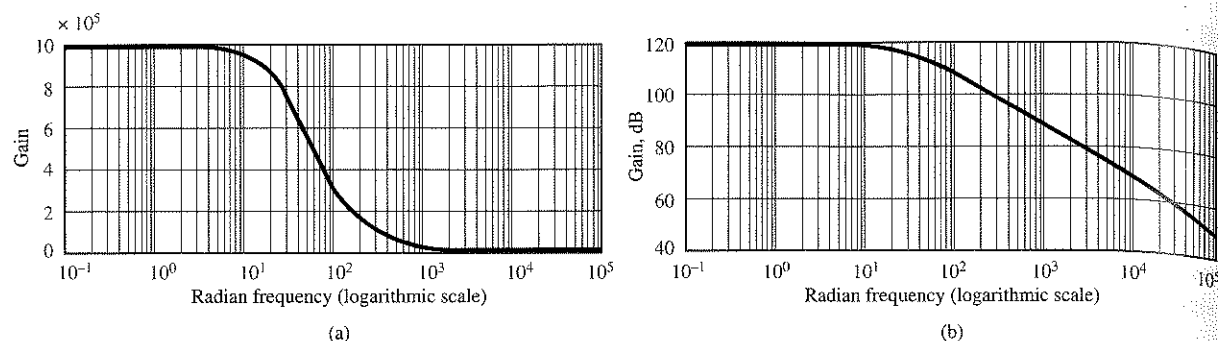


Figure 8.52 Open-loop gain of practical op-amp (a) amplitude ratio response; (b) dB response

The cutoff frequency of the op-amp open-loop gain ω_0 represents approximately the point where the amplifier response starts to drop off as a function of frequency and is analogous to the cutoff frequencies of the RC and RL circuits of Chapter 6. Figure 8.52 depicts $A(j\omega)$ in both linear and decibel plots for the fairly typical values $A_0 = 10^6$ and $\omega_0 = 10\pi$. It should be apparent from Figure 8.52 that the assumption of a very large open-loop gain becomes less and less accurate for increasing frequency. Recall the initial derivation of the closed-loop gain for the inverting amplifier: In obtaining the final result $V_o/V_S = -R_F/R_S$, it was assumed that $A \rightarrow \infty$. This assumption is clearly inadequate at the higher frequencies.

The finite bandwidth of the practical op-amp results in a fixed **gain-bandwidth product** for any given amplifier. The effect of a constant gain-bandwidth product is that as the closed-loop gain of the amplifier is increased, its 3-dB bandwidth is proportionally reduced until, in the limit, if the amplifier were used in the open-loop mode, its gain would be equal to A_0 and its 3-dB bandwidth would be equal to ω_0 . The constant gain-bandwidth product is therefore equal to the product of the open-loop gain and the open-loop bandwidth of the amplifier: $A_0\omega_0 = K$. When the amplifier is connected in a closed-loop configuration (e.g., as an inverting amplifier), its gain is typically much less than the open-loop gain and the 3-dB bandwidth of the amplifier is proportionally increased. To explain this further, Figure 8.53 depicts the case in which two different linear amplifiers (achieved through any two different negative feedback configurations) have been designed for the same op-amp. The first has closed-loop gain $G_1 = A_1$, and the second has closed-loop gain $G_2 = A_2$. The bold line in the figure indicates the open-loop frequency response, with gain A_0 and cutoff frequency ω_0 . As the gain decreases from A_0 to A_1 , the cutoff frequency increases from ω_0 to ω_1 . As the gain decreases to A_2 , the bandwidth increases to ω_2 . Thus:

The gain-bandwidth product of any given op-amp is constant.

$$A_0 \times \omega_0 = A_1 \times \omega_1 = A_2 \times \omega_2 = K$$

(8.99)

Input Offset Voltage

Another limitation of practical op-amps results because even in the absence of any external inputs, it is possible that an **offset voltage** will be present at the input of an op-amp. This voltage is usually denoted by $\pm V_{os}$, and it is caused by mismatches in

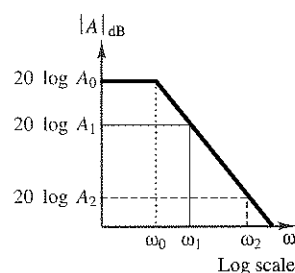


Figure 8.53

the internal circuitry of the op-amp. The offset voltage appears as a differential input voltage between the inverting and non-inverting input terminals. The presence of an additional input voltage will cause a DC bias error in the amplifier output. Typical and maximum values of V_{os} are quoted in manufacturers' data sheets. The worst-case effects due to the presence of offset voltages can therefore be predicted for any given application.

Input Bias Currents

Another non-ideal characteristic of op-amps results from the presence of small input bias currents at the inverting and non-inverting terminals. Once again, these are due to the internal construction of the input stage of an operational amplifier. Figure 8.54 illustrates the presence of nonzero input bias currents I_B going into an op-amp.

Typical values of I_{B+} and I_{B-} depend on the semiconductor technology employed in the construction of the op-amp. Op-amps with bipolar transistor input stages may see input bias currents as large as $1 \mu\text{A}$, while for FET input devices, the input bias currents are less than 1 nA . These currents depend on the internal design of the op-amp and are not necessarily equal.

One often designates the **input offset current** I_{os} as

$$I_{os} = I_{B+} - I_{B-}$$

(8.100)

The latter parameter is sometimes more convenient from the standpoint of analysis.

Output Offset Adjustment

Both the offset voltage and the input offset current contribute to an output offset voltage $V_{o,os}$. Some op-amps provide a means for minimizing $V_{o,os}$. For example, the $\mu\text{A}741$ op-amp provides a connection for this procedure. Figure 8.55 shows a typical pin configuration for an op-amp in an eight-pin dual-in-line package (DIP) and the circuit used for nulling the output offset voltage. The variable resistor is adjusted until v_{out} reaches a minimum (ideally, 0 V). Nulling the output voltage in this manner removes the effect of both input offset voltage and current on the output.

Slew Rate Limit

Another important restriction in the performance of a practical op-amp is associated with rapid changes in voltage. The op-amp can produce only a finite rate of change at its output. This limit rate is called the **slew rate**. Consider an ideal step input, where at $t = 0$ the input voltage is switched from 0 to V volts. Then we would expect the output to switch from 0 to AV volts, where A is the amplifier gain. However, v_o can change at only a finite rate; thus,

$$\left| \frac{dv_o}{dt} \right|_{\max} = S_0 \quad \text{Slew rate limitation}$$

(8.101)

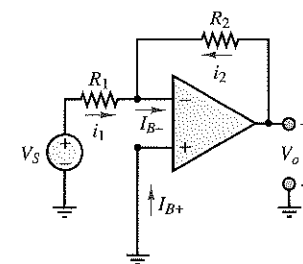


Figure 8.54

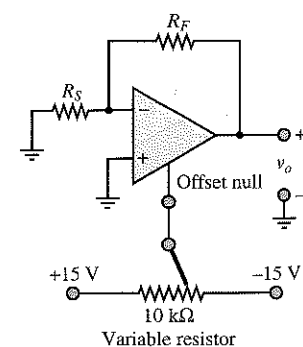
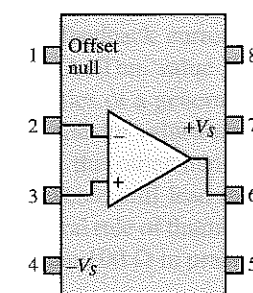


Figure 8.55 Output offset voltage adjustment

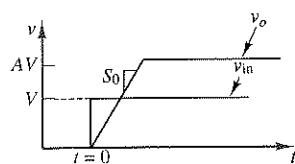


Figure 8.56 Slew rate limit in op-amps

Figure 8.56 shows the response of an op-amp to an ideal step change in input voltage. Here, S_0 , the slope of v_o , represents the slew rate.

The slew rate limitation can affect sinusoidal signals, as well as signals that display abrupt changes, as does the step voltage of Figure 8.56. This may not be obvious until we examine the sinusoidal response more closely. It should be apparent that the maximum rate of change for a sinusoid occurs at the zero crossing, as shown by Figure 8.57. To evaluate the slope of the waveform at the zero crossing, let

$$v_{in}(t) = V \sin \omega t \quad \text{such that} \quad v_o(t) = AV \sin \omega t \quad (8.102)$$

Then:

$$\frac{dv_o}{dt} = \omega AV \cos \omega t \quad (8.103)$$

The maximum slope of the sinusoidal signal will therefore occur at $\omega t = 0, \pi, 2\pi, \dots$, so that

$$\left| \frac{dv_o}{dt} \right|_{\max} = \omega AV = S_0 \quad (8.104)$$

Thus, the maximum slope of a sinusoid is proportional to both the signal frequency and the amplitude. The curve shown by a dashed line in Figure 8.57 should indicate that as ω increases, so does the slope of $v(t)$ at the zero crossings. What is the direct consequence of this result, then?

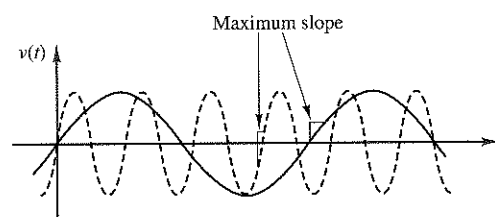


Figure 8.57 The maximum slope of a sinusoidal signal varies with the signal frequency.

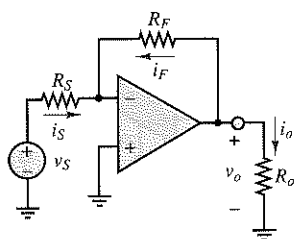


Figure 8.58

Short-Circuit Output Current

Recall the model for the op-amp shown in Figure 8.3, which depicted the internal circuit of the op-amp as an equivalent input impedance R_{in} and a controlled voltage source Av_{in} . In practice, the internal source is not ideal because it cannot provide an infinite amount of current (to the load, to the feedback connection, or to both). The immediate consequence of this non-ideal op-amp characteristic is that the maximum output current of the amplifier is limited by the so-called short-circuit output current I_{SC} :

$$|i_{out}| < I_{SC} \quad \text{Short-circuit output current limitation} \quad (8.105)$$

To further explain this point, consider that the op-amp needs to provide current to the feedback path (in order to “zero” the voltage differential at the input) and to whatever load resistance, R_o , may be connected to the output. Figure 8.58 illustrates this idea

for the case of an inverting amplifier, where I_{SC} is the load current that would be provided to a short-circuit load ($R_o = 0$).

Common-Mode Rejection Ratio (CMRR)

The concepts of common-mode and differential-mode voltages as well as the common-mode rejection ratio (CMRR) were introduced in section 8.2 and expressed mathematically by equations 8.66 to 8.70. The CMRR is an amplifier characteristic that can be found in the data sheet for any particular amplifier, such as a 741 operational amplifier.

$$\text{CMRR} = 20 \log \left| \frac{A_{DM}}{A_{CM}} \right| \quad (\text{in dB})$$



Practical Op-Amp Considerations

The results presented in the preceding pages suggest that operational amplifiers permit the design of rather sophisticated circuits in a few simple steps, by selecting appropriate resistor values. This is certainly true, provided that the circuit component selection satisfies certain criteria. A few important practical criteria for selecting op-amp circuit component values are summarized here.

1. Use standard resistor values. While any arbitrary value of gain can, in principle, be achieved by selecting the appropriate combination of resistors, the designer is often constrained to the use of standard 5 percent resistor values. For example, if a design requires a gain of 25, it might be tempting to select, say, 100- and 4-k Ω resistors to achieve $R_F/R_S = 25$ for the inverting amplifier shown in Figure 8.58. However, 4 k Ω is not a standard value; the closest 5 percent tolerance resistor value is 3.9 k Ω , leading to a gain of 25.64. Can you find a combination of standard 5 percent resistors whose ratio is closer to 25?
2. Ensure that the load current is reasonable. Assume the maximum output voltage in the step 1 example is 10 V. The feedback current required by your design with $R_F = 100$ k Ω and $R_S = 4$ k Ω would be $I_F = 10/100,000 = 0.1$ mA. This is a very reasonable value for an op-amp. If you tried to achieve the same gain
3. Avoid stray capacitance by avoiding excessively large resistances, which can cause unwanted signals to couple into the circuit through a mechanism known as *capacitive coupling*. This phenomenon is discussed in Chapter 15. Large resistances can also cause other problems. As a general rule of thumb, avoid resistor values higher than 1 M Ω in practical designs.
4. Precision designs may be warranted. If a certain design requires that the amplifier gain be set to a very accurate value, it may be appropriate to use the (more expensive) option of precision resistors: for example, 1 percent tolerance resistors are commonly available, at a premium cost. Some of the examples and homework problems explore the variability in gain due to the use of higher- and lower-tolerance resistors.



EXAMPLE 8.9 Voltage Supply Limits in an Inverting Amplifier

Problem

Compute and sketch the output voltage of the inverting amplifier of Figure 8.59.

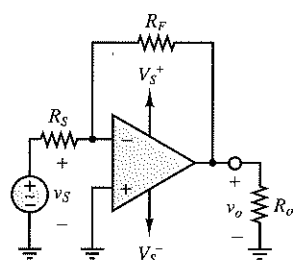


Figure 8.59

Solution

Known Quantities: Resistor and supply voltage values; input voltage.

Find: $v_o(t)$.

Schematics, Diagrams, Circuits, and Given Data: $R_o = 1 \text{ k}\Omega$; $R_F = 10 \text{ k}\Omega$; $R_s = 1 \text{ k}\Omega$; $V_S^+ = 15 \text{ V}$; $V_S^- = -15 \text{ V}$; $v_s(t) = 2 \sin(1,000t)$.

Assumptions: Assume a supply voltage-limited op-amp.

Analysis: For an ideal op-amp the output would be

$$v_o(t) = -\frac{R_F}{R_S} v_s(t) = -10 \times 2 \sin(1,000t) = -20 \sin(1,000t)$$

However, the supply voltage is limited to $\pm 15 \text{ V}$, and the op-amp output voltage will therefore saturate before reaching the theoretical peak output value of $\pm 20 \text{ V}$. Figure 8.60 depicts the output voltage waveform.

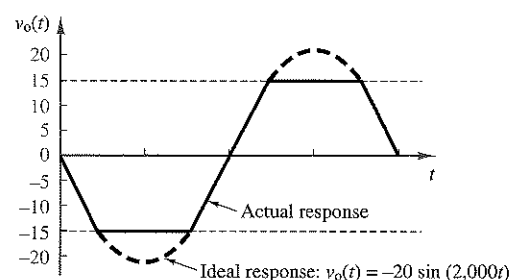


Figure 8.60 Op-amp output with voltage supply limit

Comments: In a practical op-amp, saturation would be reached at 1.5 V below the supply voltages, or at approximately $\pm 13.5 \text{ V}$.

EXAMPLE 8.10 Voltage Supply Limits in an Op-Amp Integrator

Problem

Compute and sketch the output voltage of the integrator of Figure 8.44.

Solution

Known Quantities: Resistor, capacitor, and supply voltage values; input voltage.

Find: $v_o(t)$.

Schematics, Diagrams, Circuits, and Given Data: $R_S = 10 \text{ k}\Omega$; $C_F = 20 \mu\text{F}$; $V_S^+ = 15 \text{ V}$; $V_S^- = -15 \text{ V}$; $v_s(t) = 0.5 + 0.3 \cos(10t)$.

Assumptions: Assume a supply voltage-limited op-amp. The initial condition is $v_{out}(0) = 0$.

Analysis: For an ideal op-amp integrator the output would be

$$\begin{aligned} v_{out}(t) &= -\frac{1}{R_S C_F} \int_{-\infty}^t v_s(t') dt' = -\frac{1}{0.2} \int_{-\infty}^t [0.5 + 0.3 \cos(10t')] dt' \\ &= -2.5t + 1.5 \sin(10t) \end{aligned}$$

However, the supply voltage is limited to $\pm 15 \text{ V}$, and the integrator output voltage will therefore saturate at the lower supply voltage value of -15 V as the term $2.5t$ increases with time. Figure 8.61 depicts the output voltage waveform.

Comments: Note that the DC offset in the waveform causes the integrator output voltage to increase linearly with time. The presence of even a very small DC offset will always cause integrator saturation. One solution to this problem is to include a large feedback resistor in parallel with the capacitor.

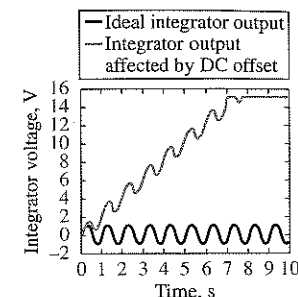


Figure 8.61 Effect of DC offset on integrator

EXAMPLE 8.11 Gain-Bandwidth Product Limit in an Op-Amp

Problem

Determine the maximum allowable closed-loop voltage gain of an op-amp if the amplifier is required to have an audio-range bandwidth of 20 kHz.

Solution

Known Quantities: Gain-bandwidth product.

Find: $G_{\max}$.

Schematics, Diagrams, Circuits, and Given Data: $A_0 = 10^6$; $\omega_0 = 2\pi \times 5 \text{ rad/s}$.

Assumptions: Assume a gain-bandwidth product limited op-amp.

Analysis: The gain-bandwidth product of the op-amp is

$$A_0 \times \omega_0 = K = 10^6 \times 2\pi \times 5 = \pi \times 10^7 \text{ rad/s}$$

The desired bandwidth is $\omega_{\max} = 2\pi \times 20,000 \text{ rad/s}$, and the maximum allowable gain will therefore be

$$G_{\max} = \frac{K}{\omega_{\max}} = \frac{\pi \times 10^7}{\pi \times 4 \times 10^4} = 250 \frac{\text{V}}{\text{V}}$$

For any closed-loop voltage gain greater than 250, the amplifier would have reduced bandwidth.

Comments: If we desired to achieve gains greater than 250 and maintain the same bandwidth, two options would be available: (1) Use a different op-amp with greater gain-bandwidth product, or (2) connect two amplifiers in cascade, each with lower gain and greater bandwidth, such that the product of the gains would be greater than 250.

To further explore the first option, you may wish to look at the device data sheets for different op-amps and verify that op-amps can be designed (at a cost!) to have substantially greater gain-bandwidth product than the amplifier used in this example.



EXAMPLE 8.12 Increasing the Gain-Bandwidth Product by Means of Amplifiers in Cascade

Problem

Determine the overall 3-dB bandwidth of the cascade amplifier of Figure 8.62.

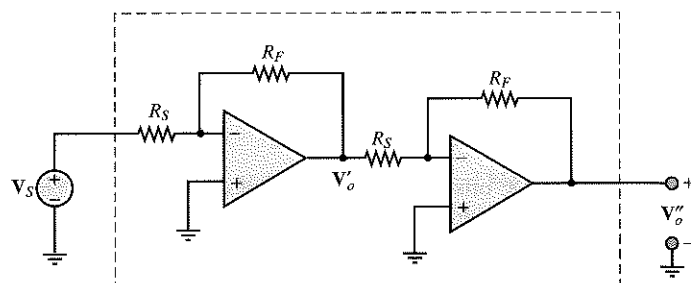


Figure 8.62 Cascade amplifier

Solution

Known Quantities: Gain-bandwidth product and gain of each amplifier.

Find: ω_{3dB} of cascade amplifier.

Schematics, Diagrams, Circuits, and Given Data: $A_0 \Omega_0 = K = 4\pi \times 10^6$ for each amplifier. $R_F/R_S = 100$ for each amplifier.

Assumptions: Assume gain-bandwidth product limited (otherwise ideal) op-amps.

Analysis: Let G_1 and ω_1 denote the gain and the 3-dB bandwidth of the first amplifier, respectively, and G_2 and ω_2 those of the second amplifier.

The 3-dB bandwidth of the first amplifier is

$$\omega_1 = \frac{K}{G_1} = \frac{4\pi \times 10^6}{10^2} = 4\pi \times 10^4 \frac{\text{rad}}{\text{s}}$$

The second amplifier will also have

$$\omega_2 = \frac{K}{G_2} = \frac{4\pi \times 10^6}{10^2} = 4\pi \times 10^4 \frac{\text{rad}}{\text{s}}$$

Thus, the approximate bandwidth of the cascade amplifier is $4\pi \times 10^4$, and the gain of the cascade amplifier is $G_1 G_2 = 100 \times 100 = 10^4$ or 80 dB.

Had we attempted to achieve the same gain with a single-stage amplifier having the same K , we would have achieved a bandwidth of only

$$\omega_3 = \frac{K}{G_3} = \frac{4\pi \times 10^6}{10^4} = 4\pi \times 10^2 \frac{\text{rad}}{\text{s}}$$

Comments: In practice, the actual 3-dB bandwidth of the cascade amplifier is not quite as large as that of each of the two stages because the gain of each amplifier starts decreasing at frequencies somewhat lower than the nominal cutoff frequency.

EXAMPLE 8.13 Effect of Input Offset Voltage on an Amplifier

Problem

Determine the effect of the input offset voltage V_{os} on the output of the amplifier shown in Figure 8.63.

Solution

Known Quantities: Nominal closed-loop voltage gain; input offset voltage.

Find: The offset voltage component in the output voltage V_o , os .

Schematics, Diagrams, Circuits, and Given Data: $A_{nom} = 100$; $V_{os} = 1.5$ mV.

Assumptions: Assume an input offset voltage-limited (otherwise ideal) op-amp.

Analysis: The amplifier is connected in a non-inverting configuration; thus its gain is

$$G_{nom} = 100 = 1 + \frac{R_F}{R_S}$$

The DC offset voltage, represented by an ideal voltage source, is represented as being directly applied to the non-inverting input; thus

$$V_{o, os} = G_{nom} V_{os} = 100 V_{os} = 150 \text{ mV}$$

Thus, we should expect the output of the amplifier to be shifted upward by 150 mV.

Comments: The input offset voltage is not, of course, an external source, but it represents a voltage offset between the inputs of the op-amp. Figure 8.55 depicts how such an offset can be zeroed. The worst-case offset voltage is usually listed in the device data sheets. Typical values are 2 mV for the 741c general-purpose op-amp and 5 mV for the FET-input TLO81.

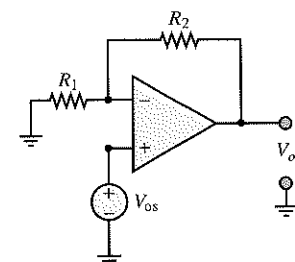


Figure 8.63 Op-amp input offset voltage

EXAMPLE 8.14 Effect of Input Offset Current on an Amplifier

Problem

Determine the effect of the input offset current I_{os} on the output of the amplifier of Figure 8.64.

Solution

Known Quantities: Resistor values; input offset current.

Find: The offset voltage component in the output voltage $v_{out, os}$.

Schematics, Diagrams, Circuits, and Given Data: $I_{os} = 1 \mu\text{A}$; $R_2 = 10 \text{ k}\Omega$.

Assumptions: Assume an input offset current-limited (otherwise ideal) op-amp.

Analysis: We calculate the inverting and non-inverting terminal voltages caused by the offset current in the absence of an external input:

$$v^+ = R_3 I_{B^+} \quad v^- = v^+ = R_3 I_{B^+}$$

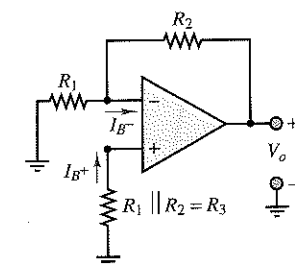


Figure 8.64

With these values we can apply KCL at the inverting node and write

$$\begin{aligned}\frac{V_o - v^-}{R_2} - \frac{v^+}{R_1} &= I_{B^-} \\ \frac{V_o}{R_2} - \frac{-R_3 I_{B^+}}{R_2} - \frac{-R_3 I_{B^+}}{R_1} &= I_{B^-} \\ V_o &= R_2 \left[-I_{B^+} R_3 \left(\frac{1}{R_2} + \frac{1}{R_1} \right) + I_{B^-} \right] = -R_2 I_{os}\end{aligned}$$

Thus, we should expect the output of the amplifier to be shifted downward by $R_2 I_{os}$, or $10^4 \times 10^{-6} = 10 \text{ mV}$ for the data given in this example.

Comments: Usually, the worst-case input offset currents (or input bias currents) are listed in the device data sheets. Values can range from 100 pA (for CMOS op-amps, for example, LMC6061) to around 200 nA for a low-cost general-purpose amplifier (for example, $\mu\text{A}741\text{c}$).



EXAMPLE 8.15 Effect of Slew Rate Limit on an Amplifier

Problem

Determine the effect of the slew rate limit S_0 on the output of an inverting amplifier for a sinusoidal input voltage of known amplitude and frequency.

Solution

Known Quantities: Slew rate limit S_0 ; amplitude and frequency of sinusoidal input voltage; amplifier closed-loop gain.

Find: Sketch the theoretically correct output and the actual output of the amplifier in the same graph.

Schematics, Diagrams, Circuits, and Given Data: $S_0 = 1 \text{ V}/\mu\text{s}$; $v_S = \sin(2\pi \times 10^5 t)$; $G = 10$.

Assumptions: Assume the op-amp is slew rate-limited, but otherwise ideal.

Analysis: Given the closed-loop voltage gain of 10, compute the theoretical output voltage to be:

$$v_o = -10 \sin(2\pi \times 10^5 t)$$

The maximum slope of the output voltage is then computed as follows:

$$\left| \frac{dv_o}{dt} \right|_{\max} = G\omega = 10 \times 2\pi \times 10^5 = 6.28 \frac{\text{V}}{\mu\text{s}}$$

Clearly, the value calculated above far exceeds the slew rate limit. Figure 8.65 depicts the approximate appearance of the waveforms that one would measure in an experiment.

Comments: Note that in this example the slew rate limit has been exceeded severely, and the output waveform is visibly distorted, to the point that it has effectively become a triangular wave. The effect of the slew rate limit is not always necessarily so dramatic and visible; thus one needs to pay attention to the specifications of a given op-amp. The slew rate limit is listed in the device data sheets. Typical values can range from $13 \text{ V}/\mu\text{s}$, for the TLO81, to around $0.5 \text{ V}/\mu\text{s}$ for a low-cost general-purpose amplifier (for example, $\mu\text{A}741\text{c}$).

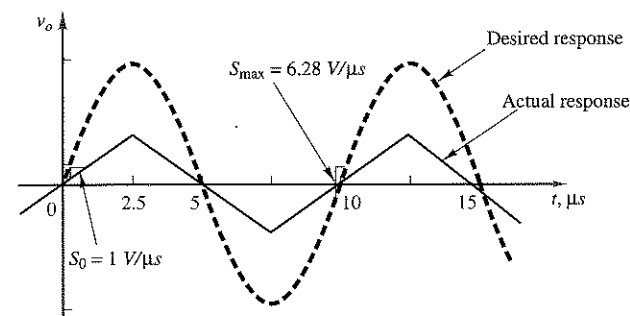


Figure 8.65 Distortion introduced by slew rate limit

EXAMPLE 8.16 Effect of Short-Circuit Current Limit on an Amplifier



Problem

Determine the effect of the short-circuit limit I_{SC} on the output of an inverting amplifier for a sinusoidal input voltage of known amplitude.

Solution

Known Quantities: Short-circuit current limit I_{SC} ; amplitude of sinusoidal input voltage; amplifier closed-loop gain.

Find: Compute the minimum allowable load resistance value $R_{o\min}$, and sketch the theoretical and actual output voltage waveforms for resistances smaller than $R_{o\min}$.

Schematics, Diagrams, Circuits, and Given Data: $I_{SC} = 50 \text{ mA}$; $v_S = 0.05 \sin(\omega t)$; $G = 100$.

Assumptions: Assume the op-amp is short-circuit current-limited, but otherwise ideal.

Analysis: Given the closed-loop voltage gain of 100, compute the theoretical output voltage to be:

$$v_o(t) = -Gv_S(t) = -5 \sin(\omega t)$$

To assess the effect of the short-circuit current limit, calculate the peak value of the output voltage since this is the condition that will require the maximum output current from the op-amp:

$$\begin{aligned}V_{o\text{peak}} &= 5 \text{ V} \\ I_{SC} &= 50 \text{ mA} \\ R_{o\min} &= \frac{V_{o\text{peak}}}{I_{SC}} = \frac{5 \text{ V}}{50 \text{ mA}} = 100 \Omega\end{aligned}$$

For any load resistance less than 100Ω , the required load current will be greater than I_{SC} . For example, if we chose a $75\text{-}\Omega$ load resistor, we would find that

$$V_{o\text{peak}} = I_{SC} \times R_o = 3.75 \text{ V}$$

That is, the output voltage cannot reach the theoretically correct 5-V peak and would be "compressed" to reach a peak voltage of only 3.75 V. This effect is depicted in Figure 8.66.

Comments: The short-circuit current limit is listed in the device data sheets. Typical values for a low-cost general-purpose amplifier (say, the 741c) are in the tens of milliamperes.

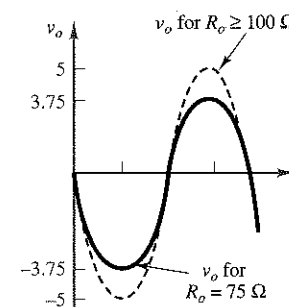


Figure 8.66 Distortion introduced by short-circuit current limit

CHECK YOUR UNDERSTANDING

How long will it take (approximately) for the integrator of Example 8.10 to saturate if the input signal has a 0.1-VDC bias [that is, $v_s(t) = 0.1 + 0.3 \cos(10t)$]?

Answer: Approximately 30 s

CHECK YOUR UNDERSTANDING

What is the maximum gain that could be achieved by the op-amp of Example 8.11 if the desired bandwidth is 100 kHz?

Answer: $A_{\max} = 50$

CHECK YOUR UNDERSTANDING

In Example 8.12, the closed-loop gain of each amplifier was assumed constant at frequencies below the cutoff frequency. In practice, this is only approximately true, since the open-loop gain A of each op-amp begins to decrease with frequency at frequencies usually much lower than the closed-loop gain cutoff frequency. The frequency response for the open-loop gain of an op-amp is well approximated by:

$$A(j\omega) = \frac{A_0}{1 + j\omega/\omega_0}$$

Use this expression to find an expression for the closed-loop gain of the cascade amplifier. (Hint: The combined gain is equal to the product of the individual closed-loop gains.) What is the actual gain in decibels at the cutoff frequency ω_0 for the cascade amplifier?

What is the 3-dB bandwidth of the cascade amplifier of Example 8.12? [Hint: The gain of the cascade amplifier is the product of the individual op-amp frequency responses. Compute the magnitude of this product, set it equal to $(1/\sqrt{2}) \times 10,000$, and solve for ω .]

Answer: 74 dB; $\omega_{3dB} = 2\pi \times 12,800$ rad/s

CHECK YOUR UNDERSTANDING

What is the maximum gain that can be accepted in the op-amp circuit of Example 8.13 if the offset is not to exceed 50 mV?

Answer: $A_{V \max} = 33.3$

CHECK YOUR UNDERSTANDING

Given the desired peak output amplitude (10 V), what is the maximum frequency that will not result in violating the slew rate limit for the op-amp of Example 8.15.

Answer: $f_{\max} = 159$ kHz

Conclusion

Operational amplifiers constitute the single most important building block in analog electronics. The contents of this chapter will be frequently referenced in later sections of this book. Upon completing this chapter, the following learning objectives should have been mastered:

1. Understand the properties of ideal amplifiers and the concepts of gain, input impedance, output impedance, and feedback. Ideal amplifiers represent fundamental building blocks of electronic instrumentation. With the concept of an ideal amplifier clearly established, one can design practical amplifiers, filters, integrators, and many other signal processing circuits. A practical op-amp closely approximates the characteristics of ideal amplifiers.
2. Understand the difference between open-loop and closed-loop op-amp configuration; and compute the gain (or complete the design of) simple inverting, non-inverting, summing, and differential amplifiers using ideal op-amp analysis. Analyze more advanced op-amp circuits, using ideal op-amp analysis, and identify important performance parameters in op-amp data sheets. Analysis of op-amp circuits is made easy by a few simplifying assumptions, which are based on the op-amp having a very large input impedance, a very small output impedance, and a large open-loop gain. The simple inverting and non-inverting amplifier configurations permit the design of very useful circuits simply by appropriately selecting and placing a few resistors.
3. Analyze and design simple active filters. Analyze and design ideal integrator and differentiator circuits. The use of capacitors in op-amp circuits extends the applications of this useful element to include filtering, integration, and differentiation.
4. Understand the structure and behavior of analog computers, and design analog computer circuits to solve simple differential equations. The properties of op-amp summing amplifiers and integrators make it possible to construct analog computers that can serve as an aid in the solution of differential equations and in the simulation of dynamic systems. While digital computer-based numerical simulations have become very popular in the last two decades, there is still a role for analog computers in some specialized applications.
5. Understand the principal physical limitations of an op-amp. It is important to understand that there are limitations in the performance of op-amp circuits that are not predicted by the simple op-amp models presented in the early sections of the chapter. In practical designs, issues related to voltage supply limits, bandwidth limits, offsets, slew rate limits, and output current limits are very important if one is to achieve the design performance of an op-amp circuit.

HOMEWORK PROBLEMS

Section 8.1: Ideal Amplifiers

8.1 The circuit shown in Figure P8.1 has a DC signal source, two stages of amplification, and a load. Determine, in decibels, the power gain $G = P_o/P_s = V_o I_o / V_s I_s$, where:

$$\begin{aligned} R_s &= 0.5 \text{ k}\Omega & R_{o3} &= 0.7 \text{ k}\Omega \\ R_{i1} &= 3.2 \text{ k}\Omega & R_{i2} &= 2.8 \text{ k}\Omega \\ R_{o1} &= 2.2 \text{ k}\Omega & R_{o2} &= 2.2 \text{ k}\Omega \\ A_1 &= 90 & H_2 &= 300 \text{ mS} \end{aligned}$$

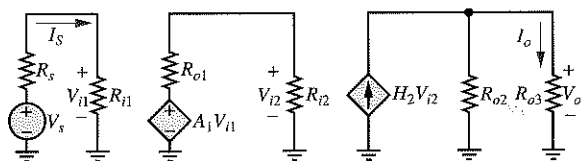


Figure P8.1

8.2 A temperature sensor in a production line under normal operating conditions produces a no-load (i.e., sensor current = 0) voltage:

$$\begin{aligned} v_s &= V_{pk} \cos(\omega t) & R_s &= 400 \Omega \\ V_{pk} &= 500 \text{ mV} & \omega &= 6.28 \text{ krad/s} \end{aligned}$$

The temperature is monitored on a display (the load) with a vertical line of light-emitting diodes. Normal conditions are indicated when a string of the bottommost diodes 2 cm in length is on. This requires that a voltage be supplied to the display input terminals where

$$v_o = V_m \cos(\omega t) \quad V_m = 6 \text{ V}$$

The signal from the sensor must be amplified. Therefore, a voltage amplifier, shown in Figure P8.2, is connected between the sensor and CRT with

$$R_i = 2 \text{ k}\Omega \quad R_o = 3 \text{ k}\Omega$$

Determine the required no-load gain of the amplifier.

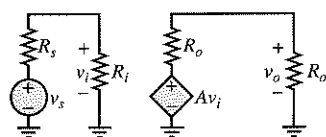


Figure P8.2

8.3 What approximations are valid for the voltages and currents shown in Figure P8.3 of an ideal operational amplifier. What conditions must be satisfied for these approximations?

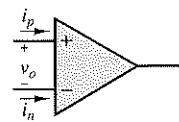


Figure P8.3

8.4 What approximations are usually made about the circuit components and parameters shown in Figure P8.4 for an ideal op-amp?

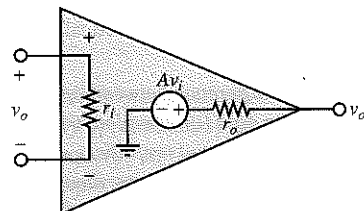


Figure P8.4

Section 8.2: The Operational Amplifier

8.5 Find v_1 in the circuits of Figure P8.5(a) and (b). In Figure P8.5(a) the 3-k Ω resistor "loads" the output; that is, v_1 is changed by attaching the 3-k Ω resistor in parallel with the lower 6-k Ω resistor. However, in Figure P8.5(b) the isolation buffer holds v_1 to $v_g/2$, regardless of the presence of the 3-k Ω resistor and its value!

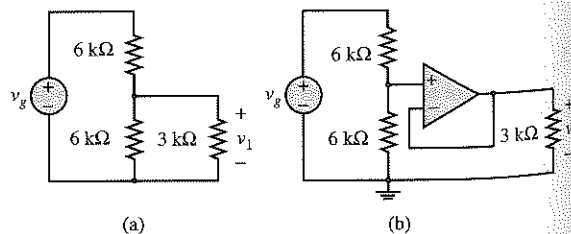


Figure P8.5

8.6 Find the current i in the circuit of Figure P8.6.

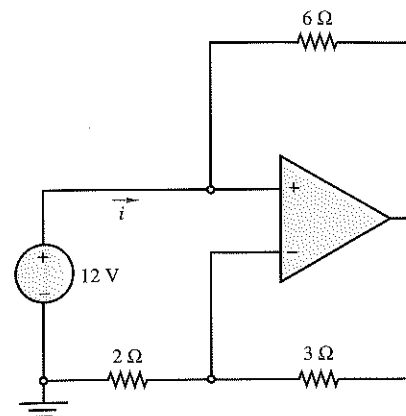


Figure P8.6

8.7 Find the voltage v_o in Figure P8.7 by (i) applying nodal analysis, and by (ii) finding the Thévenin equivalent network seen to the left of nodes a and b to form an archetypical inverting amplifier.

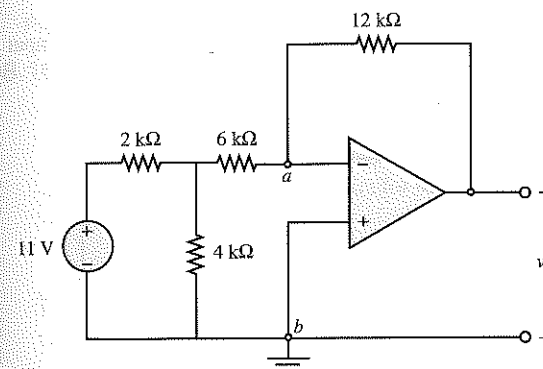


Figure P8.7

8.8 Find the Thévenin equivalent network seen between the non-inverting terminal node and the reference node in Figure P8.8.

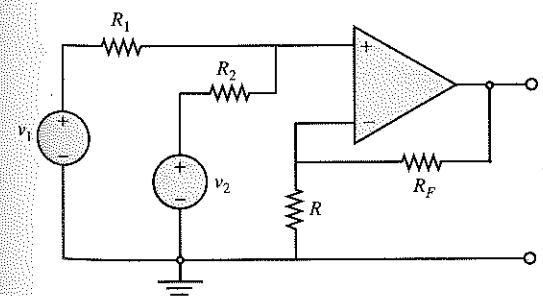


Figure P8.8

8.9 Determine an expression for the closed-loop voltage gain $G = v_o/v_i$ for the circuit of Figure P8.9. Find the input conductance i_1/v_i seen by the voltage source. Assume the op-amp is ideal.

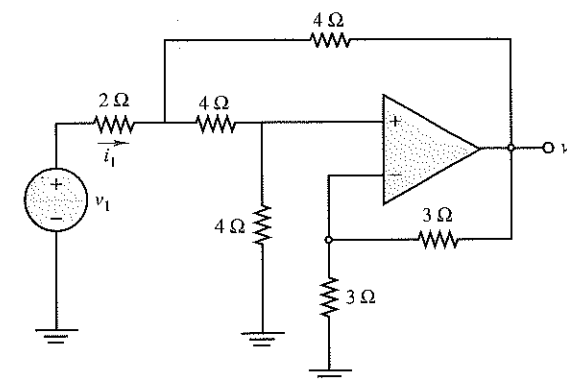


Figure P8.9

8.10 Differential amplifiers are often used in conjunction with a Wheatstone bridge, such as that shown in Figure P8.10, where each resistor is a temperature sensing element, and their change in resistance ΔR is directly proportional to their change in temperature ΔT . The constant of proportionality is the temperature coefficient $\pm\alpha$, which can be positive (PTC) or negative (NTC). Find the Thévenin equivalent network seen by the amplifier to the left of nodes a and b . Assume that $\Delta R = \pm\Delta T$ and $|\Delta R| \ll R_0$.

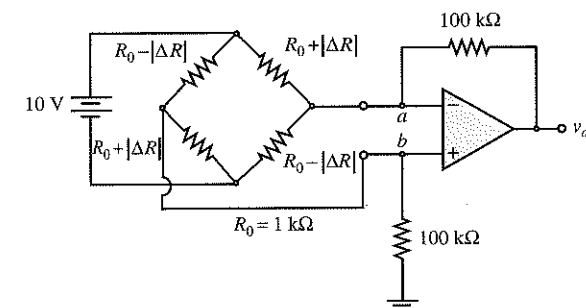


Figure P8.10

8.11 The circuit shown in Figure P8.11 is a negative impedance converter. Find the input impedance Z_{in} :

$$Z_{in} = \frac{V_1}{I_1}$$

when:

$$a. Z_o = R$$

$$b. Z_o = \frac{1}{j\omega C}$$

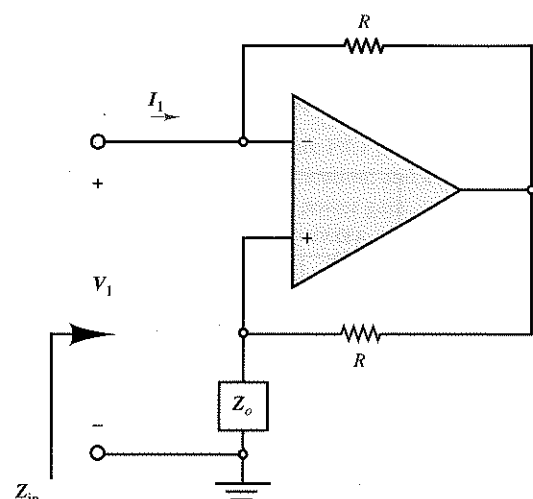


Figure P8.11

- 8.12** The circuit of Figure P8.12 demonstrates that op-amp feedback can create a resonant circuit without the use of an inductor. Assume $R_1 = R_2 = 1\ \Omega$, $C_1 = 2Q\ \text{F}$, and $C_2 = 1/2Q\ \text{F}$, where Q is the quality factor introduced in Chapter 6. Use nodal analysis to determine the voltage gain v_o/v_{in} .

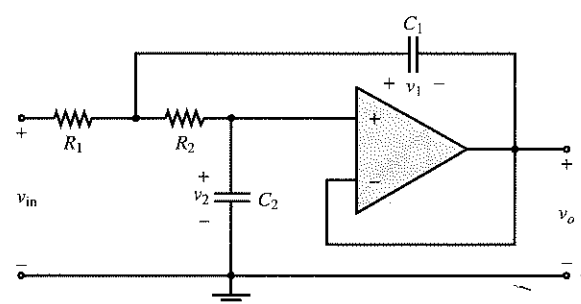


Figure P8.12

- 8.13** Inductors are difficult to use as components of integrated circuits due to the need for large coils of wire, which require significant space and tend to act as excellent antennas for ambient noise. As an alternative, a "solid-state inductor" can be constructed as shown in Figure P8.13.

- Determine the input impedance $Z_{in} = V_1/I_1$.
- What is Z_{in} when $R = 1,000\ \Omega$ and $C = 0.02\ \mu\text{F}$?

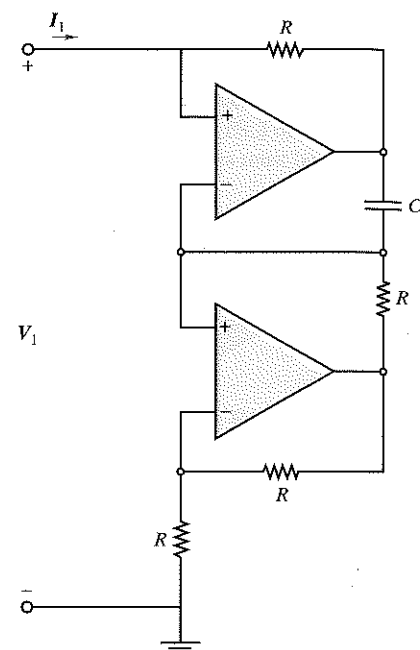


Figure P8.13

- 8.14** In the circuit of Figure P8.14, determine the input impedance $Z_{in} = V_1/I_1$.

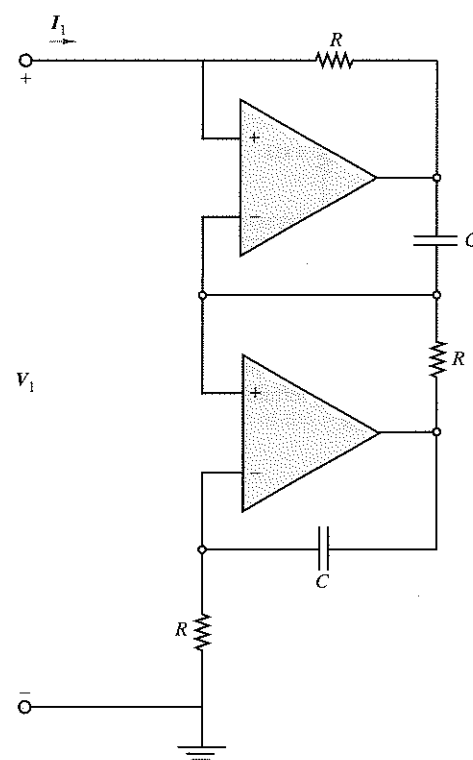


Figure P8.14

- 8.15** It is easy to construct a current source using an inverting amplifier configuration. Verify that the current I through R_o is independent of the value of R_o , assuming that the op-amp stays in its linear operating region, and find the value of I .

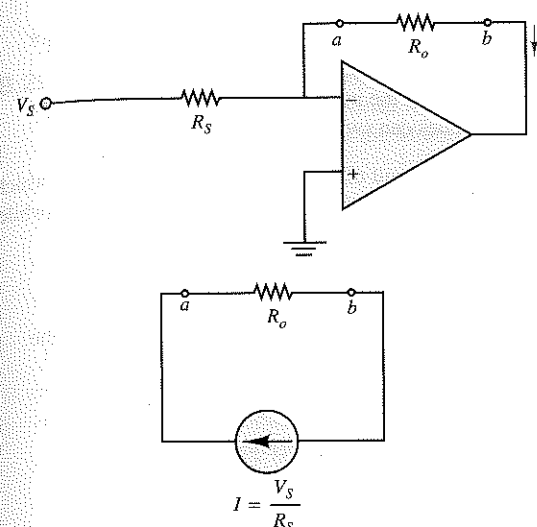


Figure P8.15

- 8.16** A "super diode" or "precision diode" circuit, which eliminates the diode offset voltage, is shown in Figure P8.16. The diode permits current directed from anode to cathode only, as indicated in the figure. Determine the output voltage $v_o(t)$ for the given input voltage $v_{in}(t)$.

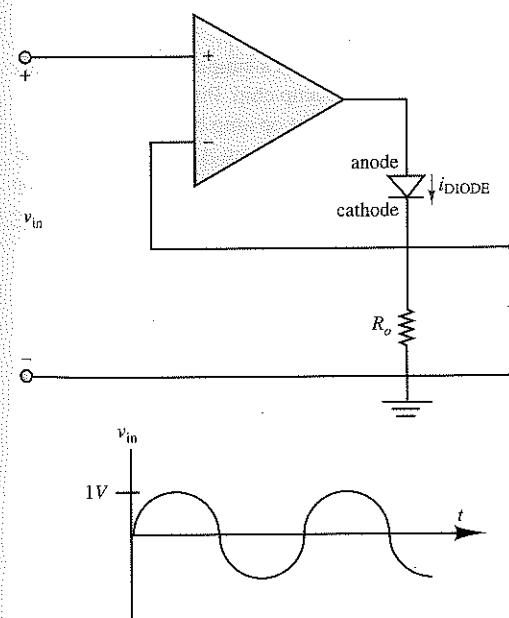


Figure P8.16

- 8.17** Determine the response function $\tilde{V}_2/\tilde{V}_1$ for the circuit of Figure P8.17.

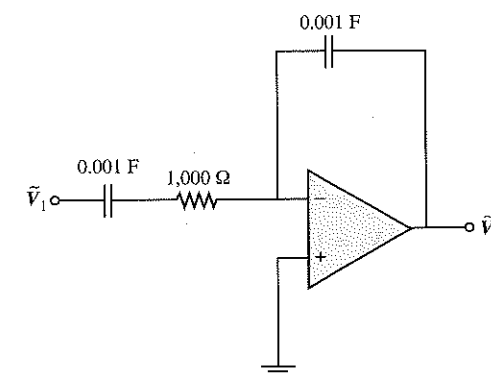


Figure P8.17

- 8.18** Time delays are often encountered in engineering systems. They can be approximated using Euler's definition as

$$e^{-sT} = \lim_{N \rightarrow \infty} \left[\frac{1}{sT/N + 1} \right]^N$$

If $T = 1$ and $N = 1$, then the approximation can be implemented by the circuit of Problem 8.17 (see Figure P8.17), with the addition of a unity gain inverting amplifier to eliminate the negative sign. Modify the circuit of Figure P8.17 as needed, and use it as many times as necessary to design an approximate time delay for $T = 1$ and $N = 4$ in Euler's definition of the exponential.

- 8.19** Show that the output voltage v_3 shown in Figure P8.8 has the form $v_3 = a_1 v_1 + a_2 v_2$, where a_1 and a_2 are constants.

- 8.20** For the circuit of Figure P8.20, find v_o .

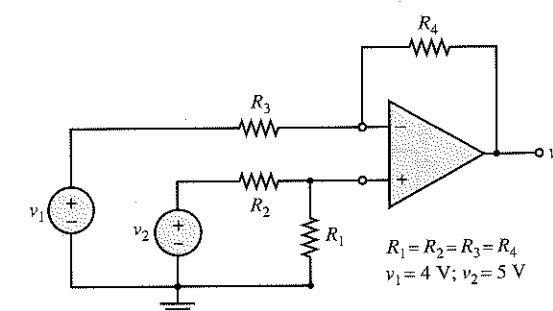


Figure P8.20

8.21 Differential amplifiers are often used in conjunction with a Wheatstone bridge, such as that shown in Figure P8.10, where each resistor is a temperature sensing element, and their change in resistance ΔR is directly proportional to their change in temperature ΔT . The constant of proportionality is the temperature coefficient $\pm\alpha$, which can be positive (PTC) or negative (NTC). Assume $|\Delta R| = K\Delta T$, where $K = \text{constant}$. Find an expression for $v_o(\Delta T)$.

8.22 Consider the circuit of Figure P8.22. Assume $\omega = 1000 \text{ Rad/s}$:

- If $V_1 - V_2 = 1 \angle 0^\circ \text{ V}$, use phasor analysis to find $|V_o|$.
- Use phasor analysis to find $\angle V_o$.

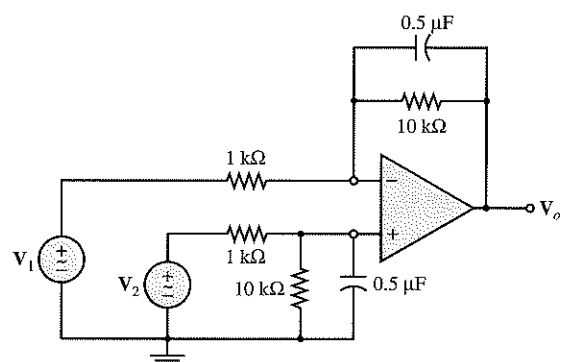


Figure P8.22

8.23 Find an expression for the voltage gain $\tilde{V}_{out}/\tilde{V}_{in}$ of the circuit of Figure P8.12. Assume $R_1 = 3 \Omega$, $R_2 = 2 \Omega$, and $C_1 = C_2 = \frac{1}{6} \text{ F}$.

8.24 In the circuit of Figure P8.24, assume $R_F = 12 \text{ k}\Omega$ and that it is critical that the voltage gain v_o/v_s remain within ± 2 percent of the nominal gain of 20. What value of R_S is needed for the nominal gain? What are the allowed maximum and minimum values of R_S ? Will a standard 5 percent tolerance resistor be adequate to satisfy this requirement? (See Table 2.3 of standard resistor values in Chapter 2.)

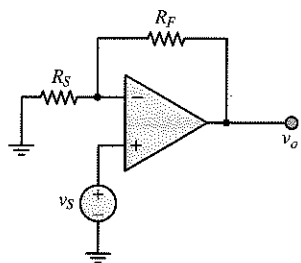


Figure P8.24

8.25 The two 5 percent tolerance resistors of an inverting amplifier (see Figure 8.8) have nominal values $R_F = 33 \text{ k}\Omega$ and $R_S = 1.5 \text{ k}\Omega$.

- What is the nominal voltage gain $G = v_o/v_s$ of the amplifier?
- What is the maximum value of G if the resistor values can swing ± 5 percent?
- What is the minimum value of G if the resistor values can swing ± 5 percent?

8.26 The circuit of Figure P8.26 is a *level shifter*, which adjusts the DC portion of the input voltage $v_1(t)$ while also amplifying the AC portion. Let: $v_1(t) = 10 + 10^{-3} \sin \omega t \text{ V}$, $R_F = 10 \text{ k}\Omega$, and $V_{batt} = 20 \text{ V}$.

- Find R_S such that no DC voltage appears at the output.
- What is $v_o(t)$, using R_S from part a?

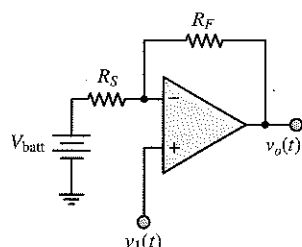


Figure P8.26

8.27 Figure P8.27 shows a simple practical amplifier that uses a 741 op-amp chip. Pin numbers are as indicated. Assume the op amp has a 2-M Ω input resistance, an open-loop gain $A = 200,000$, and an output impedance $R_o = 50 \Omega$. Find the closed-loop gain $G = v_o/v_i$.

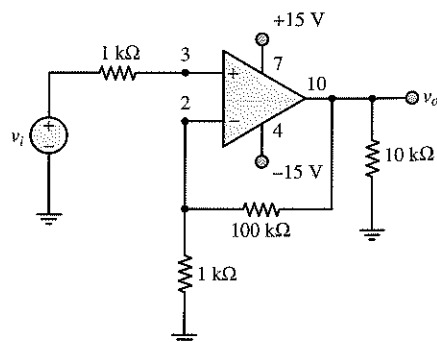


Figure P8.27

8.28 Design an inverting summing amplifier to obtain the following weighted sum of four different signal sources:

$$v_o = -(2 \sin \omega_1 t + 4 \sin \omega_2 t + 8 \sin \omega_3 t + 16 \sin \omega_4 t)$$

Assume that $R_F = 5 \text{ k}\Omega$, and determine the required source resistors.

8.29 The amplifier shown in Figure P8.29 has a signal source (v_s in series with R_s) and load R_o separated by an amplification stage built upon the Motorola MC1741C op-amp. Assume:

$$R_s = 2.2 \text{ k}\Omega \quad R_1 = 1 \text{ k}\Omega \\ R_F = 8.7 \text{ k}\Omega \quad R_o = 20 \Omega$$

The op-amp itself has a 2-M Ω input resistance, a 75- Ω output resistance, and a 200K open-loop gain. To a first approximation, the op-amp would be modeled as ideal. A better model would include the effects of the parameters listed above.

- Assume the op-amp is not ideal, and derive an expression for the input resistance $r_i = v_i/i_i$ of the overall amplifier, where $v_i = v_s - i_i R_s$.
- Determine the value of that input resistance, and compare it to the input resistance derived for an ideal op-amp.

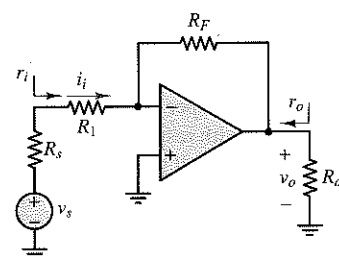


Figure P8.29

8.30 In the circuit shown in Figure P8.30, assume $R_1 = 40 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $R_F = 150 \text{ k}\Omega$, and $v_s = 0.01 + 0.005 \cos(\omega t) \text{ V}$. Determine an expression for the output voltage v_o and its value. Assume an ideal op-amp.

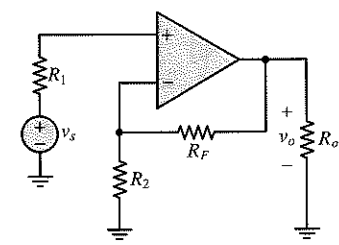


Figure P8.30

8.31 For the circuit shown in Figure P8.31, assume $v_s = 0.3 + 0.2 \cos(\omega t)$, $R_s = 4 \Omega$, and $R_o = 15 \Omega$. Determine the output voltage v_o for an ideal op-amp and also for a Motorola MC1741C op-amp with characteristics as given in Problem P8.29.

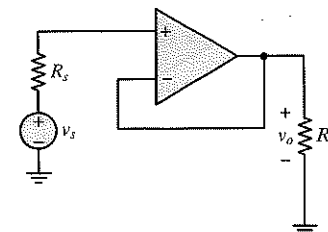


Figure P8.31

8.32 For the circuit shown in Figure P8.32, assume:

$$v_{S1} = 2.9 \times 10^{-3} \cos(\omega t) \text{ V} \\ v_{S2} = 3.1 \times 10^{-3} \cos(\omega t) \text{ V} \\ R_1 = 1 \text{ k}\Omega \quad R_2 = 3.3 \text{ k}\Omega \\ R_3 = 10 \text{ k}\Omega \quad R_4 = 18 \text{ k}\Omega$$

Determine an expression and value for the output voltage v_o .

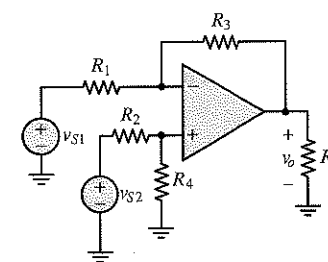


Figure P8.32

8.33 For the circuit shown in Figure P8.33, assume $v_{S1} = -2 \text{ V}$, $v_{S2} = 2 \sin(2\pi \cdot 2,000t) \text{ V}$, $R_1 = 100 \text{ k}\Omega$, $R_2 = 50 \text{ k}\Omega$, and $R_F = 150 \text{ k}\Omega$. Determine the output voltage v_o .

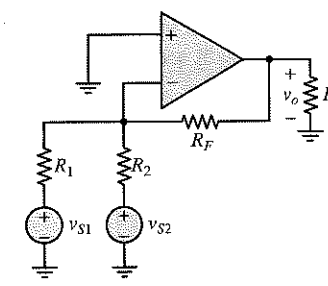


Figure P8.33

8.34 For the circuit shown in Figure P8.33, assume: $v_{S1} = v_{S2} = 5 \text{ mV}$, $R_1 = 50 \Omega$, $R_2 = 2 \text{ k}\Omega$, and $R_F = 2 \text{ k}\Omega$. The non-ideal MC1741C op-amp has a $2\text{-m}\Omega$ input resistance, a $75\text{-}\Omega$ output resistance, and an open-loop gain of 200K . Determine:

- An expression for the output voltage v_o .
- The voltage gain for each of the two input signals.

8.35 In the circuit shown in Figure P8.35, assume ideal op-amps to determine the output voltage V_o . All resistances are equal and $V_{in} = 4\angle 0^\circ \text{ V}$.

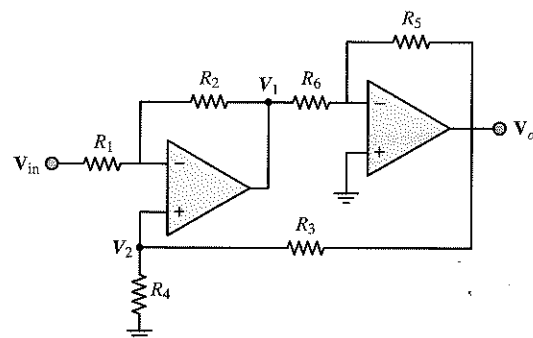


Figure P8.35

8.36 In the circuit shown in Figure P8.36, assume $V_2 = 8\angle 0^\circ \text{ V}$ and find the input voltage V_{in} such that $V_o = 0$. Assume ideal op-amps.

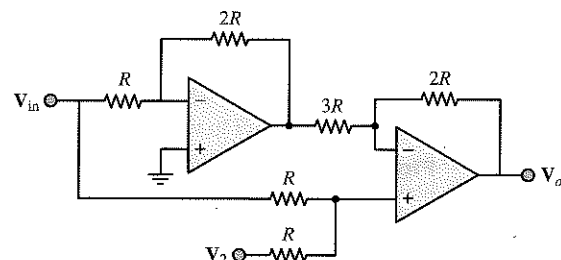


Figure P8.36

8.37 In the circuit shown in Figure P8.32, assume:

$$\begin{aligned} v_{S1} &= 1.3 \text{ V} & v_{S2} &= 1.9 \text{ V} \\ R_1 &= R_2 = 4.7 \text{ k}\Omega \\ R_3 &= R_4 = 10 \text{ k}\Omega & R_o &= 1.8 \text{ k}\Omega \end{aligned}$$

Determine:

- The output voltage v_o .
- The common-mode component of v_o .
- The differential-mode component of v_o .

8.38 In the circuit shown in Figure P8.38, determine the output voltage V_o . Let $R_1 = 10 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, $R_3 = 15 \text{ k}\Omega$, $R_4 = 10 \text{ k}\Omega$, $R_F = 50 \text{ k}\Omega$, and $V_{in} = 6 \text{ V}$.

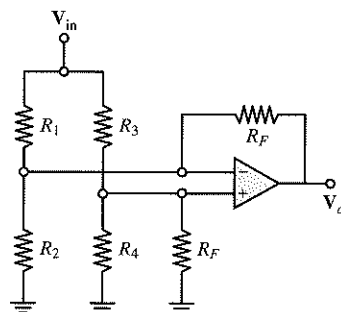


Figure P8.38

8.39 A linear potentiometer R_P is used to sense and generate a voltage v_y proportional to the y -coordinate of an xy inkjet printer head. A reference signal v_R is supplied by the software controlling the printer. The difference between these voltages is amplified to drive a motor. The motor changes the position of the printer head until that difference equals zero. For proper operation, the motor voltage must be 10 times the difference between the signal and reference voltage. For rotation in the proper direction, the motor voltage must be negative with respect to v_y . In addition, i_P must be negligibly small to avoid loading the pot and causing an erroneous signal voltage.

- Design an op-amp circuit that satisfies these specifications. Redraw Figure P8.39, replacing the dotted line box with your amplifier circuit. Be sure to indicate component values.
- Mark the pin numbers on your redrawn figure for an 8-pin single $\mu\text{A}741\text{C}$ op-amp chip.

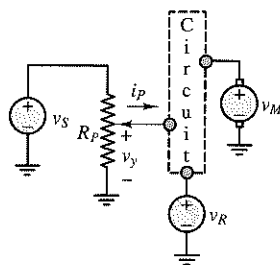


Figure P8.39

8.40 Compute the current I_{batt} delivered by the battery in Figure P8.40. Assume: $R_{S1} = 30 \text{ k}\Omega$, $R_{S2} = 30 \text{ k}\Omega$, $R_{F1} = 100 \text{ k}\Omega$, $R_{F2} = 60 \text{ k}\Omega$, $R_1 = 5 \text{ k}\Omega$, $R_2 = 7 \text{ k}\Omega$, and $V_{\text{batt}} = 3 \text{ V}$.

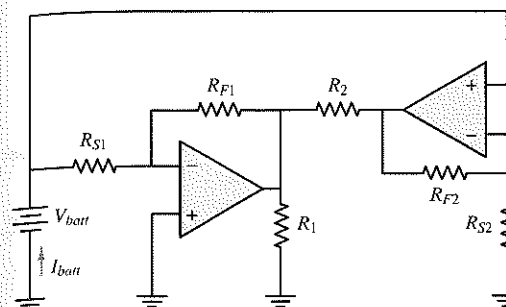


Figure P8.40

8.41 Figure P8.41 shows a simple voltage-to-current converter. Show that the current I_o through the light-emitting diode (LED), and therefore its brightness, is proportional to the source voltage V_s as long as $V_s > 0$. The LED permits current in the direction shown only.

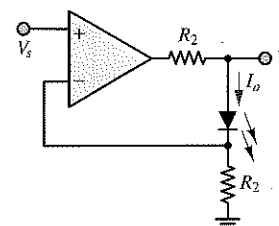


Figure P8.41

8.42 Figure P8.42 shows a simple current-to-voltage converter. Show that the voltage V_o is proportional to the current generated by the cadmium sulfide (CdS) solar cell. Also show that the transimpedance of the circuit V_o/I_s is $-R$!

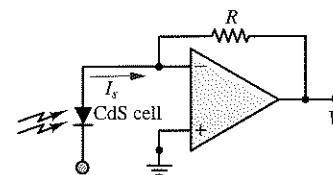


Figure P8.42

8.43 An op-amp voltmeter circuit as in Figure P8.43 is required to measure a maximum input of $V_s = 15 \text{ mV}$. The op-amp input current is $I_B = 0.25 \mu\text{A}$. The ammeter is designed for full-scale deflection when

$I_m = 80 \mu\text{A}$ and $r_m = 8 \text{ k}\Omega$. Determine suitable values for R_3 and R_4 so that the full-scale deflection of the ammeter corresponds to $V_s = 15 \text{ mV}$. What is the significance of I_B for a non-ideal op-amp?

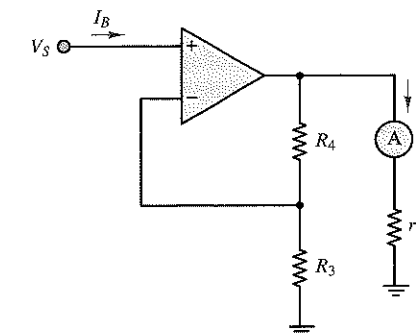


Figure P8.43

8.44 Find an expression for the voltage gain v_o/v_s in Figure P8.44. What is the gain when $R_{S1} = R_{S2} = 2.5 \text{ k}\Omega$ and $R_{F1} = R_{F2} = 9.0 \text{ k}\Omega$?

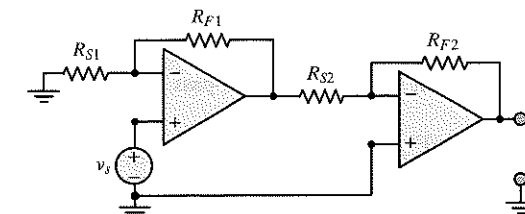


Figure P8.44

8.45 Select appropriate components using standard 5 percent resistors to obtain a voltage gain $v_o/v_s \approx 80$ for the circuit of Figure P8.44. How closely can you approximate the desired gain? Compute the expected error.

8.46 Repeat Problem 8.45 but compute the maximum and minimum possible voltage gains if the resistor values are allowed to swing ± 5 percent.

8.47 The circuit shown in Figure P8.47 can function as a precision ammeter. Assume that the voltmeter has a range of 0 to 10 V and an internal resistance of $20 \text{ k}\Omega$. The full-scale reading of the ammeter is intended to be 1 mA. Find the resistance R such that the voltmeter reading is 10 V when $i_{in} = 1 \text{ mA}$.

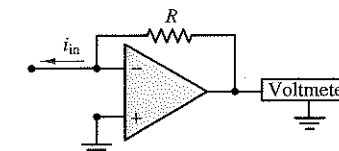


Figure P8.47

8.48 Select appropriate components using standard 5 percent resistors to obtain a voltage gain $v_o/v_s = 20$ for the circuit of Figure P8.30. How closely can you approximate the desired gain? Compute the expected error.

8.49 Repeat Problem 8.48 but compute the maximum and minimum possible voltage gains if the resistor values are allowed to swing ± 5 percent.

8.50 Select appropriate components using standard 1 percent resistors to obtain a differential gain of approximately 15 in the circuit of Figure P8.32. Assume that $R_3 = R_4$ and $R_1 = R_2$. How closely can you approximate the desired gain? Compute the expected error.

8.51 Repeat Problem 8.50 but compute the maximum and minimum possible voltage gains if the resistor values are allowed to swing ± 1 percent. Also compute the maximum common-mode output for the same allowed ± 1 percent swing. Pick the nominal resistor values so that $R_3 = R_4$ and $R_1 = R_2$.

Section 8.3: Active Filters

8.52 The circuit shown in Figure P8.52 with input V_s and output V_o is an active high-pass filter. Assume:

$$C = 1 \mu\text{F} \quad R = 10 \text{ k}\Omega \quad R_o = 1 \text{ k}\Omega$$

Determine:

- The voltage gain $|V_o/V_s|$ (in dB) in the passband.
- The cutoff frequency.

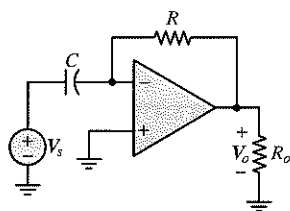


Figure P8.52

8.53 The op-amp circuit shown in Figure P8.53 is used as a high-pass filter. Assume:

$$C = 0.2 \mu\text{F} \quad R_o = 222 \Omega$$

$$R_1 = 1.5 \text{ k}\Omega \quad R_2 = 5.5 \text{ k}\Omega$$

Determine:

- The voltage gain $|V_o/V_s|$ (in dB), in the passband.
- The cutoff frequency.

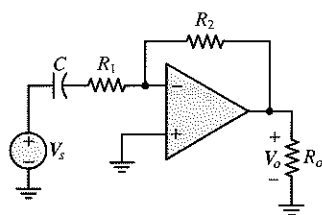


Figure P8.53

8.54 The op-amp circuit shown in Figure P8.53 is used as a high-pass filter. Assume:

$$C = 200 \text{ pF} \quad R_o = 1 \text{ k}\Omega$$

$$R_1 = 10 \text{ k}\Omega \quad R_2 = 220 \text{ k}\Omega$$

Determine:

- The voltage gain $|V_o/V_s|$ (in dB), in the passband.
- The cutoff frequency.

8.55 The circuit shown in Figure P8.55 is an active filter. Assume:

$$C = 120 \text{ pF} \quad R_o = 180 \text{ k}\Omega$$

$$R_1 = 3 \text{ k}\Omega \quad R_2 = 50 \text{ k}\Omega$$

Determine the break frequencies and $|V_o/V_i|$ at very low and at very high frequencies.

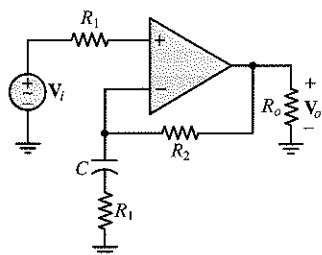


Figure P8.55

8.56 The circuit shown in Figure P8.56 is an active filter. Assume:

$$C = 15 \text{ nF} \quad R_o = 4 \text{ k}\Omega$$

$$R_1 = 1.2 \text{ k}\Omega \quad R_2 = 5.6 \text{ k}\Omega$$

$$R_3 = 62 \text{ k}\Omega$$

Determine:

- An expression for the voltage gain in standard form:
- $$H_v(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$
- The break frequencies.
 - The passband gain.
 - The Bode magnitude and phase plots of V_o/V_i .

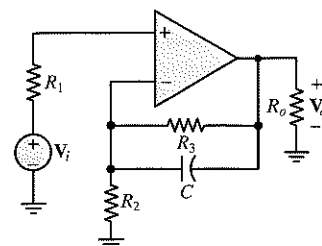


Figure P8.56

8.57 The op-amp circuit shown in Figure P8.57 is used as a low-pass filter. Assume:

$$C = 0.8 \mu\text{F} \quad R_o = 1 \text{ k}\Omega$$

$$R_1 = 5 \text{ k}\Omega \quad R_2 = 15 \text{ k}\Omega$$

Determine:

- An expression in standard form for the voltage gain V_o/V_s .
- The gain, in dB, in the passband and at the cutoff frequency.

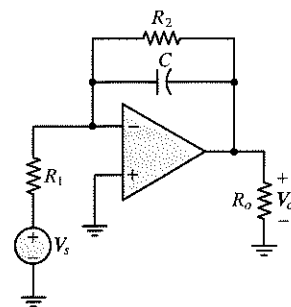


Figure P8.57

8.58 The op-amp circuit shown in Figure P8.57 is a low-pass filter. Assume:

$$R_1 = 2.2 \text{ k}\Omega \quad R_2 = 68 \text{ k}\Omega$$

$$C = 0.47 \text{ nF} \quad R_o = 1 \text{ k}\Omega$$

Determine:

- An expression in standard form for the voltage gain V_o/V_s .
- The gain, in dB, in the passband and at the cutoff frequency.

8.59 The circuit shown in Figure P8.59 is a bandpass filter. Assume:

$$R_1 = R_2 = 10 \text{ k}\Omega \quad R_o = 4.7 \text{ k}\Omega$$

$$C_1 = C_2 = 0.1 \mu\text{F}$$

Determine:

- The voltage gain $|V_o/V_i|$ in the passband.
- The resonant frequency.
- The break frequencies.
- The quality factor Q .
- The Bode magnitude and phase plots of V_o/V_i .

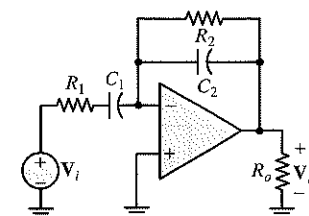


Figure P8.59

8.60 The op-amp circuit shown in Figure P8.57 is a low-pass filter. Assume:

$$R_1 = 12 \text{ k}\Omega \quad R_2 = 4.7 \text{ k}\Omega \quad R_o = 3.3 \text{ k}\Omega$$

$$C = 0.7 \text{ nF}$$

Determine:

- An expression in standard form for the voltage gain V_o/V_s .
- The gain, in dB, in the passband and at the cutoff frequency.
- Would such small resistance values cause a practical op-amp to behave in a significantly non-ideal manner?

8.61 The circuit shown in Figure P8.59 is a bandpass filter. Assume:

$$R_1 = 2.2 \text{ k}\Omega \quad R_2 = 100 \text{ k}\Omega$$

$$C_1 = 2.2 \mu\text{F} \quad C_2 = 1 \text{ nF}$$

Determine the passband gain.

8.62 Derive the frequency response function V_o/V_{in} for the circuit shown in Figure P8.62.

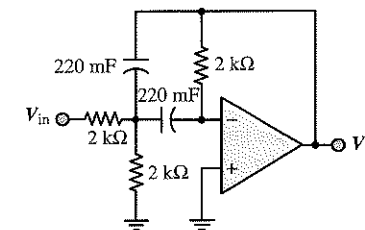


Figure P8.62

8.63 The circuit shown in Figure P8.63 can be used as a low-pass filter.

- Derive the frequency response V_o/V_{in} of the circuit.
- If $R_1 = R_2 = 100 \text{ k}\Omega$ and $C = 0.1 \text{ }\mu\text{F}$, compute the attenuation, in dB, of V_o/V_{in} at $\omega = 1,000 \text{ rad/s}$.
- Compute the amplitude and phase of V_o/V_{in} at $\omega = 2,500 \text{ rad/s}$.
- Find the range of frequencies over which the attenuation of V_o/V_{in} is less than 1 dB.

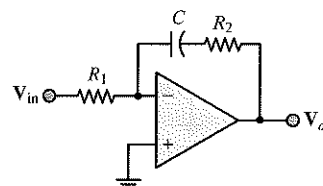


Figure P8.63

8.64 Determine a symbolic expression in standard form for the voltage gain V_o/V_{in} in Figure P8.64. What kind of a filter does the voltage gain represent?

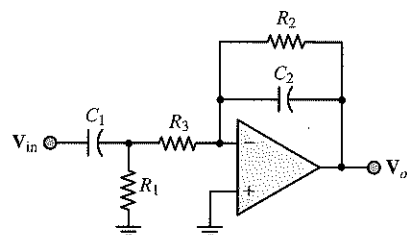


Figure P8.64

8.65 For the circuit of Figure P8.65, sketch the amplitude response of V_2/V_1 , indicating the half-power frequencies. Assume the op-amp is ideal.

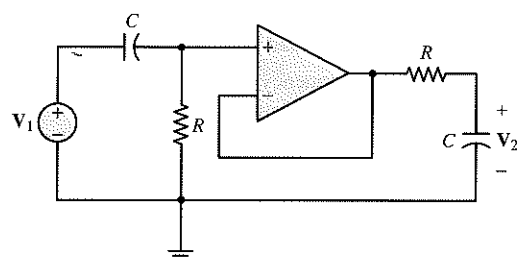


Figure P8.65

8.66 Determine a symbolic expression for the voltage gain V_o/V_{S1} of Figure P8.66. What kind of a filter does the gain represent?

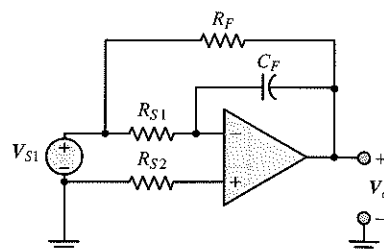


Figure P8.66

8.67 Determine a symbolic expression for the voltage gain V_o/V_S of Figure P8.67. What kind of a filter does the gain represent?

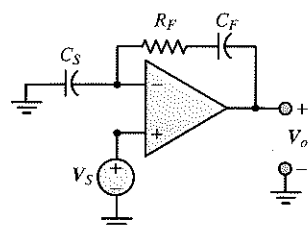


Figure P8.67

Section 8.4: Integrators and Differentiators

8.68 The circuit shown in Figure P8.68(a) produces an output voltage v_o which is either the integral or the derivative of the source voltage v_s shown in Figure P8.68(b) multiplied by some gain. Assume:

$$C = 1.5 \text{ }\mu\text{F} \quad R = 5 \text{ k}\Omega \quad R_o = 1.5 \text{ k}\Omega$$

For the given source voltage, determine the output voltage as a function of time and plot it.

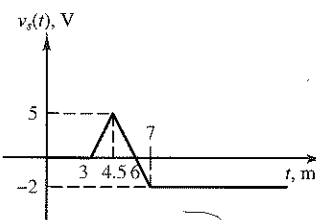
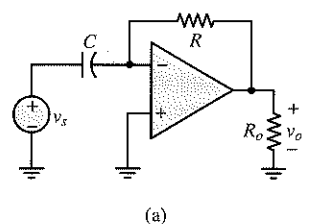


Figure P8.68

8.69 The circuit shown in Figure P8.69(a) produces an output voltage v_o which is either the integral or the derivative of the source voltage v_s shown in Figure P8.69(b) multiplied by some gain. Assume:

$$C = 0.5 \text{ }\mu\text{F} \quad R = 8 \text{ k}\Omega \quad R_o = 2 \text{ k}\Omega$$

For the given source voltage, determine the output voltage as a function of time and plot it.

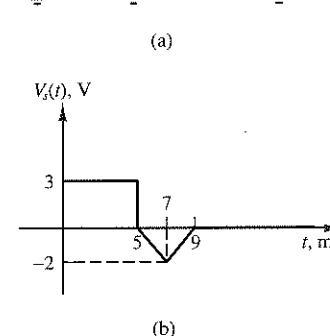
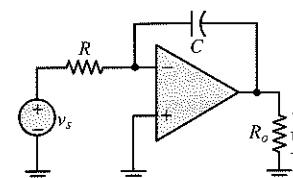


Figure P8.69

8.70 The circuit shown in Figure P8.70 is an integrator. The capacitor is initially uncharged, and the source voltage is

$$v_{in}(t) = 10^{-2} V + \sin(2,000\pi t) V$$

- At $t = 0$, the switch S_1 is closed. How long does it take before clipping occurs at the output if $R_s = 10 \text{ k}\Omega$ and $C_F = 0.008 \text{ }\mu\text{F}$?
- At what times does the integration of the DC input cause the op-amp to saturate fully?

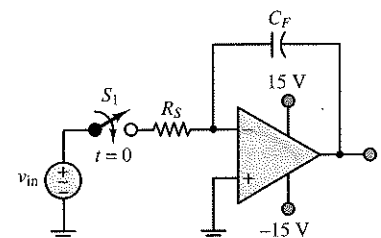


Figure P8.70

8.71 A practical integrator is shown in Figure 8.35. Note that the resistor in parallel with the feedback capacitor provides a path for the capacitor to discharge DC voltage. Usually, the time constant $R_F C_F$ is chosen to be large enough not to interfere with the integration.

- If $R_S = 10 \text{ k}\Omega$, $R_F = 2 \text{ M}\Omega$, $C_F = 0.008 \text{ }\mu\text{F}$, and $v_S(t) = 10 \text{ V} + \sin(2,000\pi t) \text{ V}$, find $v_o(t)$, using phasor analysis.
- Repeat part a if $R_F = 200 \text{ k}\Omega$ and if $R_F = 20 \text{ k}\Omega$.
- Compare the time constants $R_F C_F$ with the period of the waveform for parts a and b. What can you say about the time constant and the ability of the circuit to integrate?

8.72 The circuit of Figure 8.40 is a practical differentiator. Assume an ideal op-amp, and $v_S(t) = 10^4 \sin(2,000\pi t) \text{ mV}$, $C_S = 100 \text{ }\mu\text{F}$, $C_F = 0.008 \text{ }\mu\text{F}$, $R_F = 2 \text{ M}\Omega$, and $R_S = 10 \text{ k}\Omega$.

- Determine the voltage gain V_o/V_S .
- Sum the DC and AC components of $v_o(t)$ to find the total output voltage.

8.73 Derive the differential equation in $x(t)$ for the circuit of Figure P8.73.

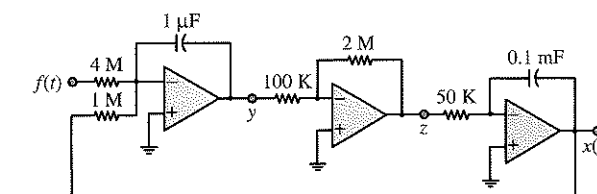


Figure P8.73

8.74 Construct a circuit corresponding to the following differential equation:

$$\frac{d^2 x}{dt^2} + 100 \frac{dx}{dt} + 10x = -5f(t)$$

Section 8.5: Physical Limitations of Operational Amplifiers

8.75 Consider the non-inverting amplifier of Figure 8.9. Find the error introduced in v_o when the op-amp has an input offset voltage of 2 mV. Assume the input bias currents are zero and $R_1 = R_F = 4.7 \text{ k}\Omega$. Assume that the offset voltage appears as shown in Figure 8.63.

8.76 In the circuit shown in Figure P8.76, sketch the output voltage $v_o(t)$ for the two input voltages $v_1(t)$ and $v_2(t)$. Assume $R_1 = 120 \text{ k}\Omega$, $R_2 = 150 \text{ k}\Omega$, and $C = 2 \text{ nF}$. Also assume the op-amp slew rate limit is $S_0 = 1.0 \text{ V}/\mu\text{s}$ and the capacitor is initially uncharged.

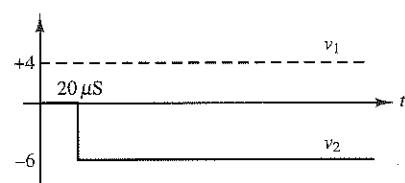
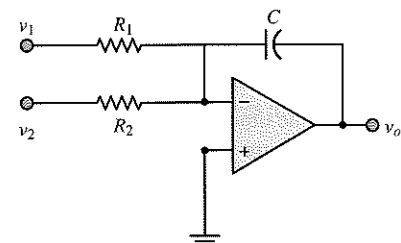


Figure P8.76

8.77 Consider a standard inverting amplifier, as shown in Figure P8.77. Assume that the offset voltage can be neglected and that the two input bias currents are equal. Find the value of R_x that eliminates the error in the output voltage due to the bias currents.

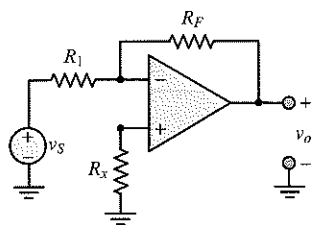


Figure P8.77

8.78 Determine the effect of the slew rate limit $S_0 = 0.5 \text{ V}/\mu\text{s}$ on the output of the isolation buffer for each of the following sinusoidal input voltages:

- $v_{in} = 0.8 \sin(2\pi \cdot 6,000t) \text{ V}$
- $v_{in} = 0.9 \sin(2\pi \cdot 7,500t) \text{ V}$
- $v_{in} = 0.9 \sin(2\pi \cdot 15,000t) \text{ V}$

8.79 In the circuit shown in Figure P8.79, derive the output voltage $v_o(t)$ as a function of $v_{in}(t)$. Compute the effect of the slew rate on the maximum slope of the output voltage. Assume v_{in} is zero but then undergoes a step increase of amplitude ΔV .

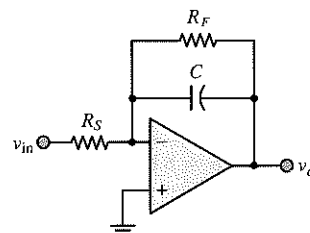


Figure P8.79

8.80 Determine the effect of the slew rate limit $S_0 = 0.5 \text{ V}/\mu\text{s}$ on the output of a non-inverting amplifier with closed-loop voltage gain $G = 10$ for a symmetric square wave v_{in} . Sketch the output waveform for each following case:

- v_{in} switches between $\pm 0.5 \text{ V}$ and $f = 500 \text{ Hz}$.
- v_{in} switches between $\pm 1.25 \text{ V}$ and $f = 5 \text{ kHz}$.
- v_{in} switches between $\pm 0.5 \text{ V}$ and $f = 25 \text{ kHz}$.

8.81 Consider a differential amplifier. We desire the common-mode output to be less than 1 percent of the differential-mode output. Find the minimum decibel common-mode rejection ratio to fulfill this requirement if the differential-mode gain $A_{dm} = 1,000$. Let

$$\begin{aligned} v_1 &= \sin(2,000\pi t) + 0.1 \sin(120\pi t) \text{ V} \\ v_2 &= \sin(2,000\pi t + 180^\circ) + 0.1 \sin(120\pi t) \text{ V} \\ v_o &= A_{dm}(v_1 - v_2) + A_{cm} \frac{v_1 + v_2}{2} \end{aligned}$$

8.82 Square wave testing can be used with operational amplifiers to estimate the *slew rate*, which is defined as the maximum rate at which the output can change (in volts per microsecond). Input and output waveforms for a non-inverting op-amp circuit are shown in Figure P8.82. As indicated, the rise time t_R of the output waveform is defined as the time it takes for that waveform to increase from 10 percent to 90 percent of its final value, or

$$t_R \triangleq t_B - t_A = -\tau(\ln 0.1 - \ln 0.9) = 2.2\tau$$

where τ is the circuit time constant. Estimate the slew rate for the op-amp.

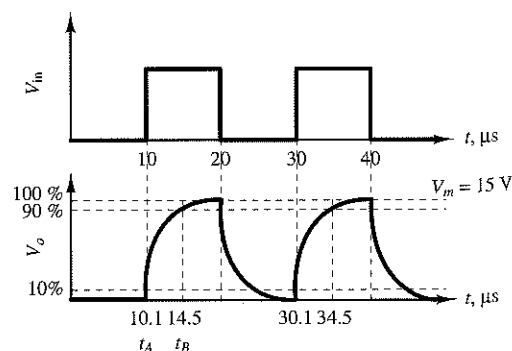


Figure P8.82

8.83 The non-ideal op-amp used in the inverting amplifier of Figure 8.8 has an open-loop voltage gain $A = 250 \times 10^3$. Assume that v^- is small but non-zero. The input terminal currents i_{in} can still be assumed zero. Apply equation 8.23 to find:

$$\frac{v_o}{v_s} = \frac{-R_F/R_S}{1 + (1/A)[(R_F + R_S)/R_S]}$$

- If $R_S = 10 \text{ k}\Omega$ and $R_F = 1 \text{ M}\Omega$, find the closed-loop voltage gain $G = v_o/v_s$.
- Repeat part a for $R_F = 10 \text{ M}\Omega$.
- Repeat part a for $R_F = 100 \text{ M}\Omega$.
- Evaluate G as $A \rightarrow \infty$ for parts a to c.

8.84 The non-ideal op-amp used in the non-inverting amplifier of Figure P8.84 has an open-loop voltage gain $A = 250 \times 10^3$. Assume $v_{in} = v^- + \Delta v$, where Δv is small but non-zero, as suggested in equation 8.23. The input terminal currents i_{in} can still be assumed zero. Find:

- The closed-loop gain v_o/v_{in} for $R_F = R_S = 7.5 \text{ k}\Omega$;
- The closed-loop gain v_o/v_{in} for $R_F = R_S = 7.5 \text{ k}\Omega$.

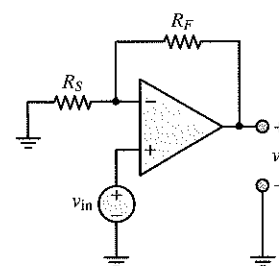


Figure P8.84

8.85 Given the unity-gain bandwidth for an ideal op-amp equal to 5.0 MHz , find the voltage gain at a frequency of $f = 500 \text{ kHz}$.

8.86 The open-loop gain A of real (non-ideal) op-amps is very large at low frequencies but decreases markedly as frequency increases. As a result, the closed-loop gain of op-amp circuits can be strongly dependent on frequency. Determine the relationship between a finite and frequency-dependent open-loop gain $A(\omega)$ and the closed-loop gain $G(\omega)$ of an inverting amplifier as a function of frequency. Plot G versus ω . Notice that $-R_F/R_S$ is the low-frequency closed-loop gain.

8.87 A sinusoidal sound (pressure) wave $p(t)$ impinges upon a condenser microphone of sensitivity S (mV/kPa). The voltage output of the microphone v_s is amplified by two cascaded inverting amplifiers to

produce an amplified signal v_o . Determine the peak amplitude of the sound wave (in dB) if $v_o = 5 \text{ V}_{\text{RMS}}$. Estimate the maximum peak magnitude of the sound wave in order that v_o not contain any saturation effects of the op-amps.

8.88 For the circuit shown in Figure P8.88, assume a non-ideal op-amp and:

$$\begin{aligned} v_{S1} &= 2.8 + 0.01 \cos(\omega t) \text{ V} \\ v_{S2} &= 3.5 - 0.007 \cos(\omega t) \text{ V} \\ A_1 &= -13 \quad A_2 = 10 \quad \omega = 4 \text{ krad/s} \end{aligned}$$

where A_1 and A_2 are the open-loop voltage gains associated with inputs v_{S1} and v_{S2} , respectively. Determine:

- Common- and differential-mode input signals.
- Common- and differential-mode closed-loop gains.
- Common- and differential-mode components of the output voltage.
- Total output voltage.
- Common-mode rejection ratio.

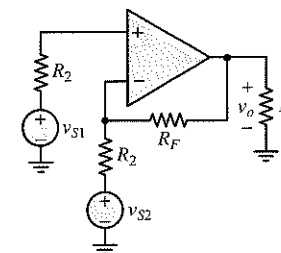


Figure P8.88

8.89 For the circuit shown in Figure P8.88, assume a non-ideal op-amp and:

$$\begin{aligned} v_{S1} &= 3.5 + 0.01 \cos(\omega t) \text{ V} \\ v_{S2} &= 3.5 - 0.01 \cos(\omega t) \text{ V} \\ A_c &= 10 \text{ dB} \quad A_d = 20 \text{ dB} \quad \omega = 4 \text{ krad/s} \end{aligned}$$

where A_c and A_d are the common- and differential-mode open-loop voltage gains, respectively. Determine:

- Common- and differential-mode input voltages.
- The voltage gains for v_{S1} and v_{S2} .
- The common-mode component and differential-mode component of the output voltage.
- The common-mode rejection ratio (CMRR), in dB.

8.90 In the circuit shown in Figure P8.90, the two voltage sources are temperature sensors with $T =$ temperature (Kelvin) and

$$v_{S1} = kT_1 \quad v_{S2} = kT_2$$

where

$$k = 120 \mu\text{V/K}$$

$$R_1 = R_3 = R_4 = 5 \text{ k}\Omega$$

$$R_2 = 3 \text{ k}\Omega \quad R_o = 600 \Omega$$

If

$$T_1 = 310 \text{ K} \quad T_2 = 335 \text{ K}$$

determine

- The voltage gains for the two input voltages.
- The common-mode and differential-mode input voltages.
- The common-mode and differential-mode gains.
- The common-mode component and the differential-mode component of the output voltage.
- The common-mode rejection ratio (CMRR), in dB.

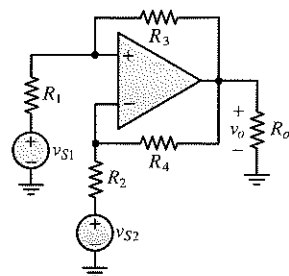


Figure P8.90

8.91 In the differential amplifier shown in Figure P8.90,

$$v_{S1} = 13 \text{ mV} \quad v_{S2} = 9 \text{ mV}$$

$$v_o = v_{oc} + v_{od}$$

$$v_{oc} = 33 \text{ mV} \quad (\text{common-mode output voltage})$$

$$v_{od} = 18 \text{ V} \quad (\text{differential-mode output voltage})$$

Determine

- The common-mode gain.
- The differential-mode gain.
- The common-mode rejection ratio, in dB.

8.92 The ideal charge amplifier discussed in the Focus on Measurements box “Charge Amplifiers” will saturate in the presence of any DC offsets. Figure P8.92 presents a practical charge amplifier in which the user is provided with a choice of three time constants— RC_F , $10RC_F$, and $100RC_F$ —which can be selected by means of a switch. Assume that $R = 0.1 \text{ M}\Omega$, and $C_F = 0.1 \mu\text{F}$. Analyze the frequency response of the practical charge amplifier for each time constant, and determine the lowest input signal frequency that can be amplified without excessive distortion for each case. Can this circuit amplify a DC signal?

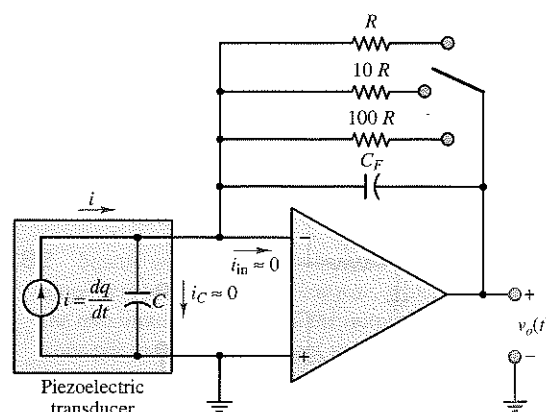


Figure P8.92

CHAPTER

9

SEMICONDUCTORS AND DIODES

Marvelous advances have taken place in the field of **solid state electronics** ever since the invention of the *diode* and *transistor*. Modern analog and digital electronic systems are possible because these discrete electronic elements have been integrated into complex devices and systems. Although discrete electronic elements have been replaced in many applications by *integrated circuits* (e.g., operational amplifiers), it is nonetheless important to understand how these elements function. The aim of Part II of this textbook is to explore the behavior and applications of diodes, transistors, and other *electronic devices*.

This chapter explains the workings of the semiconductor diode, a device that finds use in many practical circuits used in electric power systems and in high- and low-power electronic circuits. While the i - v characteristic of a diode is inherently non-linear, simple *linear models* can be used to approximate the diode characteristic and thus produce *linear circuits* that can be analyzed using the analytical tools developed in earlier chapters.